Cubic Time Dilation Model for Enhanced GNSS Accuracy and Security

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Executive Summary

Global Navigation Satellite Systems (GNSS), including GPS, GLONASS, Galileo, and BeiDou, are critical for global navigation, communication, and timing across military, scientific, and civilian domains. The precision and security of these systems rely heavily on accurate timekeeping and the correction of relativistic effects, particularly time dilation.

This document introduces a significant advancement in time dilation correction for GNSS: the cubic time dilation model. Through rigorous testing—validated by 1,000 simulations—this model has demonstrated superior accuracy compared to the traditional time dilation model. Moreover, the cubic model offers substantial benefits for GNSS security by enhancing timing precision, thereby providing a stronger defense against signal spoofing and other security threats. Implementing this model is essential for advancing GNSS accuracy, reliability, and security.

Introduction

Relativistic time dilation is a cornerstone of GNSS operations. Traditionally, time dilation corrections have relied on the well-established formula $\sqrt{1-\frac{v^2}{c^2}}$, derived from Einstein's theory of special relativity. While effective, this traditional model has limitations when applied to the extremely high precision required in modern GNSS systems.

Recent advancements in our understanding of relativistic effects have led to the development of a cubic time dilation model, $\left(1-\frac{v^2}{c^2}\right)^{-1/3}$. This model, through comprehensive testing, has been shown to reduce positional errors across a wide array of scenarios and to strengthen the overall security framework of GNSS systems.

Theoretical Foundation

Traditional Time Dilation

Einstein's theory of special relativity predicts time dilation as an object's velocity approaches the speed of light. This phenomenon is mathematically described by the equation:

$$\Delta t' = \Delta t \cdot \sqrt{1-\frac{v^2}{c^2}}$$

where Δt represents the proper time interval as observed from a reference frame at rest relative to the moving object, v denotes the velocity of the object, and c is the speed of light in a vacuum.

This formula has been the standard for correcting the timing signals emitted by GNSS satellites, ensuring the accuracy of positioning data provided to users on Earth.

Cubic Time Dilation

The cubic time dilation model refines the traditional approach by introducing a higher-order correction. It modifies the standard Lorentz factor, which is typically expressed as:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

In the cubic model, the Lorentz factor is adjusted to:

$$\gamma_{cubic} = \left(1 - \frac{v^2}{c^2}\right)^{-1/3}$$

This alteration is designed to more accurately reflect the relativistic effects experienced by objects moving at significant fractions of the speed of light, particularly in contexts requiring extreme precision, such as GNSS.

Mathematical Derivation and Justification

The cubic time dilation model is rooted in an expanded understanding of the Lorentz factor. By incorporating higher-order corrections, this model captures more subtle relativistic effects that are critical in high-precision systems.

The Lorentz factor γ can be expanded in a Taylor series for small velocities ($v \ll c$) as follows:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = 1 + \frac{1}{2}\frac{v^2}{c^2} + \frac{3}{8}\frac{v^4}{c^4} + \frac{5}{16}\frac{v^6}{c^6} + \cdots$$

The first term, 1, represents the classical, non-relativistic scenario, while subsequent terms introduce corrections that become increasingly important as the velocity v approaches the speed of light.

The traditional time dilation model truncates this expansion after the first correction term, approximating the Lorentz factor as:

$$\gamma \approx 1 + \frac{1}{2}\frac{v^2}{c^2}$$

While adequate for many applications, this approximation overlooks higher-order terms that can significantly influence time dilation, especially in contexts where precision is paramount, such as GNSS.

The cubic model addresses this by modifying the exponent from $\frac{1}{2}$ (the square root) to $\frac{1}{3}$ (the cubic root), resulting in:

$$\gamma_{cubic} = \left(1 - \frac{v^2}{c^2}\right)^{-1/3} \approx 1 + \frac{1}{3}\frac{v^2}{c^2} + \frac{2}{9}\frac{v^4}{c^4} + \cdots$$

This adjustment allows the cubic model to retain more of the series expansion's higher-order terms, offering a more precise correction for time dilation.

By incorporating these higher-order terms, the cubic model provides a more accurate representation of time dilation, particularly in scenarios where even minute errors can accumulate and lead to significant discrepancies in positioning data. This is critical in GNSS, where the precision of timekeeping directly impacts the accuracy of location data.

To validate the cubic model's effectiveness, it is instructive to compare its predictions with those of the traditional model across various velocities. Consider a scenario where a satellite moves at $v = 3 \times 10^3$ m/s:

- **Traditional Model**:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \approx 1 + \frac{1}{2}\frac{(3 \times 10^3)^2}{(3 \times 10^8)^2} \approx 1 + 1.5 \times 10^{-10}$$

- **Cubic Model**:

$$\gamma_{cubic} = \left(1 - \frac{v^2}{c^2}\right)^{-1/3} \approx 1 + \frac{1}{3}\frac{(3 \times 10^3)^2}{(3 \times 10^8)^2} + \frac{2}{9}\frac{(3 \times 10^3)^4}{(3 \times 10^8)^4} \approx 1 + 10^{-10}$$

Although the differences between the models might appear small, these corrections become increasingly significant over extended periods and numerous satellite orbits, particularly in high-precision GNSS applications.

The cubic model's enhanced precision not only improves positional accuracy but also strengthens GNSS security. By minimizing timing errors, the cubic model makes it more challenging for adversaries to exploit vulnerabilities in signal timing for spoofing or other attacks. Additionally, the cubic model offers a refined baseline for detecting anomalies, making it a critical component in securing GNSS operations.

This model remains fully consistent with the framework of special relativity, extending the application of relativistic principles to ensure higher-order effects are appropriately considered in scenarios where they are non-negligible. As a result, the cubic model is both mathematically rigorous and practically superior in environments demanding the highest levels of precision and security.

Validation and Results

A series of 1,000 simulations were conducted to rigorously evaluate the cubic model across multiple GNSS scenarios. These simulations were designed to comprehensively cover all relevant aspects of GNSS operation, ensuring the cubic model's robustness and applicability under a wide range of conditions.

The simulations encompassed:

- **Satellite Systems**: The cubic model was tested across all major GNSS platforms, including GPS, GLONASS, Galileo, and BeiDou, demonstrating consistent accuracy improvements regardless of the satellite system.
- **Atmospheric Conditions**: The model's resilience was confirmed under various atmospheric conditions, including ionospheric and tropospheric delays, which are known to affect signal propagation and time dilation corrections.
- **Orbital Scenarios**: The cubic model was evaluated in both circular and elliptical orbits with varying eccentricities, providing a comprehensive understanding of its performance across different orbital paths and velocities.
- **Satellite Anomalies**: Stress tests involving unexpected velocity changes and orbital shifts demonstrated the cubic model's robustness in handling anomalies that may occur due to unforeseen events or maneuvers.
- **Gravitational Effects**: Temporal changes in Earth's gravity field, along with gravitational perturbations from the Moon, were incorporated into the simulations to validate the cubic model's effectiveness under varying gravitational influences.
- **Cross-Platform Integration**: The cubic model's ability to integrate seamlessly with multiple GNSS systems was tested, confirming its capability to enhance overall accuracy when used in conjunction with systems like GLONASS, Galileo, and BeiDou.
- **Velocity Extremes**: Simulations included both low-velocity conditions (such as lower Earth orbits) and high-velocity scenarios (including higher orbits and deep space missions), ensuring the model's adaptability across a broad range of velocities.
- **System Maintenance**: The cubic model's performance was validated during satellite maintenance scenarios, including station-keeping maneuvers and repositioning within the satellite constellation, confirming its reliability during routine operations.
- **Ground-Based Observer Variations**: The model was also tested from the perspective of ground-based observers at different latitudes and altitudes, accounting for variations due to Earth's rotation and confirming that the cubic model maintains accuracy across diverse observational conditions.

Mars Missions, Near-Space, and Deep-Space Applications

The cubic time dilation model is especially relevant for missions beyond Earth, including Mars, near-space, and deep-space missions. As spacecraft venture

further from Earth and experience different gravitational fields and velocities, the relativistic effects on timekeeping become increasingly significant. The cubic model's ability to account for higher-order corrections makes it particularly well-suited for these environments.

- **Mars Missions**: The journey to Mars involves varying velocities as spacecraft accelerate, decelerate, and maneuver in different gravitational fields. The cubic model provides more accurate time dilation corrections, ensuring precise navigation and communication between Earth and Mars.
- **Near-Space Operations**: Operations in near-space, such as those involving satellites in geostationary orbits or lunar missions, benefit from the cubic model's enhanced accuracy in time dilation. The model ensures that even subtle relativistic effects are corrected, maintaining the precision necessary for these missions.
- **Deep-Space Exploration**: For deep-space missions, where spacecraft may travel at high velocities over long periods and through varying gravitational fields, the cubic model's corrections become even more critical. The model reduces the risk of cumulative timing errors that could lead to significant navigation inaccuracies, thereby enhancing the safety and success of these missions.

Implementation Strategy

The implementation of the cubic time dilation model in GNSS and other space applications involves several key steps to ensure its success:

Empirical Validation

Pilot testing will be conducted in a controlled GNSS environment to gather empirical data, ensuring the cubic model's effectiveness is validated against real-world conditions. Continuous comparisons with the traditional time dilation model will be made throughout this phase to confirm the cubic model's superiority in reducing positional errors. The testing period will extend to cover various operational scenarios and seasonal variations, providing a comprehensive assessment of the model's performance.

Data-driven adjustments will be applied to mitigate any discrepancies between theoretical predictions and practical outcomes. Insights gained from real-world data will be used to fine-tune the model, ensuring that the benefits observed in simulations are fully realized in practice.

Compatibility with Existing Systems

To ensure seamless integration, the model will undergo a thorough compatibility assessment with existing GNSS software and hardware. This assessment

will identify potential challenges in incorporating the cubic model into current infrastructure. If necessary, incremental updates will be deployed to avoid significant disruptions, allowing for a smooth transition. The integration plan will also include a detailed schedule to minimize any operational downtime.

Higher-Order Effects

Further theoretical research will explore the interaction between the cubic model and other relativistic effects, such as gravitational time dilation and frame-dragging. This research will ensure that the cubic model's introduction does not inadvertently complicate the broader relativistic framework within which GNSS and space exploration operate. Should any unforeseen interactions arise, adjustments will be made to either the model or its application to maintain accuracy without compromising system integrity.

Computational Overhead

An evaluation of the computational impact of the cubic model will be performed to ensure that it does not impose an excessive processing load on GNSS systems. Optimization techniques, such as algorithmic refinements or hardware acceleration, will be employed to minimize any additional computational demands. These steps will ensure that the benefits of the cubic model can be realized without compromising the system's overall efficiency.

Adoption and Stakeholder Engagement

Recognizing that introducing a new time dilation model may encounter resistance, a comprehensive communication strategy will be developed. This strategy will emphasize the clear advantages of the cubic model, supported by rigorous simulation results and empirical data. Engaging key stakeholders early in the process will be crucial, ensuring that their concerns are addressed and that they are confident in the transition. Educational materials and training sessions will be provided to facilitate the adoption process.

Sensitivity to Input Data

To address potential sensitivity to input data, the model will incorporate error-checking mechanisms and data validation procedures. These mechanisms will ensure that only the most accurate and reliable data are used in the model's calculations. Additionally, the model will be tested under varying levels of input accuracy to understand its robustness and to implement any necessary adjustments that will mitigate the impact of less precise data.

Impact on Extreme Environments

For Mars, near-space, and deep-space missions, where spacecraft encounter extreme conditions and varying gravitational fields, the cubic model's higher-order

corrections ensure precise time dilation adjustments. The model's application in these environments will be closely monitored, with further research conducted to explore any unique challenges presented by these missions. Should the cubic model demonstrate significant advantages in these scenarios, it may be further refined to optimize performance in such extreme environments.

Conclusion

The cubic time dilation model represents a significant advancement in the field of GNSS and space exploration, offering a more accurate and secure approach to timekeeping. By incorporating higher-order relativistic corrections, the cubic model provides superior accuracy over traditional time dilation models, making it particularly valuable in high-precision environments, including Earth-based GNSS, Mars missions, and deep-space exploration. The comprehensive testing and validation of the cubic model across a wide range of scenarios, including 1,000 simulations, confirm its robustness and applicability. This model not only enhances positional accuracy but also strengthens the security of GNSS systems by reducing the risk of timing errors and improving the detection of anomalies.

The successful implementation of the cubic model will require careful planning, empirical validation, and stakeholder engagement. However, the benefits it offers in terms of precision, reliability, and security make it a necessary evolution for GNSS and space exploration systems. The cubic model stands ready to meet the demands of modern and future navigation and timing systems, ensuring their continued accuracy and integrity in an increasingly complex and high-stakes environment.

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Mathematical Formulation of Time Dilation Models

Traditional Time Dilation Model

Given the orbital velocity v and the speed of light $c = 299,792.458$ km/s, the traditional time dilation factor is expressed as:

$$\text{Time Dilation Factor}_{\text{traditional}} = \sqrt{1 - \frac{v^2}{c^2}}$$

For high velocities approaching c, the time dilation factor approaches zero, leading to increasingly significant corrections.

Cubic Time Dilation Model

The cubic model modifies the traditional time dilation factor by applying a cubic root:

$$\text{Time Dilation Factor}_{\text{cubic}} = \left(1 - \frac{v^2}{c^2}\right)^{\frac{1}{3}}$$

This cubic modification results in a smoother decay of the time dilation factor as v approaches c, providing a more gradual correction.

Advantages at Extreme Velocities

Smooth Decay and Avoidance of Overcorrection

At extreme velocities, the traditional model can produce aggressive corrections due to the rapid decrease of the factor:

$$\text{Traditional}: \quad \lim_{v \to c} \sqrt{1 - \frac{v^2}{c^2}} \to 0$$

In contrast, the cubic model offers a more conservative and smoother correction:

$$\text{Cubic}: \quad \lim_{v \to c} \left(1 - \frac{v^2}{c^2}\right)^{\frac{1}{3}} \to 0$$

The cubic model's approach reduces the risk of overcorrection, making it more stable for high-velocity applications.

Higher-Order Terms and Stability

Traditional Model Expansion

The Taylor expansion for the traditional model up to fourth-order terms is:

$$\text{Time Dilation Factor}_{\text{traditional}} = 1 - \frac{1}{2}\frac{v^2}{c^2} - \frac{1}{8}\frac{v^4}{c^4} - \frac{1}{16}\frac{v^6}{c^6} - \frac{5}{128}\frac{v^8}{c^8}$$

This expansion shows that at higher velocities, additional correction terms become increasingly significant.

Cubic Model Expansion

Similarly, the Taylor expansion for the cubic model up to fourth-order terms is:

$$\text{Time Dilation Factor}_{\text{cubic}} = 1 - \frac{1}{3}\frac{v^2}{c^2} - \frac{2}{9}\frac{v^4}{c^4} - \frac{5}{27}\frac{v^6}{c^6} - \frac{10}{81}\frac{v^8}{c^8}$$

This expansion indicates that the cubic model also accounts for higher-order corrections but does so in a more gradual manner compared to the traditional model.

Satellites & How to Derive

Satellite Name	Mean Motion (revs/day)	Eccentricity	Semi-Major Axis (km)	Orbital Velocity (km/s)	Altitude (km)	Inclination (°)	Right Ascension (°)	Argument of Perigee (°)	Time Dilation Factor
BEIDOU-206	1.86228212	0.0009995	26560	3.603	21560	55.0	102.5	180.0	1.0000003
BEIDOU-212	1.86231458	0.0003615	26560	3.606	21560	55.0	105.3	180.0	1.0000003
GPS BIII-1 (PRN 04)	2.00558911	0.0030086	26560	3.868	20200	55.0	250.1	90.0	1.0000005
GSAT0201 (GALILEO 5)	1.85519401	0.1613034	26560	3.548	21560	56.0	0.0	270.0	0.9999999
BEIDOU-3 M17 (C36)	1.86228212	0.0009995	26560	3.603	21560	55.0	102.5	180.0	1.0000003
BEIDOU-3 M14 (C33)	1.86231458	0.0003615	26560	3.606	21560	55.0	105.3	180.0	1.0000003
GLONASS-K1 (PRN 03)	2.12469129	0.0020812	25464	3.874	19130	64.8	85.1	90.0	1.0000004
GALILEO-FOC FM16 (E19)	1.70550945	0.0031053	29601	3.391	23222	56.0	78.2	180.0	0.9999998
BEIDOU-3 M19 (C38)	1.86237925	0.0003279	26560	3.607	21560	55.0	109.4	180.0	1.0000003
GPS BIIA-10 (PRN 14)	2.00559574	0.0045617	26560	3.864	20200	55.0	251.7	90.0	1.0000005
GSAT0202 (GALILEO 6)	1.85513948	0.1604823	26560	3.550	21560	56.0	82.3	270.0	0.9999999
GLONASS-M (PRN 16)	2.12469095	0.0020896	25464	3.872	19130	64.8	85.1	90.0	1.0000004
BEIDOU-3 M13 (C32)	1.86237118	0.0003587	26560	3.606	21560	55.0	104.7	180.0	1.0000003
GPS BIII-3 (PRN 09)	2.00559613	0.0031420	26560	3.865	20200	55.0	248.9	90.0	1.0000005
GALILEO-FOC FM15 (E20)	1.70551029	0.0031146	29601	3.392	23222	56.0	78.6	180.0	0.9999998
BEIDOU-3 M18 (C37)	1.86232164	0.0003568	26560	3.605	21560	55.0	107.1	180.0	1.0000003

Table 1: Combined GNSS Time Dilation and Orbital Parameters

Mean Motion (n)

The mean motion n is related to the orbital period T by

$$n = \frac{2\pi}{T}$$

From Kepler's third law, the orbital period T is given by

$$T = 2\pi\sqrt{\frac{a^3}{\mu}}$$

where a is the semi-major axis, and $\mu = 398600\,\text{km}^3/\text{s}^2$ is the standard gravitational parameter for Earth.

Substituting T into the expression for n,

$$n = \sqrt{\frac{\mu}{a^3}}$$

If $a = 26560\,\text{km}$, then

$$n = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{(26560\,\text{km})^3}} = \sqrt{\frac{398600}{18730727680000}}\,\text{rad/s}$$

Convert to revolutions per day,

$$n = \frac{1}{2\pi} \times \sqrt{\frac{398600 \times (86400)^2}{(26560)^3}}\,\text{revs/day}$$

$$n \approx \frac{1}{2\pi} \times \sqrt{\frac{2.9851 \times 10^{15}}{1.8761 \times 10^{10}}}\,\text{revs/day} \approx 1.862\,\text{revs/day}$$

Orbital Velocity (v)

Orbital velocity at a given point in orbit is given by

$$v = \sqrt{\mu \left(\frac{2}{r} - \frac{1}{a} \right)}$$

where r is the distance from the center of Earth at a specific point in the orbit, and a is the semi-major axis. For a circular orbit where $r = a$, this simplifies to

$$v = \sqrt{\frac{\mu}{a}}$$

Given $a = 26560\,\text{km}$ and $\mu = 398600\,\text{km}^3/\text{s}^2$,

$$v = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{26560\,\text{km}}} \approx \sqrt{15.013}\,\text{km/s} \approx 3.873\,\text{km/s}$$

Time Dilation Factor

The relativistic time dilation due to velocity is given by the Lorentz factor

$$\gamma = \sqrt{1 - \frac{v^2}{c^2}}$$

where c is the speed of light. Assuming $v = 3.873\,\text{km/s}$,

$$\gamma \approx \sqrt{1 - \left(\frac{3.873 \times 10^3}{3 \times 10^8} \right)^2} \approx 1 - 8.35 \times 10^{-12}$$

Gravitational time dilation due to altitude is expressed as

$$\text{Gravitational Time Dilation Factor} = 1 - \frac{GM}{Rc^2}$$

where G is the gravitational constant, M is Earth's mass, and R is the distance from Earth's center.

For typical satellite orbits, these two effects—relativistic and gravitational—combine to produce a total time dilation factor that is slightly less than one. The reason for this lies in the balance between the satellite's velocity and its altitude. As a satellite moves at high velocities to maintain orbit, the Lorentz factor causes time to slow down slightly relative to an observer on Earth. However, because the satellite is also farther from Earth's center of mass, the gravitational time dilation effect partially counteracts this by causing time to run faster due to the weaker gravitational field at altitude.

Since the orbital velocities of satellites are significant but still much less than the speed of light, and the altitude is not enough to completely negate the Lorentz effect, the combined time dilation effect results in a net factor just under one. This subtle difference must be accounted for in precise satellite operations, such as those involving GPS systems, where even minute variations in time can lead to significant positional errors.

$$\text{Total Time Dilation Factor} \approx \gamma - \frac{GM}{Rc^2}$$

For typical satellite orbits, this factor is slightly less than 1.

Now calculate each satellite for Mean Motion (n) , Orbital Velocity (v) , and Time Dilation (γ) values.

Satellite: BEIDOU-3 M14 (C33)

Mean Motion (n)

$$n_{BEIDOU-3\,M14} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{(26560\,\text{km})^3}}$$

$$n_{BEIDOU-3\,M14} = \sqrt{\frac{398600}{1.8761 \times 10^{13}}}\,\text{rad/s}$$

$$n_{BEIDOU-3\,M14} = \sqrt{2.1259 \times 10^{-8}}\,\text{rad/s} \approx 1.458 \times 10^{-4}\,\text{rad/s}$$

Convert to revolutions per day:

$$n_{BEIDOU-3\,M14} = \frac{1.458 \times 10^{-4}\,\text{rad/s} \times 86400\,\text{s/day}}{2\pi} \approx 1.862\,\text{revs/day}$$

Orbital Velocity (v)

$$v_{BEIDOU-3\,M14} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{26560\,\text{km}}}$$

$$v_{BEIDOU-3\,M14} = \sqrt{15.013\,\text{km}^2/\text{s}^2} \approx 3.874\,\text{km/s}$$

Time Dilation Factor ()

$$\gamma_{BEIDOU-3\,M14} = \sqrt{1 - \frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}} - \frac{398600 \times 10^9\,\text{m}^3/\text{s}^2}{(21560 + 6371) \times 10^3\,\text{m} \times (3 \times 10^8\,\text{m/s})^2}$$

$$\gamma_{BEIDOU-3\,M14} \approx \sqrt{1 - 1.672 \times 10^{-10}} - 2.793 \times 10^{-10}$$

$$\gamma_{BEIDOU-3\,M14} \approx 1 - 8.36 \times 10^{-11} - 2.793 \times 10^{-10} \approx 1 - 3.63 \times 10^{-10}$$

Satellite: GLONASS-K1 (PRN 03)

Mean Motion (n)

$$n_{GLONASS-K1} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{(25464\,\text{km})^3}}$$

$$n_{GLONASS-K1} = \sqrt{\frac{398600}{1.647 \times 10^{13}}}\,\text{rad/s}$$

$$n_{GLONASS-K1} = \sqrt{2.42 \times 10^{-8}}\,\text{rad/s} \approx 1.55 \times 10^{-4}\,\text{rad/s}$$

Convert to revolutions per day:

$$n_{GLONASS-K1} = \frac{1.55 \times 10^{-4}\,\text{rad/s} \times 86400\,\text{s/day}}{2\pi} \approx 2.12\,\text{revs/day}$$

Orbital Velocity (v)

$$v_{GLONASS-K1} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{25464\,\text{km}}}$$

$$v_{GLONASS-K1} = \sqrt{15.66\,\text{km}^2/\text{s}^2} \approx 3.96\,\text{km/s}$$

Time Dilation Factor ()

$$\gamma_{GLONASS-K1} = \sqrt{1 - \frac{(3.96 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}} - \frac{398600 \times 10^9\,\text{m}^3/\text{s}^2}{(19130 + 6371) \times 10^3\,\text{m} \times (3 \times 10^8\,\text{m/s})^2}$$

$$\gamma_{GLONASS-K1} \approx \sqrt{1 - 1.76 \times 10^{-10}} - 3.31 \times 10^{-10}$$

$$\gamma_{GLONASS-K1} \approx 1 - 8.8 \times 10^{-11} - 3.31 \times 10^{-10} \approx 1 - 4.2 \times 10^{-10}$$

Satellite: GALILEO-FOC FM16 (E19)

Mean Motion (n)

$$n_{GALILEO-FOC\,FM16} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{(29601\,\text{km})^3}}$$

$$n_{GALILEO-FOC\,FM16} = \sqrt{\frac{398600}{2.59 \times 10^{13}}}\,\text{rad/s}$$

$$n_{GALILEO-FOC\,FM16} = \sqrt{1.54 \times 10^{-8}}\,\text{rad/s} \approx 1.24 \times 10^{-4}\,\text{rad/s}$$

Convert to revolutions per day:

$$n_{GALILEO-FOC\,FM16} = \frac{1.24 \times 10^{-4}\,\text{rad/s} \times 86400\,\text{s/day}}{2\pi} \approx 1.705\,\text{revs/day}$$

Orbital Velocity (v)

$$v_{GALILEO-FOC\,FM16} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{29601\,\text{km}}}$$

$$v_{GALILEO-FOC\,FM16} = \sqrt{13.47\,\text{km}^2/\text{s}^2} \approx 3.67\,\text{km/s}$$

Time Dilation Factor ()

$$\gamma_{GALILEO-FOC\,FM16} = \sqrt{1 - \frac{(3.67 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}} - \frac{398600 \times 10^9\,\text{m}^3/\text{s}^2}{(23222 + 6371) \times 10^3\,\text{m} \times (3 \times 10^8\,\text{m/s})^2}$$

$$\gamma_{GALILEO-FOC\,FM16} \approx \sqrt{1 - 1.5 \times 10^{-10}} - 2.98 \times 10^{-10}$$

$$\gamma_{GALILEO-FOC\,FM16} \approx 1 - 7.5 \times 10^{-11} - 2.98 \times 10^{-10} \approx 1 - 3.73 \times 10^{-10}$$

Satellite: BEIDOU-3 M19 (C38)

Mean Motion (n)

$$n_{BEIDOU-3\,M19} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{(26560\,\text{km})^3}}$$

$$n_{BEIDOU-3\,M19} = \sqrt{\frac{398600}{1.8761 \times 10^{13}}}\,\text{rad/s}$$

$$n_{BEIDOU-3\,M19} = \sqrt{2.1259 \times 10^{-8}}\,\text{rad/s} \approx 1.458 \times 10^{-4}\,\text{rad/s}$$

Convert to revolutions per day:

$$n_{BEIDOU-3\,M19} = \frac{1.458 \times 10^{-4}\,\text{rad/s} \times 86400\,\text{s/day}}{2\pi} \approx 1.862\,\text{revs/day}$$

Orbital Velocity (v)

$$v_{BEIDOU-3\,M19} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{26560\,\text{km}}}$$

$$v_{BEIDOU-3\,M19} = \sqrt{15.013\,\text{km}^2/\text{s}^2} \approx 3.874\,\text{km/s}$$

Time Dilation Factor ()

$$\gamma_{BEIDOU-3\,M19} = \sqrt{1 - \frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}} - \frac{398600 \times 10^9\,\text{m}^3/\text{s}^2}{(21560 + 6371) \times 10^3\,\text{m} \times (3 \times 10^8\,\text{m/s})^2}$$

$$\gamma_{BEIDOU-3\,M19} \approx \sqrt{1 - 1.672 \times 10^{-10}} - 2.793 \times 10^{-10}$$

$$\gamma_{BEIDOU-3\,M19} \approx 1 - 8.36 \times 10^{-11} - 2.793 \times 10^{-10} \approx 1 - 3.63 \times 10^{-10}$$

Satellite: GPS BIIA-10 (PRN 14)

Mean Motion (n)

$$n_{GPS\,BIIA-10} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{(26560\,\text{km})^3}}$$

$$n_{GPS\,BIIA-10} = \sqrt{\frac{398600}{1.8761 \times 10^{13}}}\,\text{rad/s}$$

$$n_{GPS\,BIIA-10} = \sqrt{2.1259 \times 10^{-8}}\,\text{rad/s} \approx 1.458 \times 10^{-4}\,\text{rad/s}$$

Convert to revolutions per day:

$$n_{GPS\,BIIA-10} = \frac{1.458 \times 10^{-4}\,\text{rad/s} \times 86400\,\text{s/day}}{2\pi} \approx 1.862\,\text{revs/day}$$

Orbital Velocity (v)

$$v_{GPS\,BIIA-10} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{26560\,\text{km}}}$$

$$v_{GPS\,BIIA-10} = \sqrt{15.013\,\text{km}^2/\text{s}^2} \approx 3.874\,\text{km/s}$$

Time Dilation Factor ()

$$\gamma_{GPS\,BIIA-10} = \sqrt{1 - \frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}} - \frac{398600 \times 10^9\,\text{m}^3/\text{s}^2}{(20200 + 6371) \times 10^3\,\text{m} \times (3 \times 10^8\,\text{m/s})^2}$$

$$\gamma_{GPS\,BIIA-10} \approx \sqrt{1 - 1.672 \times 10^{-10}} - 2.964 \times 10^{-10}$$

$$\gamma_{GPS\,BIIA-10} \approx 1 - 8.36 \times 10^{-11} - 2.964 \times 10^{-10} \approx 1 - 3.80 \times 10^{-10}$$

Satellite: GSAT0202 (GALILEO 6)

Mean Motion (n)

$$n_{GSAT0202} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{(26560\,\text{km})^3}}$$

$$n_{GSAT0202} = \sqrt{\frac{398600}{1.8761 \times 10^{13}}}\,\text{rad/s}$$

$$n_{GSAT0202} = \sqrt{2.1259 \times 10^{-8}}\,\text{rad/s} \approx 1.458 \times 10^{-4}\,\text{rad/s}$$

Convert to revolutions per day:

$$n_{GSAT0202} = \frac{1.458 \times 10^{-4}\,\text{rad/s} \times 86400\,\text{s/day}}{2\pi} \approx 1.862\,\text{revs/day}$$

Orbital Velocity (v)

$$v_{GSAT0202} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{26560\,\text{km}}}$$

$$v_{GSAT0202} = \sqrt{15.013\,\text{km}^2/\text{s}^2} \approx 3.874\,\text{km/s}$$

Time Dilation Factor ()

$$\gamma_{GSAT0202} = \sqrt{1 - \frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}} - \frac{398600 \times 10^9\,\text{m}^3/\text{s}^2}{(21560 + 6371) \times 10^3\,\text{m} \times (3 \times 10^8\,\text{m/s})^2}$$

$$\gamma_{GSAT0202} \approx \sqrt{1 - 1.672 \times 10^{-10}} - 2.793 \times 10^{-10}$$

$$\gamma_{GSAT0202} \approx 1 - 8.36 \times 10^{-11} - 2.793 \times 10^{-10} \approx 1 - 3.63 \times 10^{-10}$$

Satellite: GLONASS-M (PRN 16)

Mean Motion (n)

$$n_{GLONASS-M} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{(25464\,\text{km})^3}}$$

$$n_{GLONASS-M} = \sqrt{\frac{398600}{1.647 \times 10^{13}}}\,\text{rad/s}$$

$$n_{GLONASS-M} = \sqrt{2.42 \times 10^{-8}}\,\text{rad/s} \approx 1.55 \times 10^{-4}\,\text{rad/s}$$

Convert to revolutions per day:

$$n_{GLONASS-M} = \frac{1.55 \times 10^{-4}\,\text{rad/s} \times 86400\,\text{s/day}}{2\pi} \approx 2.12\,\text{revs/day}$$

Orbital Velocity (v)

$$v_{GLONASS-M} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{25464\,\text{km}}}$$

$$v_{GLONASS-M} = \sqrt{15.66\,\text{km}^2/\text{s}^2} \approx 3.96\,\text{km/s}$$

Time Dilation Factor ()

$$\gamma_{GLONASS-M} = \sqrt{1 - \frac{(3.96 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}} - \frac{398600 \times 10^9\,\text{m}^3/\text{s}^2}{(19130 + 6371) \times 10^3\,\text{m} \times (3 \times 10^8\,\text{m/s})^2}$$

$$\gamma_{GLONASS-M} \approx \sqrt{1 - 1.76 \times 10^{-10}} - 3.31 \times 10^{-10}$$

$$\gamma_{GLONASS-M} \approx 1 - 8.8 \times 10^{-11} - 3.31 \times 10^{-10} \approx 1 - 4.2 \times 10^{-10}$$

Satellite: BEIDOU-3 M13 (C32)

Mean Motion (n)

$$n_{BEIDOU-3\,M13} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{(26560\,\text{km})^3}}$$

$$n_{BEIDOU-3\,M13} = \sqrt{\frac{398600}{1.8761 \times 10^{13}}}\,\text{rad/s}$$

$$n_{BEIDOU-3\,M13} = \sqrt{2.1259 \times 10^{-8}}\,\text{rad/s} \approx 1.458 \times 10^{-4}\,\text{rad/s}$$

Convert to revolutions per day:

$$n_{BEIDOU-3\,M13} = \frac{1.458 \times 10^{-4}\,\text{rad/s} \times 86400\,\text{s/day}}{2\pi} \approx 1.862\,\text{revs/day}$$

Orbital Velocity (v)

$$v_{BEIDOU-3\,M13} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{26560\,\text{km}}}$$

$$v_{BEIDOU-3\,M13} = \sqrt{15.013\,\text{km}^2/\text{s}^2} \approx 3.874\,\text{km/s}$$

Time Dilation Factor ()

$$\gamma_{BEIDOU-3\,M13} = \sqrt{1 - \frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}} - \frac{398600 \times 10^9\,\text{m}^3/\text{s}^2}{(21560 + 6371) \times 10^3\,\text{m} \times (3 \times 10^8\,\text{m/s})^2}$$

$$\gamma_{BEIDOU-3\,M13} \approx \sqrt{1 - 1.672 \times 10^{-10}} - 2.793 \times 10^{-10}$$

$$\gamma_{BEIDOU-3\,M13} \approx 1 - 8.36 \times 10^{-11} - 2.793 \times 10^{-10} \approx 1 - 3.63 \times 10^{-10}$$

Satellite: GPS BIII-3 (PRN 09)

Mean Motion (n)

$$n_{GPS\,BIII-3} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{(26560\,\text{km})^3}}$$

$$n_{GPS\,BIII-3} = \sqrt{\frac{398600}{1.8761 \times 10^{13}}}\,\text{rad/s}$$

$$n_{GPS\,BIII-3} = \sqrt{2.1259 \times 10^{-8}}\,\text{rad/s} \approx 1.458 \times 10^{-4}\,\text{rad/s}$$

Convert to revolutions per day:

$$n_{GPS\,BIII-3} = \frac{1.458 \times 10^{-4}\,\text{rad/s} \times 86400\,\text{s/day}}{2\pi} \approx 1.862\,\text{revs/day}$$

Orbital Velocity (v)

$$v_{GPS\,BIII-3} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{26560\,\text{km}}}$$

$$v_{GPS\,BIII-3} = \sqrt{15.013\,\text{km}^2/\text{s}^2} \approx 3.874\,\text{km/s}$$

Time Dilation Factor ()

$$\gamma_{GPS\,BIII-3} = \sqrt{1 - \frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}} - \frac{398600 \times 10^9\,\text{m}^3/\text{s}^2}{(20200 + 6371) \times 10^3\,\text{m} \times (3 \times 10^8\,\text{m/s})^2}$$

$$\gamma_{GPS\,BIII-3} \approx \sqrt{1 - 1.672 \times 10^{-10}} - 2.964 \times 10^{-10}$$

$$\gamma_{GPS\,BIII-3} \approx 1 - 8.36 \times 10^{-11} - 2.964 \times 10^{-10} \approx 1 - 3.80 \times 10^{-10}$$

Satellite: GALILEO-FOC FM15 (E20)

Mean Motion (n)

$$n_{GALILEO-FOC\,FM15} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{(29601\,\text{km})^3}}$$

$$n_{GALILEO-FOC\,FM15} = \sqrt{\frac{398600}{2.59 \times 10^{13}}}\,\text{rad/s}$$

$$n_{GALILEO-FOC\,FM15} = \sqrt{1.54 \times 10^{-8}}\,\text{rad/s} \approx 1.24 \times 10^{-4}\,\text{rad/s}$$

Convert to revolutions per day:

$$n_{GALILEO-FOC\,FM15} = \frac{1.24 \times 10^{-4}\,\text{rad/s} \times 86400\,\text{s/day}}{2\pi} \approx 1.705\,\text{revs/day}$$

Orbital Velocity (v)

$$v_{GALILEO-FOC\,FM15} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{29601\,\text{km}}}$$

$$v_{GALILEO-FOC\,FM15} = \sqrt{13.47\,\text{km}^2/\text{s}^2} \approx 3.67\,\text{km/s}$$

Time Dilation Factor ()

$$\gamma_{GALILEO-FOC\,FM15} = \sqrt{1 - \frac{(3.67 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}} - \frac{398600 \times 10^9\,\text{m}^3/\text{s}^2}{(23222 + 6371) \times 10^3\,\text{m} \times (3 \times 10^8\,\text{m/s})^2}$$

$$\gamma_{GALILEO-FOC\,FM15} \approx \sqrt{1 - 1.5 \times 10^{-10}} - 2.98 \times 10^{-10}$$

$$\gamma_{GALILEO-FOC\,FM15} \approx 1 - 7.5 \times 10^{-11} - 2.98 \times 10^{-10} \approx 1 - 3.73 \times 10^{-10}$$

Satellite: BEIDOU-3 M18 (C37)

Mean Motion (n)

$$n_{BEIDOU-3\,M18} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{(26560\,\text{km})^3}}$$

$$n_{BEIDOU-3\,M18} = \sqrt{\frac{398600}{1.8761 \times 10^{13}}}\,\text{rad/s}$$

$$n_{BEIDOU-3\,M18} = \sqrt{2.1259 \times 10^{-8}}\,\text{rad/s} \approx 1.458 \times 10^{-4}\,\text{rad/s}$$

Convert to revolutions per day:

$$n_{BEIDOU-3\,M18} = \frac{1.458 \times 10^{-4}\,\text{rad/s} \times 86400\,\text{s/day}}{2\pi} \approx 1.862\,\text{revs/day}$$

Orbital Velocity (v)

$$v_{BEIDOU-3\,M18} = \sqrt{\frac{398600\,\text{km}^3/\text{s}^2}{26560\,\text{km}}}$$

$$v_{BEIDOU-3\,M18} = \sqrt{15.013\,\text{km}^2/\text{s}^2} \approx 3.874\,\text{km/s}$$

Time Dilation Factor ()

$$\gamma_{BEIDOU-3\,M18} = \sqrt{1 - \frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}} - \frac{398600 \times 10^9\,\text{m}^3/\text{s}^2}{(21560 + 6371) \times 10^3\,\text{m} \times (3 \times 10^8\,\text{m/s})^2}$$

$$\gamma_{BEIDOU-3\,M18} \approx \sqrt{1 - 1.672 \times 10^{-10}} - 2.793 \times 10^{-10}$$

$$\gamma_{BEIDOU-3\,M18} \approx 1 - 8.36 \times 10^{-11} - 2.793 \times 10^{-10} \approx 1 - 3.63 \times 10^{-10}$$

Apply the Cubic Time Dilation Factor transforms.

Cubic Time Dilation Calculations for Each Satellite

$$\gamma_{\text{cubic}} = \sqrt[3]{\frac{1}{1 - \frac{v^2}{c^2}}}$$

Satellite: BEIDOU-3 M14 (C33)
Where, $v = 3.874 \times 10^3\,\text{m/s}$ (orbital velocity) $c = 3 \times 10^8\,\text{m/s}$ (speed of light)
Substitute the values into the cubic time dilation equation

$$\gamma_{\text{cubic}} = \sqrt[3]{\frac{1}{1 - \frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}}}$$

Calculate the velocity term

$$\frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2} = 1.668 \times 10^{-10}$$

Substitute this result back into the cubic time dilation factor

$$\gamma_{\text{cubic}} = \sqrt[3]{\frac{1}{1 - 1.668 \times 10^{-10}}}$$

Simplify the expression

$$\gamma_{\text{cubic}} \approx \sqrt[3]{1 + 1.668 \times 10^{-10}} \approx 1 + \frac{1.668 \times 10^{-10}}{3} = 1 + 5.56 \times 10^{-11}$$

The cubic time dilation factor for BEIDOU-3 M14 (C33) is

$$\gamma_{\text{cubic}} \approx 1 + 5.56 \times 10^{-11}$$

Satellite: GLONASS-K1 (PRN 03)
Start with the cubic time dilation factor

$$\gamma_{\text{cubic}} = \sqrt[3]{\frac{1}{1 - \frac{v^2}{c^2}}}$$

Where, $v = 3.96 \times 10^3\,\text{m/s}$ (orbital velocity) $c = 3 \times 10^8\,\text{m/s}$ (speed of light)
Substitute the values into the cubic time dilation equation

$$\gamma_{\text{cubic}} = \sqrt[3]{\frac{1}{1 - \frac{(3.96 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}}}$$

Calculate the velocity term

$$\frac{(3.96 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2} = 1.742 \times 10^{-10}$$

Substitute this result back into the cubic time dilation factor

$$\gamma_{\text{cubic}} = \sqrt[3]{\frac{1}{1 - 1.742 \times 10^{-10}}}$$

Simplify the expression

$$\gamma_{\text{cubic}} \approx \sqrt[3]{1 + 1.742 \times 10^{-10}} \approx 1 + \frac{1.742 \times 10^{-10}}{3} = 1 + 5.81 \times 10^{-11}$$

The cubic time dilation factor for GLONASS-K1 (PRN 03) is

$$\gamma_{\text{cubic}} \approx 1 + 5.81 \times 10^{-11}$$

Satellite: GALILEO-FOC FM15 (E20)
Start with the cubic time dilation factor

$$\gamma_{\text{cubic}} = \sqrt[3]{\frac{1}{1 - \frac{v^2}{c^2}}}$$

Where, $v = 3.67 \times 10^3\,\text{m/s}$ (orbital velocity) $c = 3 \times 10^8\,\text{m/s}$ (speed of light)
Substitute the values into the cubic time dilation equation

$$\gamma_{\text{cubic}} = \sqrt[3]{\frac{1}{1 - \frac{(3.67 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}}}$$

Calculate the velocity term

$$\frac{(3.67 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2} = 1.497 \times 10^{-10}$$

Substitute this result back into the cubic time dilation factor

$$\gamma_{\text{cubic}} = \sqrt[3]{\frac{1}{1 - 1.497 \times 10^{-10}}}$$

Simplify the expression

$$\gamma_{\text{cubic}} \approx \sqrt[3]{1 + 1.497 \times 10^{-10}} \approx 1 + \frac{1.497 \times 10^{-10}}{3} = 1 + 4.99 \times 10^{-11}$$

The cubic time dilation factor for GALILEO-FOC FM15 (E20) is

$$\gamma_{\text{cubic}} \approx 1 + 4.99 \times 10^{-11}$$

Satellite: BEIDOU-3 M18 (C37)
Start with the cubic time dilation factor

$$\gamma_{\text{cubic}} = \sqrt[3]{\frac{1}{1 - \frac{v^2}{c^2}}}$$

Where, $v = 3.874 \times 10^3\,\text{m/s}$ (orbital velocity) $c = 3 \times 10^8\,\text{m/s}$ (speed of light)
Substitute the values into the cubic time dilation equation

$$\gamma_{\text{cubic}} = \sqrt[3]{\frac{1}{1 - \frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2}}}$$

Calculate the velocity term

$$\frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2} = 1.668 \times 10^{-10}$$

Substitute this result back into the cubic time dilation factor

$$\gamma_{\text{cubic}} = \sqrt[3]{\frac{1}{1 - 1.668 \times 10^{-10}}}$$

Simplify the expression

$$\gamma_{\text{cubic}} \approx \sqrt[3]{1 + 1.668 \times 10^{-10}} \approx 1 + \frac{1.668 \times 10^{-10}}{3} = 1 + 5.56 \times 10^{-11}$$

The cubic time dilation factor for BEIDOU-3 M18 (C37) is

$$\gamma_{\text{cubic}} \approx 1 + 5.56 \times 10^{-11}$$

Traditional Time Dilation Calculations (Second-Order Expansion) for All Satellites

Start with the traditional time dilation factor

$$\gamma_{\text{traditional}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$$

Where, $v = a \times 10^b\,\text{m/s}$ (orbital velocity) $c = 3 \times 10^8\,\text{m/s}$ (speed of light)

Recall the following transformation,

Equivalence of $-\frac{1}{2}$ Exponent and Square Root

In algebra, the expression $x^{-\frac{1}{2}}$ is equivalent to the reciprocal of the square root of x:

$$x^{-\frac{1}{2}} = \frac{1}{\sqrt{x}}$$

This equivalence is applied in the traditional time dilation factor:

$$\gamma_{\text{traditional}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

The $-\frac{1}{2}$ exponent indicates taking the square root and then the reciprocal.

Which leads into

Taylor Expansion of Time Dilation

The traditional time dilation factor, $\gamma_{\text{traditional}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$, is a key expression in special relativity, describing how time slows down as an object's velocity v approaches the speed of light c. To make this expression more practical for calculations, especially in systems where the velocity v is much less than c, we expand it into a Taylor series around $v/c = 0$.

The second-order Taylor expansion of the time dilation factor is given by

$$\gamma_{\text{traditional}} \approx 1 + \frac{1}{2}\left(\frac{v^2}{c^2}\right) + \frac{3}{8}\left(\frac{v^4}{c^4}\right)$$

This expansion allows us to approximate the time dilation effect more efficiently, capturing the essential behavior of the function while simplifying the computation.

Rationale for Using the Second-Order Form

The choice to use the second-order expansion is driven by the need to balance accuracy with computational efficiency. The first-order expansion $\gamma \approx 1 + \frac{1}{2}\left(\frac{v^2}{c^2}\right)$ provides a basic approximation but omits the next significant term that becomes relevant at higher velocities, though still small compared to c.

Including the second-order term $\frac{3}{8}\left(\frac{v^4}{c^4}\right)$ refines the approximation, improving the accuracy without introducing significant additional complexity. This makes the second-order expansion particularly suitable for systems like GNSS, where the velocity of satellites is high but still small compared to the speed of light.

The second-order approximation offers enhanced precision, ensuring that calculations remain accurate enough for real-time applications such as satellite navigation and timing while avoiding the computational burden of higher-order terms.

Moving forward..

Satellite: BEIDOU-3 M14 (C33)
Start with the traditional time dilation factor

$$\gamma_{\text{traditional}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$$

Expand the traditional time dilation factor to second order

$$\gamma_{\text{traditional}} \approx 1 + \frac{1}{2}\left(\frac{v^2}{c^2}\right) + \frac{3}{8}\left(\frac{v^4}{c^4}\right)$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2} = \frac{15.015 \times 10^6\,\text{m}^2/\text{s}^2}{9 \times 10^{16}\,\text{m}^2/\text{s}^2} = 1.668 \times 10^{-10}$$

Calculate the second term

$$\left(\frac{v^4}{c^4}\right) = \left(1.668 \times 10^{-10}\right)^2 = 2.782 \times 10^{-20}$$

Substitute these into the expanded expression

$$\gamma_{\text{traditional}} \approx 1 + \frac{1.668 \times 10^{-10}}{2} + \frac{3 \times 2.782 \times 10^{-20}}{8}$$

Simplify the expression

$$\gamma_{\text{traditional}} \approx 1 + 8.34 \times 10^{-11} + 1.043 \times 10^{-20} \approx 1 + 8.34 \times 10^{-11}$$

The traditional time dilation factor for BEIDOU-3 M14 (C33) is

$$\gamma_{\text{traditional}} \approx 1 + 8.34 \times 10^{-11}$$

Satellite: GLONASS-K1 (PRN 03)
Start with the traditional time dilation factor

$$\gamma_{\text{traditional}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$$

Where, $v = 3.96 \times 10^3\,\text{m/s}$ (orbital velocity) $c = 3 \times 10^8\,\text{m/s}$ (speed of light)
Expand the traditional time dilation factor to second order

$$\gamma_{\text{traditional}} \approx 1 + \frac{1}{2}\left(\frac{v^2}{c^2}\right) + \frac{3}{8}\left(\frac{v^4}{c^4}\right)$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.96 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2} = \frac{15.6816 \times 10^6\,\text{m}^2/\text{s}^2}{9 \times 10^{16}\,\text{m}^2/\text{s}^2} = 1.742 \times 10^{-10}$$

Calculate the second term

$$\left(\frac{v^4}{c^4}\right) = \left(1.742 \times 10^{-10}\right)^2 = 3.033 \times 10^{-20}$$

Substitute these into the expanded expression

$$\gamma_{\text{traditional}} \approx 1 + \frac{1.742 \times 10^{-10}}{2} + \frac{3 \times 3.033 \times 10^{-20}}{8}$$

Simplify the expression

$$\gamma_{\text{traditional}} \approx 1 + 8.71 \times 10^{-11} + 1.137 \times 10^{-20} \approx 1 + 8.71 \times 10^{-11}$$

The traditional time dilation factor for GLONASS-K1 (PRN 03) is

$$\gamma_{\text{traditional}} \approx 1 + 8.71 \times 10^{-11}$$

Satellite: GALILEO-FOC FM15 (E20)
Start with the traditional time dilation factor

$$\gamma_{\text{traditional}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$$

Where, $v = 3.67 \times 10^3\,\text{m/s}$ (orbital velocity) $c = 3 \times 10^8\,\text{m/s}$ (speed of light)
Expand the traditional time dilation factor to second order

$$\gamma_{\text{traditional}} \approx 1 + \frac{1}{2}\left(\frac{v^2}{c^2}\right) + \frac{3}{8}\left(\frac{v^4}{c^4}\right)$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.67 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2} = \frac{13.4689 \times 10^6\,\text{m}^2/\text{s}^2}{9 \times 10^{16}\,\text{m}^2/\text{s}^2} = 1.497 \times 10^{-10}$$

Calculate the second term

$$\left(\frac{v^4}{c^4}\right) = \left(1.497 \times 10^{-10}\right)^2 = 2.241 \times 10^{-20}$$

Substitute these into the expanded expression

$$\gamma_{\text{traditional}} \approx 1 + \frac{1.497 \times 10^{-10}}{2} + \frac{3 \times 2.241 \times 10^{-20}}{8}$$

Simplify the expression

$$\gamma_{\text{traditional}} \approx 1 + 7.49 \times 10^{-11} + 8.403 \times 10^{-21} \approx 1 + 7.49 \times 10^{-11}$$

The traditional time dilation factor for GALILEO-FOC FM15 (E20) is

$$\gamma_{\text{traditional}} \approx 1 + 7.49 \times 10^{-11}$$

Satellite: BEIDOU-3 M18 (C37)
Start with the traditional time dilation factor

$$\gamma_{\text{traditional}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$$

Where, $v = 3.874 \times 10^3\,\text{m/s}$ (orbital velocity) $c = 3 \times 10^8\,\text{m/s}$ (speed of light)
Expand the traditional time dilation factor to second order

$$\gamma_{\text{traditional}} \approx 1 + \frac{1}{2}\left(\frac{v^2}{c^2}\right) + \frac{3}{8}\left(\frac{v^4}{c^4}\right)$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2} = \frac{15.015 \times 10^6\,\text{m}^2/\text{s}^2}{9 \times 10^{16}\,\text{m}^2/\text{s}^2} = 1.668 \times 10^{-10}$$

Calculate the second term

$$\left(\frac{v^4}{c^4}\right) = \left(1.668 \times 10^{-10}\right)^2 = 2.782 \times 10^{-20}$$

Substitute these into the expanded expression

$$\gamma_{\text{traditional}} \approx 1 + \frac{1.668 \times 10^{-10}}{2} + \frac{3 \times 2.782 \times 10^{-20}}{8}$$

Simplify the expression

$$\gamma_{\text{traditional}} \approx 1 + 8.34 \times 10^{-11} + 1.043 \times 10^{-20} \approx 1 + 8.34 \times 10^{-11}$$

The traditional time dilation factor for BEIDOU-3 M18 (C37) is

$$\gamma_{\text{traditional}} \approx 1 + 8.34 \times 10^{-11}$$

Second-Order Expansion for the Cubic Model for Each Satellite

Satellite: BEIDOU-3 M14 (C33)

Start with the cubic time dilation factor

$$\gamma_{\text{cubic}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{3}}$$

Expand to second order

$$\gamma_{\text{cubic}} \approx 1 + \frac{1}{3}\left(\frac{v^2}{c^2}\right) + \frac{2}{9}\left(\frac{v^4}{c^4}\right)$$

For BEIDOU-3 M14, the velocity is given by

$$v = 3.874 \times 10^3\,\text{m/s}, \quad c = 3 \times 10^8\,\text{m/s}$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2} = 1.668 \times 10^{-10}$$

Calculate the second term

$$\left(\frac{v^4}{c^4}\right) = \left(1.668 \times 10^{-10}\right)^2 = 2.782 \times 10^{-20}$$

Substitute these into the expanded cubic time dilation factor

$$\gamma_{\text{cubic}} \approx 1 + \frac{1.668 \times 10^{-10}}{3} + \frac{2 \times 2.782 \times 10^{-20}}{9}$$

Simplify the expression

$$\gamma_{\text{cubic}} \approx 1 + 5.56 \times 10^{-11} + 6.18 \times 10^{-21} \approx 1 + 5.56 \times 10^{-11}$$

The second-order cubic time dilation factor for BEIDOU-3 M14 is

$$\gamma_{\text{cubic}} \approx 1 + 5.56 \times 10^{-11}$$

Satellite: GLONASS-K1 (PRN 03)

Start with the cubic time dilation factor

$$\gamma_{\text{cubic}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{3}}$$

Expand to second order

$$\gamma_{\text{cubic}} \approx 1 + \frac{1}{3}\left(\frac{v^2}{c^2}\right) + \frac{2}{9}\left(\frac{v^4}{c^4}\right)$$

For GLONASS-K1, the velocity is given by

$$v = 3.96 \times 10^3\,\text{m/s}, \quad c = 3 \times 10^8\,\text{m/s}$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.96 \times 10^3\,\mathrm{m/s})^2}{(3 \times 10^8\,\mathrm{m/s})^2} = 1.742 \times 10^{-10}$$

Calculate the second term

$$\left(\frac{v^4}{c^4}\right) = \left(1.742 \times 10^{-10}\right)^2 = 3.033 \times 10^{-20}$$

Substitute these into the expanded cubic time dilation factor

$$\gamma_{\text{cubic}} \approx 1 + \frac{1.742 \times 10^{-10}}{3} + \frac{2 \times 3.033 \times 10^{-20}}{9}$$

Simplify the expression

$$\gamma_{\text{cubic}} \approx 1 + 5.81 \times 10^{-11} + 6.74 \times 10^{-21} \approx 1 + 5.81 \times 10^{-11}$$

The second-order cubic time dilation factor for GLONASS-K1 is

$$\gamma_{\text{cubic}} \approx 1 + 5.81 \times 10^{-11}$$

Satellite: GALILEO-FOC FM15 (E20)
Start with the cubic time dilation factor

$$\gamma_{\text{cubic}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{3}}$$

Expand to second order

$$\gamma_{\text{cubic}} \approx 1 + \frac{1}{3}\left(\frac{v^2}{c^2}\right) + \frac{2}{9}\left(\frac{v^4}{c^4}\right)$$

For GALILEO-FOC FM15, the velocity is given by

$$v = 3.67 \times 10^3\,\mathrm{m/s}, \quad c = 3 \times 10^8\,\mathrm{m/s}$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.67 \times 10^3\,\mathrm{m/s})^2}{(3 \times 10^8\,\mathrm{m/s})^2} = 1.497 \times 10^{-10}$$

Calculate the second term

$$\left(\frac{v^4}{c^4}\right) = \left(1.497 \times 10^{-10}\right)^2 = 2.241 \times 10^{-20}$$

Substitute these into the expanded cubic time dilation factor

$$\gamma_{\text{cubic}} \approx 1 + \frac{1.497 \times 10^{-10}}{3} + \frac{2 \times 2.241 \times 10^{-20}}{9}$$

Simplify the expression

$$\gamma_{\text{cubic}} \approx 1 + 4.99 \times 10^{-11} + 4.98 \times 10^{-21} \approx 1 + 4.99 \times 10^{-11}$$

The second-order cubic time dilation factor for GALILEO-FOC FM15 is

$$\gamma_{\text{cubic}} \approx 1 + 4.99 \times 10^{-11}$$

Satellite: BEIDOU-3 M18 (C37)
Start with the cubic time dilation factor

$$\gamma_{\text{cubic}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{3}}$$

Expand to second order

$$\gamma_{\text{cubic}} \approx 1 + \frac{1}{3}\left(\frac{v^2}{c^2}\right) + \frac{2}{9}\left(\frac{v^4}{c^4}\right)$$

For BEIDOU-3 M18, the velocity is given by

$$v = 3.874 \times 10^3\,\text{m/s}, \quad c = 3 \times 10^8\,\text{m/s}$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2} = 1.668 \times 10^{-10}$$

Calculate the second term

$$\left(\frac{v^4}{c^4}\right) = \left(1.668 \times 10^{-10}\right)^2 = 2.782 \times 10^{-20}$$

Substitute these into the expanded cubic time dilation factor

$$\gamma_{\text{cubic}} \approx 1 + \frac{1.668 \times 10^{-10}}{3} + \frac{2 \times 2.782 \times 10^{-20}}{9}$$

Simplify the expression

$$\gamma_{\text{cubic}} \approx 1 + 5.56 \times 10^{-11} + 6.18 \times 10^{-21} \approx 1 + 5.56 \times 10^{-11}$$

The second-order cubic time dilation factor for BEIDOU-3 M18 is

$$\gamma_{\text{cubic}} \approx 1 + 5.56 \times 10^{-11}$$

Commentary on High-Order Expansions

In both the traditional and cubic time dilation models, expanding the expressions into a Taylor series allows us to approximate the effects of velocity on time dilation with greater precision. While it is theoretically possible to continue this expansion indefinitely, or to include as many as 8 to 12 terms, practical considerations necessitate a different approach.

Higher-order terms in the expansion do contribute to increased accuracy, especially at velocities that are a significant fraction of the speed of light. However, each additional term increases the computational complexity, requiring more processing power and time. For most practical applications, such as in GNSS systems, the added precision from including terms beyond the fourth order does not justify the significant increase in computational resources.

Capping the expansion at the fourth order strikes a balance between accuracy and computational efficiency. This approach provides sufficient precision for current technological needs without overwhelming the available computational resources. The choice of a fourth-order expansion is particularly relevant in real-time systems where processing speed is critical.

In the future, as computational power continues to advance, it may become feasible to include higher-order terms or even approach an infinite series expansion. However, with current technological constraints, the fourth-order expansion offers a practical solution that meets the needs of modern satellite navigation and positioning systems without imposing undue burdens on the hardware and software.

Fourth-Order Expansion for the Traditional Model

Satellite: BEIDOU-3 M14 (C33)

Start with the traditional time dilation factor

$$\gamma_{\text{traditional}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$$

Expand to fourth order

$$\gamma_{\text{traditional}} \approx 1 + \frac{1}{2}\left(\frac{v^2}{c^2}\right) + \frac{3}{8}\left(\frac{v^4}{c^4}\right) + \frac{5}{16}\left(\frac{v^6}{c^6}\right) + \frac{35}{128}\left(\frac{v^8}{c^8}\right)$$

For BEIDOU-3 M14, the velocity is given by

$$v = 3.874 \times 10^3 \,\text{m/s}, \quad c = 3 \times 10^8 \,\text{m/s}$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.874 \times 10^3 \,\text{m/s})^2}{(3 \times 10^8 \,\text{m/s})^2} = \frac{15.015 \times 10^6 \,\text{m}^2/\text{s}^2}{9 \times 10^{16} \,\text{m}^2/\text{s}^2} = 1.668 \times 10^{-10}$$

Calculate the second term

$$\frac{v^4}{c^4} = \frac{(1.668 \times 10^{-10})^2 \,\text{m}^4/\text{s}^4}{(9 \times 10^{16})^2 \,\text{m}^4/\text{s}^4} = 2.782 \times 10^{-20}$$

Calculate the third term

$$\frac{v^6}{c^6} = \frac{(1.668 \times 10^{-10})^3 \,\text{m}^6/\text{s}^6}{(9 \times 10^{16})^3 \,\text{m}^6/\text{s}^6} = 4.635 \times 10^{-30}$$

Calculate the fourth term

$$\frac{v^8}{c^8} = \frac{(1.668 \times 10^{-10})^4 \,\text{m}^8/\text{s}^8}{(9 \times 10^{16})^4 \,\text{m}^8/\text{s}^8} = 7.717 \times 10^{-40}$$

Substitute these into the expanded traditional time dilation factor

$$\gamma_{\text{traditional}} \approx 1 + \frac{1.668 \times 10^{-10}}{2} + \frac{3 \times 2.782 \times 10^{-20}}{8} + \frac{5 \times 4.635 \times 10^{-30}}{16} + \frac{35 \times 7.717 \times 10^{-40}}{128}$$

Simplify the expression

$$\gamma_{\text{traditional}} \approx 1 + 8.34 \times 10^{-11} + 1.043 \times 10^{-20} + 1.448 \times 10^{-30} + 2.111 \times 10^{-40}$$

The fourth-order traditional time dilation factor for BEIDOU-3 M14 is

$$\gamma_{\text{traditional}} \approx 1 + 8.34 \times 10^{-11}$$

Satellite: GLONASS-K1 (PRN 03)

Start with the traditional time dilation factor

$$\gamma_{\text{traditional}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$$

Expand to fourth order

$$\gamma_{\text{traditional}} \approx 1 + \frac{1}{2}\left(\frac{v^2}{c^2}\right) + \frac{3}{8}\left(\frac{v^4}{c^4}\right) + \frac{5}{16}\left(\frac{v^6}{c^6}\right) + \frac{35}{128}\left(\frac{v^8}{c^8}\right)$$

For GLONASS-K1, the velocity is given by

$$v = 3.96 \times 10^3 \,\text{m/s}, \quad c = 3 \times 10^8 \,\text{m/s}$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.96 \times 10^3\,\mathrm{m/s})^2}{(3 \times 10^8\,\mathrm{m/s})^2} = \frac{15.6816 \times 10^6\,\mathrm{m^2/s^2}}{9 \times 10^{16}\,\mathrm{m^2/s^2}} = 1.742 \times 10^{-10}$$

Calculate the second term

$$\frac{v^4}{c^4} = \frac{(1.742 \times 10^{-10})^2\,\mathrm{m^4/s^4}}{(9 \times 10^{16})^2\,\mathrm{m^4/s^4}} = 3.033 \times 10^{-20}$$

Calculate the third term

$$\frac{v^6}{c^6} = \frac{(1.742 \times 10^{-10})^3\,\mathrm{m^6/s^6}}{(9 \times 10^{16})^3\,\mathrm{m^6/s^6}} = 5.282 \times 10^{-30}$$

Calculate the fourth term

$$\frac{v^8}{c^8} = \frac{(1.742 \times 10^{-10})^4\,\mathrm{m^8/s^8}}{(9 \times 10^{16})^4\,\mathrm{m^8/s^8}} = 9.199 \times 10^{-40}$$

Substitute these into the expanded traditional time dilation factor

$$\gamma_{\text{traditional}} \approx 1 + \frac{1.742 \times 10^{-10}}{2} + \frac{3 \times 3.033 \times 10^{-20}}{8} + \frac{5 \times 5.282 \times 10^{-30}}{16} + \frac{35 \times 9.199 \times 10^{-40}}{128}$$

Simplify the expression

$$\gamma_{\text{traditional}} \approx 1 + 8.71 \times 10^{-11} + 1.137 \times 10^{-20} + 1.651 \times 10^{-30} + 2.524 \times 10^{-40}$$

The fourth-order traditional time dilation factor for GLONASS-K1 is

$$\gamma_{\text{traditional}} \approx 1 + 8.71 \times 10^{-11}$$

Satellite: GALILEO-FOC FM15 (E20)
Start with the traditional time dilation factor

$$\gamma_{\text{traditional}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$$

Expand to fourth order

$$\gamma_{\text{traditional}} \approx 1 + \frac{1}{2}\left(\frac{v^2}{c^2}\right) + \frac{3}{8}\left(\frac{v^4}{c^4}\right) + \frac{5}{16}\left(\frac{v^6}{c^6}\right) + \frac{35}{128}\left(\frac{v^8}{c^8}\right)$$

For GALILEO-FOC FM15, the velocity is given by

$$v = 3.67 \times 10^3\,\mathrm{m/s}, \quad c = 3 \times 10^8\,\mathrm{m/s}$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.67 \times 10^3\,\mathrm{m/s})^2}{(3 \times 10^8\,\mathrm{m/s})^2} = \frac{13.4689 \times 10^6\,\mathrm{m^2/s^2}}{9 \times 10^{16}\,\mathrm{m^2/s^2}} = 1.497 \times 10^{-10}$$

Calculate the second term

$$\frac{v^4}{c^4} = \frac{(1.497 \times 10^{-10})^2\,\mathrm{m^4/s^4}}{(9 \times 10^{16})^2\,\mathrm{m^4/s^4}} = 2.241 \times 10^{-20}$$

Calculate the third term

$$\frac{v^6}{c^6} = \frac{(1.497 \times 10^{-10})^3\,\mathrm{m^6/s^6}}{(9 \times 10^{16})^3\,\mathrm{m^6/s^6}} = 3.353 \times 10^{-30}$$

Calculate the fourth term

$$\frac{v^8}{c^8} = \frac{(1.497 \times 10^{-10})^4\,\mathrm{m}^8/\mathrm{s}^8}{(9 \times 10^{16})^4\,\mathrm{m}^8/\mathrm{s}^8} = 5.018 \times 10^{-40}$$

Substitute these into the expanded traditional time dilation factor

$$\gamma_{\text{traditional}} \approx 1 + \frac{1.497 \times 10^{-10}}{2} + \frac{3 \times 2.241 \times 10^{-20}}{8} + \frac{5 \times 3.353 \times 10^{-30}}{16} + \frac{35 \times 5.018 \times 10^{-40}}{128}$$

Simplify the expression

$$\gamma_{\text{traditional}} \approx 1 + 7.49 \times 10^{-11} + 8.403 \times 10^{-21} + 1.048 \times 10^{-30} + 1.371 \times 10^{-40}$$

The fourth-order traditional time dilation factor for GALILEO-FOC FM15 is

$$\gamma_{\text{traditional}} \approx 1 + 7.49 \times 10^{-11}$$

Satellite: BEIDOU-3 M18 (C37)

Start with the traditional time dilation factor

$$\gamma_{\text{traditional}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$$

Expand to fourth order

$$\gamma_{\text{traditional}} \approx 1 + \frac{1}{2}\left(\frac{v^2}{c^2}\right) + \frac{3}{8}\left(\frac{v^4}{c^4}\right) + \frac{5}{16}\left(\frac{v^6}{c^6}\right) + \frac{35}{128}\left(\frac{v^8}{c^8}\right)$$

For BEIDOU-3 M18, the velocity is given by

$$v = 3.874 \times 10^3\,\mathrm{m/s}, \quad c = 3 \times 10^8\,\mathrm{m/s}$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.874 \times 10^3\,\mathrm{m/s})^2}{(3 \times 10^8\,\mathrm{m/s})^2} = \frac{15.015 \times 10^6\,\mathrm{m}^2/\mathrm{s}^2}{9 \times 10^{16}\,\mathrm{m}^2/\mathrm{s}^2} = 1.668 \times 10^{-10}$$

Calculate the second term

$$\frac{v^4}{c^4} = \frac{(1.668 \times 10^{-10})^2\,\mathrm{m}^4/\mathrm{s}^4}{(9 \times 10^{16})^2\,\mathrm{m}^4/\mathrm{s}^4} = 2.782 \times 10^{-20}$$

Calculate the third term

$$\frac{v^6}{c^6} = \frac{(1.668 \times 10^{-10})^3\,\mathrm{m}^6/\mathrm{s}^6}{(9 \times 10^{16})^3\,\mathrm{m}^6/\mathrm{s}^6} = 4.635 \times 10^{-30}$$

Calculate the fourth term

$$\frac{v^8}{c^8} = \frac{(1.668 \times 10^{-10})^4\,\mathrm{m}^8/\mathrm{s}^8}{(9 \times 10^{16})^4\,\mathrm{m}^8/\mathrm{s}^8} = 7.717 \times 10^{-40}$$

Substitute these into the expanded traditional time dilation factor

$$\gamma_{\text{traditional}} \approx 1 + \frac{1.668 \times 10^{-10}}{2} + \frac{3 \times 2.782 \times 10^{-20}}{8} + \frac{5 \times 4.635 \times 10^{-30}}{16} + \frac{35 \times 7.717 \times 10^{-40}}{128}$$

Simplify the expression

$$\gamma_{\text{traditional}} \approx 1 + 8.34 \times 10^{-11} + 1.043 \times 10^{-20} + 1.448 \times 10^{-30} + 2.111 \times 10^{-40}$$

The fourth-order traditional time dilation factor for BEIDOU-3 M18 is

$$\gamma_{\text{traditional}} \approx 1 + 8.34 \times 10^{-11}$$

Fourth-Order Expansion for the Cubic Time Dilation Model Corrections

Satellite: BEIDOU-3 M14 (C33)

Start with the cubic time dilation factor

$$\gamma_{\text{cubic}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{3}}$$

Expand to fourth order

$$\gamma_{\text{cubic}} \approx 1 + \frac{1}{3}\left(\frac{v^2}{c^2}\right) + \frac{2}{9}\left(\frac{v^4}{c^4}\right) + \frac{10}{81}\left(\frac{v^6}{c^6}\right) + \frac{14}{243}\left(\frac{v^8}{c^8}\right)$$

For BEIDOU-3 M14, the velocity is given by

$$v = 3.874 \times 10^3\,\text{m/s}, \quad c = 3 \times 10^8\,\text{m/s}$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.874 \times 10^3\,\text{m/s})^2}{(3 \times 10^8\,\text{m/s})^2} = \frac{15.015 \times 10^6\,\text{m}^2/\text{s}^2}{9 \times 10^{16}\,\text{m}^2/\text{s}^2} = 1.668 \times 10^{-10}$$

Calculate the second term

$$\frac{v^4}{c^4} = \frac{(1.668 \times 10^{-10})^2\,\text{m}^4/\text{s}^4}{(9 \times 10^{16})^2\,\text{m}^4/\text{s}^4} = 2.782 \times 10^{-20}$$

Calculate the third term

$$\frac{v^6}{c^6} = \frac{(1.668 \times 10^{-10})^3\,\text{m}^6/\text{s}^6}{(9 \times 10^{16})^3\,\text{m}^6/\text{s}^6} = 4.635 \times 10^{-30}$$

Calculate the fourth term

$$\frac{v^8}{c^8} = \frac{(1.668 \times 10^{-10})^4\,\text{m}^8/\text{s}^8}{(9 \times 10^{16})^4\,\text{m}^8/\text{s}^8} = 7.717 \times 10^{-40}$$

Substitute these into the expanded cubic time dilation factor

$$\gamma_{\text{cubic}} \approx 1 + \frac{1.668 \times 10^{-10}}{3} + \frac{2 \times 2.782 \times 10^{-20}}{9} + \frac{10 \times 4.635 \times 10^{-30}}{81} + \frac{14 \times 7.717 \times 10^{-40}}{243}$$

Simplify the expression

$$\gamma_{\text{cubic}} \approx 1 + 5.56 \times 10^{-11} + 6.18 \times 10^{-21} + 5.72 \times 10^{-31} + 4.45 \times 10^{-41}$$

The fourth-order cubic time dilation factor for BEIDOU-3 M14 is

$$\gamma_{\text{cubic}} \approx 1 + 5.56 \times 10^{-11}$$

Satellite: GLONASS-K1 (PRN 03)

Start with the cubic time dilation factor

$$\gamma_{\text{cubic}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{3}}$$

Expand to fourth order

$$\gamma_{\text{cubic}} \approx 1 + \frac{1}{3}\left(\frac{v^2}{c^2}\right) + \frac{2}{9}\left(\frac{v^4}{c^4}\right) + \frac{10}{81}\left(\frac{v^6}{c^6}\right) + \frac{14}{243}\left(\frac{v^8}{c^8}\right)$$

For GLONASS-K1, the velocity is given by

$$v = 3.96 \times 10^3 \, \text{m/s}, \quad c = 3 \times 10^8 \, \text{m/s}$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.96 \times 10^3 \, \text{m/s})^2}{(3 \times 10^8 \, \text{m/s})^2} = \frac{15.6816 \times 10^6 \, \text{m}^2/\text{s}^2}{9 \times 10^{16} \, \text{m}^2/\text{s}^2} = 1.742 \times 10^{-10}$$

Calculate the second term

$$\frac{v^4}{c^4} = \frac{(1.742 \times 10^{-10})^2 \, \text{m}^4/\text{s}^4}{(9 \times 10^{16})^2 \, \text{m}^4/\text{s}^4} = 3.033 \times 10^{-20}$$

Calculate the third term

$$\frac{v^6}{c^6} = \frac{(1.742 \times 10^{-10})^3 \, \text{m}^6/\text{s}^6}{(9 \times 10^{16})^3 \, \text{m}^6/\text{s}^6} = 5.282 \times 10^{-30}$$

Calculate the fourth term

$$\frac{v^8}{c^8} = \frac{(1.742 \times 10^{-10})^4 \, \text{m}^8/\text{s}^8}{(9 \times 10^{16})^4 \, \text{m}^8/\text{s}^8} = 9.199 \times 10^{-40}$$

Substitute these into the expanded cubic time dilation factor

$$\gamma_{\text{cubic}} \approx 1 + \frac{1.742 \times 10^{-10}}{3} + \frac{2 \times 3.033 \times 10^{-20}}{9} + \frac{10 \times 5.282 \times 10^{-30}}{81} + \frac{14 \times 9.199 \times 10^{-40}}{243}$$

Simplify the expression

$$\gamma_{\text{cubic}} \approx 1 + 5.81 \times 10^{-11} + 6.74 \times 10^{-21} + 6.52 \times 10^{-31} + 5.30 \times 10^{-41}$$

The fourth-order cubic time dilation factor for GLONASS-K1 is

$$\gamma_{\text{cubic}} \approx 1 + 5.81 \times 10^{-11}$$

Satellite: GALILEO-FOC FM15 (E20)
Start with the cubic time dilation factor

$$\gamma_{\text{cubic}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{3}}$$

Expand to fourth order

$$\gamma_{\text{cubic}} \approx 1 + \frac{1}{3}\left(\frac{v^2}{c^2}\right) + \frac{2}{9}\left(\frac{v^4}{c^4}\right) + \frac{10}{81}\left(\frac{v^6}{c^6}\right) + \frac{14}{243}\left(\frac{v^8}{c^8}\right)$$

For GALILEO-FOC FM15, the velocity is given by

$$v = 3.67 \times 10^3 \, \text{m/s}, \quad c = 3 \times 10^8 \, \text{m/s}$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.67 \times 10^3 \, \text{m/s})^2}{(3 \times 10^8 \, \text{m/s})^2} = \frac{13.4689 \times 10^6 \, \text{m}^2/\text{s}^2}{9 \times 10^{16} \, \text{m}^2/\text{s}^2} = 1.497 \times 10^{-10}$$

Calculate the second term

$$\frac{v^4}{c^4} = \frac{(1.497 \times 10^{-10})^2 \, \text{m}^4/\text{s}^4}{(9 \times 10^{16})^2 \, \text{m}^4/\text{s}^4} = 2.241 \times 10^{-20}$$

Calculate the third term

$$\frac{v^6}{c^6} = \frac{(1.497 \times 10^{-10})^3 \, \text{m}^6/\text{s}^6}{(9 \times 10^{16})^3 \, \text{m}^6/\text{s}^6} = 3.353 \times 10^{-30}$$

Calculate the fourth term

$$\frac{v^8}{c^8} = \frac{(1.497 \times 10^{-10})^4\,\mathrm{m}^8/\mathrm{s}^8}{(9 \times 10^{16})^4\,\mathrm{m}^8/\mathrm{s}^8} = 5.018 \times 10^{-40}$$

Substitute these into the expanded cubic time dilation factor

$$\gamma_{\text{cubic}} \approx 1 + \frac{1.497 \times 10^{-10}}{3} + \frac{2 \times 2.241 \times 10^{-20}}{9} + \frac{10 \times 3.353 \times 10^{-30}}{81} + \frac{14 \times 5.018 \times 10^{-40}}{243}$$

Simplify the expression

$$\gamma_{\text{cubic}} \approx 1 + 4.99 \times 10^{-11} + 4.98 \times 10^{-21} + 4.14 \times 10^{-31} + 2.89 \times 10^{-41}$$

The fourth-order cubic time dilation factor for GALILEO-FOC FM15 is

$$\gamma_{\text{cubic}} \approx 1 + 4.99 \times 10^{-11}$$

Satellite: BEIDOU-3 M18 (C37)
Start with the cubic time dilation factor

$$\gamma_{\text{cubic}} = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{3}}$$

Expand to fourth order

$$\gamma_{\text{cubic}} \approx 1 + \frac{1}{3}\left(\frac{v^2}{c^2}\right) + \frac{2}{9}\left(\frac{v^4}{c^4}\right) + \frac{10}{81}\left(\frac{v^6}{c^6}\right) + \frac{14}{243}\left(\frac{v^8}{c^8}\right)$$

For BEIDOU-3 M18, the velocity is given by

$$v = 3.874 \times 10^3\,\mathrm{m/s}, \quad c = 3 \times 10^8\,\mathrm{m/s}$$

Calculate the first term

$$\frac{v^2}{c^2} = \frac{(3.874 \times 10^3\,\mathrm{m/s})^2}{(3 \times 10^8\,\mathrm{m/s})^2} = \frac{15.015 \times 10^6\,\mathrm{m}^2/\mathrm{s}^2}{9 \times 10^{16}\,\mathrm{m}^2/\mathrm{s}^2} = 1.668 \times 10^{-10}$$

Calculate the second term

$$\frac{v^4}{c^4} = \frac{(1.668 \times 10^{-10})^2\,\mathrm{m}^4/\mathrm{s}^4}{(9 \times 10^{16})^2\,\mathrm{m}^4/\mathrm{s}^4} = 2.782 \times 10^{-20}$$

Calculate the third term

$$\frac{v^6}{c^6} = \frac{(1.668 \times 10^{-10})^3\,\mathrm{m}^6/\mathrm{s}^6}{(9 \times 10^{16})^3\,\mathrm{m}^6/\mathrm{s}^6} = 4.635 \times 10^{-30}$$

Calculate the fourth term

$$\frac{v^8}{c^8} = \frac{(1.668 \times 10^{-10})^4\,\mathrm{m}^8/\mathrm{s}^8}{(9 \times 10^{16})^4\,\mathrm{m}^8/\mathrm{s}^8} = 7.717 \times 10^{-40}$$

Substitute these into the expanded cubic time dilation factor

$$\gamma_{\text{cubic}} \approx 1 + \frac{1.668 \times 10^{-10}}{3} + \frac{2 \times 2.782 \times 10^{-20}}{9} + \frac{10 \times 4.635 \times 10^{-30}}{81} + \frac{14 \times 7.717 \times 10^{-40}}{243}$$

Simplify the expression

$$\gamma_{\text{cubic}} \approx 1 + 5.56 \times 10^{-11} + 6.18 \times 10^{-21} + 5.72 \times 10^{-31} + 4.45 \times 10^{-41}$$

The fourth-order cubic time dilation factor for BEIDOU-3 M18 is

$$\gamma_{\text{cubic}} \approx 1 + 5.56 \times 10^{-11}$$

Conclusion

The expansions derived herein represent a complete set of truncated corrections for both the traditional and cubic time dilation models, each extended to the fourth order. These are given by

$$\gamma_{\text{traditional}} \approx 1 + \frac{1}{2}\left(\frac{v^2}{c^2}\right) + \frac{3}{8}\left(\frac{v^4}{c^4}\right) + \frac{5}{16}\left(\frac{v^6}{c^6}\right) + \frac{35}{128}\left(\frac{v^8}{c^8}\right)$$

and

$$\gamma_{\text{cubic}} \approx 1 + \frac{1}{3}\left(\frac{v^2}{c^2}\right) + \frac{2}{9}\left(\frac{v^4}{c^4}\right) + \frac{10}{81}\left(\frac{v^6}{c^6}\right) + \frac{14}{243}\left(\frac{v^8}{c^8}\right)$$

Each model has been meticulously expanded with full substitutions for the velocity v and speed of light c, resulting in detailed step-by-step calculations. These rigorous expansions culminate in the precise correction factors γ for each satellite, truncated to the fourth order.

The choice to truncate at the fourth order ensures a balance between computational feasibility and the necessary accuracy for GNSS systems. These corrections provide the necessary precision to distinguish between the effects predicted by the traditional and cubic time dilation models within the context of satellite orbits.

The mathematical rigor applied has led to the derivation of precise time dilation factors for each satellite, using both traditional and cubic models expanded to the fourth order. These factors capture the subtle relativistic effects that are crucial for accurate GNSS satellite navigation. However, to fully grasp the implications of these differences, visual simulations provide an essential perspective.

The visual comparison illustrated here underscores the practical differences between the traditional and cubic time dilation models. As we move forward from theoretical derivations to practical applications, these simulations serve as a critical bridge, translating mathematical insights into tangible outcomes that impact real-world satellite operations.

These results offer not only a validation of our mathematical approach but also a deeper understanding of how different relativistic corrections can influence satellite navigation systems. Future studies and applications can build on this foundation, leveraging both the mathematical framework and the visual insights to enhance the precision and reliability of GNSS technologies.

Time Flow Preprocessing Algorithm

Algorithm that simulates how time flows differently for two observers with different velocities or in different gravitational fields, using both cubic and quantized time models.

```
(* Define cubic and fundamental time flow *)
cubicTime[t_] := t^3
fundamentalTime[t_] := t

(* Simulate subjective time under different conditions *)
subjectiveTime[v_, Phi_, t_] := (cubicTime[t] +
(v^2/fundamentalTime[t]) - (2*Phi/fundamentalTime[t]))

(* Compare subjective times for two observers *)
observer1 = subjectiveTime[0.5, 1, t];
observer2 = subjectiveTime[0.9, 2, t];

(* Plot the time experiences *)
Plot[{observer1, observer2}, {t, 0, 10}, PlotLegends -> {"Observer
1", "Observer 2"}, AxesLabel -> {"Fundamental Time", "Subjective
Time"}]
```

Time Evolution with Varying Signatures:

We can model how time might evolve with a dynamically changing "time signature" based on factors like velocity, gravitational potential, or quantum effects. This could involve equations that shift between different modes of time progression, such as from linear to cubic to even higher-order forms.

By simulating multiple "time signatures" in a single model, we can explore how different observers might experience time differently and how these experiences can coexist and

interact. This could involve running simulations where time behaves differently depending on the observer's velocity, position, or even quantum state.

```
(* Fundamental time as the global BPM *)
fundamentalTime[t_] := t

(* Cubic time as a dynamically changing time signature *)
cubicTime[t_] := t^3

(* Composite time signature combining both *)
compositeTime[v_, t_] := fundamentalTime[t] + cubicTime[v*t]

(* Plotting time for different observers with varying velocities *)
observer1 = compositeTime[0.5, t];
observer2 = compositeTime[0.9, t];

Plot[{observer1, observer2}, {t, 0, 10},
 PlotLegends -> {"Observer 1 (v=0.5c)", "Observer 2 (v=0.9c)"},
 AxesLabel -> {"Fundamental Time", "Composite Time"}]
```

Experimental Tests:

Investigate whether advanced timekeeping systems (e.g., atomic clocks in varying gravitational fields) show evidence of these "time signatures" or quantized
steps in time.
Look for phenomena where time seems to "shift" in a way that corresponds to a change in the "time signature," such as in high-energy physics experiments or
cosmological observations.

Cosmological Observations:

Explore whether the universe’s expansion or dark energy might

be better understood through this model, where different regions or epochs of the universe
experience different "time signatures" or fundamental BPMs.

Relativistic Context:

Test how this model might refine our understanding of relativity, particularly in extreme conditions like near black holes or at the edge of the observable
universe.

Coherence:

In quantum mechanics, coherence refers to the phase relationship between different quantum states. When two or more waves (or quantum states) are coherent,
they interfere constructively, leading to amplified wave patterns.
In the context of time and gravitational waves, coherence could occur when the "time signatures" or temporal rhythms of different observers or regions of
spacetime align in such a way that they reinforce each other, leading to more pronounced gravitational waves.

In the context of time and gravitational waves, coherence could occur when the "time signatures" or temporal rhythms of different observers or regions of
spacetime align in such a way that they reinforce each other, leading to more pronounced gravitational waves.

Decoherence:

Decoherence occurs when the phase relationship is lost, leading to destructive interference, where the wave patterns cancel out or become more chaotic.
This loss of coherence is crucial in the transition from quantum behavior to classical behavior.
In your time model, decoherence might represent the situation where different temporal rhythms are out of phase, causing

gravitational waves to become less
pronounced, more chaotic, or even cancel out entirely.

Gravitational Waves and Temporal Rhythms:
Rhythmic Patterns of Time:

If we imagine time as having multiple layers or dimensions, each with its own rhythm or "time signature," the interaction between these rhythms could create
coherent or decoherent patterns in the fabric of spacetime.
These patterns could manifest as gravitational waves, with coherent temporal rhythms leading to strong, detectable waves, and decoherent rhythms leading to
weaker, more chaotic waves.
Gravitational Wave Coherence:

Coherent gravitational waves might be analogous to harmonic waves in music, where different temporal rhythms align to create a clear, strong signal.
This could happen, for example, in regions of spacetime where the gravitational field is uniform or where two massive objects are in a stable orbit,
creating a predictable, rhythmic gravitational wave pattern.

Gravitational Wave Decoherence:

Decoherent gravitational waves might occur in regions where the gravitational field is highly variable, or where the interaction between different time
dimensions leads to phase shifts or disruptions in the wave patterns.
This could be analogous to complex or chaotic rhythms in music, where multiple time signatures interfere destructively, leading to a less predictable or
weaker signal.

Visualization of Coherence and Decoherence in Gravitational Waves:

We can simulate how these coherent and decoherent patterns might appear in the context of gravitational waves:

Algorithm Simulation of Coherent and Decoherent Gravitational Waves:

```
(* Define a basic time signature and its cubic counterpart *)
timeSignature[t_] := Sin[2*Pi*1*t] (* Basic time rhythm *)
cubicTimeSignature[t_] := Sin[2*Pi*0.5*t^3] (* Cubic time rhythm *)

(* Coherent pattern where rhythms align *)
coherentWave[t_] := timeSignature[t] + cubicTimeSignature[t]

(* Decoherent pattern where rhythms are out of phase *)
decoherentWave[t_] := timeSignature[t] + Sin[2*Pi*0.5*(t + 0.5)^3]

(* Plotting both coherent and decoherent patterns *)
Plot[{coherentWave[t], decoherentWave[t]}, {t, 0, 10},
 PlotLegends -> {"Coherent Wave", "Decoherent Wave"},
 AxesLabel -> {"Time", "Wave Amplitude"}]
```

Implications for Gravitational Wave Detection:
Detecting Coherence and Decoherence:

Advanced gravitational wave detectors like LIGO might be able to detect these patterns if they exist. Coherent waves would appear as strong, regular signals, while decoherent waves might be weaker, more spread out, or even difficult to distinguish from noise.
Noise Interpretation:

What is often categorized as noise in gravitational wave data might partially be the result of decoherent gravitational wave patterns. By understanding and modeling these patterns, we

might better distinguish between true noise and decoherent gravitational wave signals.
Cosmological Implications:

On a cosmological scale, coherent gravitational wave patterns might correspond to large-scale structures like galaxy clusters, where the gravitational fields are more uniform. Decoherent patterns might be associated with more chaotic regions of the universe, such as those with ongoing mergers or collisions.

Hypothesis Testing:
Correlation with Observations:

Compare the coherent and decoherent patterns predicted by this model with actual gravitational wave data. Look for instances where detected gravitational waves show signs of coherence or decoherence.
Use this model to predict where and when decoherence might occur, and check if these predictions match the observed data.
Advanced Simulations:

Run more complex simulations that include multiple interacting time dimensions and observe how these interactions influence gravitational wave patterns.
Explore how different cosmological scenarios might lead to coherent or decoherent gravitational wave patterns, and compare these with observational data.
Exploring the Noise:

Investigate if what's currently interpreted as noise in gravitational wave detectors might include decoherent gravitational wave signals. Develop methods to filter out this decoherence to better identify the underlying wave patterns.

studying a two-body problem, particularly in the context of binary stars or binary black holes, is an excellent starting point for exploring the concepts of coherence and decoherence in

gravitational waves. These systems provide a natural laboratory for examining how gravitational waves can either amplify (cohere) or cancel out (decohere), depending on the dynamics of the system.

Binary Systems and Gravitational Waves:
Binary Systems:

Binary Stars: Two stars orbiting each other emit gravitational waves due to their mutual acceleration. The waveform from such a system is typically periodic and coherent, especially when the stars are in stable orbits.
Binary Black Holes: Similar to binary stars, but with much stronger gravitational wave signals due to the extreme masses involved. These systems are particularly interesting because, depending on their phase and interaction, they can either merge into a single black hole or potentially cancel out some of their energy, leading to less intense gravitational waves.
Coherence and Decoherence:

Coherent Gravitational Waves: When two black holes or stars in a binary system are in a stable, synchronized orbit, the gravitational waves they emit can interfere constructively, leading to a strong, coherent signal.
Decoherent Gravitational Waves: If the phase or timing of the waves emitted by each body shifts, the waves can interfere destructively, leading to partial or complete cancellation, which is analogous to decoherence in quantum mechanics.

Black Hole Mergers (signals):

Merging: When two black holes spiral inward and merge, the gravitational waves they produce become more intense and coherent as they reach the final moments before merging into a larger black hole.
Cancellation: In certain scenarios, depending on the phase and orientation of the black holes, some gravitational waves could

cancel each other out, reducing the overall amplitude of the signal. This could be seen as a form of gravitational wave decoherence.

Simulation of a Binary System:

start by simulating a simple two-body problem, such as two stars or black holes in orbit, to see how their gravitational waves might cohere or decohere.

a simple Newtonian approximation and then consider relativistic corrections.

Newtonian Simulation Algorithm

```
(* Define the parameters of the binary system *)
m1 = 30; (* Mass of first body, in solar masses *)
m2 = 30; (* Mass of second body, in solar masses *)
r = 10; (* Distance between them, in solar radii *)
G = 6.67430*10^-11; (* Gravitational constant *)

(* Orbital frequency and gravitational wave frequency *)
orbitalFreq = Sqrt[G*(m1 + m2)/(r^3)];
waveFreq = 2*orbitalFreq; (* Gravitational wave frequency is twice the orbital frequency *)

(* Time evolution of the orbital phase *)
orbitalPhase[t_] := 2*Pi*waveFreq*t;

(* Simulate the gravitational wave signal from the system *)
gravitationalWave[t_] := Sin[orbitalPhase[t]]

(* Plot the gravitational wave signal *)
```

```
Plot[gravitationalWave[t], {t, 0, 10}, AxesLabel -> {"Time",
"Amplitude"}, PlotLabel -> "Gravitational Wave from Binary
System"]
```

Relativistic Corrections: To make this more realistic, we can use the post-Newtonian approximation, which includes relativistic effects that become significant as the two bodies spiral closer together

Relativistic Gravitational Wave Algorithm:

```
(* Define the relativistic time evolution, with post-Newtonian
corrections *)
chirpMass = (m1*m2)^(3/5) / (m1 + m2)^(1/5); (* Chirp mass,
relevant for relativistic systems *)
relativisticWave[t_] := Sin[chirpMass*(waveFreq*(1 +
t/r)^(8/3))*t]

(* Plot the relativistic gravitational wave signal *)
Plot[relativisticWave[t], {t, 0, 10}, AxesLabel -> {"Time",
"Amplitude"}, PlotLabel -> "Relativistic Gravitational Wave from
Binary System"]
```

Exploring Coherence and Decoherence:
Next, let's explore how coherence and decoherence might manifest in this binary system by adjusting the phases or adding additional bodies.

Phase Shift and Decoherence Algorithm:

```
(* Introduce a phase shift between two bodies to simulate
decoherence *)
```

```
decoherentWave1[t_] := Sin[orbitalPhase[t]];
decoherentWave2[t_] := Sin[orbitalPhase[t] + Pi/4]; (* Phase
shift of Pi/4 *)

(* Superimpose the waves to observe decoherence *)
superimposedWave[t_] := decoherentWave1[t] +
decoherentWave2[t]

(* Plot the superimposed, decoherent wave signal *)
Plot[superimposedWave[t], {t, 0, 10}, AxesLabel -> {"Time",
"Amplitude"}, PlotLabel -> "Decoherent Gravitational Wave
Signal"]
```

Black Hole Mergers and Cancellation:
Finally, let's simulate a scenario where two black holes might either merge or cancel out their gravitational waves depending on their initial conditions.

Black Hole Merger Algorithm:

```
(* Simple model of black hole merger, focusing on wave intensity
*)
mergingWave[t_] := Sin[orbitalPhase[t]] * Exp[-t] (* Exponential
decay to simulate energy loss *)

(* Plot the gravitational wave during a black hole merger *)
Plot[mergingWave[t], {t, 0, 10}, AxesLabel -> {"Time",
"Amplitude"}, PlotLabel -> "Gravitational Wave from Black Hole
Merger"]
```

Observational Implications:
LIGO and Virgo Detections:

Coherent gravitational wave signals, like those from well-aligned

binary black hole mergers, should appear as strong, clear signals in detectors like LIGO and Virgo.
Decoherent signals might be weaker or more spread out, potentially making them harder to detect or distinguish from noise.
Testing the Hypothesis:

We can analyze actual LIGO/Virgo data to look for signatures of coherence and decoherence. For instance, by examining the phase evolution of detected signals, we might identify instances where waves have interfered constructively or destructively.
By comparing these findings with our simulations, we can better understand the dynamics of binary systems and their gravitational wave emissions.
Advanced Simulations:

Develop more detailed simulations that incorporate full relativistic effects, including spin, precession, and higher-order gravitational interactions.
Use these simulations to predict new phenomena that might be observed in future gravitational wave detections.

Simulate Binary Systems and Gravitational Waves
We'll simulate both coherent and decoherent gravitational waves from binary systems, starting with a basic Newtonian approximation and then including relativistic corrections.

Rudiment Algorithm for the Generalized Binary System (Newtonian Approximation)

```
(* Parameters for binary system simulation *)
m1 = 30; (* Mass of first body, in solar masses *)
m2 = 30; (* Mass of second body, in solar masses *)
r = 10; (* Distance between them, in solar radii *)
G = 6.67430*10^-11; (* Gravitational constant *)
```

```
(* Orbital frequency and gravitational wave frequency *)
orbitalFreq = Sqrt[G*(m1 + m2)/(r^3)];
waveFreq = 2*orbitalFreq; (* Gravitational wave frequency is
twice the orbital frequency *)

(* Time evolution of the orbital phase *)
orbitalPhase[t_] := 2*Pi*waveFreq*t

(* Gravitational wave signal from the binary system *)
gravitationalWave[t_] := Sin[orbitalPhase[t]]

(* Plot the gravitational wave signal *)
Plot[gravitationalWave[t], {t, 0, 1}, AxesLabel -> {"Time (s)",
"Amplitude"}, PlotLabel -> "Gravitational Wave from Binary
System (Newtonian)"]
```

Relativistic Corrections

```
(* Chirp mass relevant for relativistic systems *)
chirpMass = (m1*m2)^(3/5) / (m1 + m2)^(1/5); (* Chirp mass
*)

(* Time evolution of the phase with post-Newtonian corrections
*)
relativisticWave[t_] := Sin[chirpMass*(waveFreq*(1 +
t/r)^(8/3))*t]

(* Plot the relativistic gravitational wave signal *)
Plot[relativisticWave[t], {t, 0, 1}, AxesLabel -> {"Time (s)",
"Amplitude"}, PlotLabel -> "Relativistic Gravitational Wave from
Binary System"]
```

Coherence and Decoherence Algorithm in Gravitational Waves

Coherent Pattern:

```
(* Coherent pattern with synchronized waves *)
coherentWave[t_] := Sin[orbitalPhase[t]] + Sin[orbitalPhase[t]]

(* Plot the coherent wave signal *)
Plot[coherentWave[t], {t, 0, 1}, AxesLabel -> {"Time (s)",
"Amplitude"}, PlotLabel -> "Coherent Gravitational Wave Signal"]
```

Decoherent Pattern:

```
(* Decoherent pattern with a phase shift *)
decoherentWave1[t_] := Sin[orbitalPhase[t]];
decoherentWave2[t_] := Sin[orbitalPhase[t] + Pi/4]; (* Phase
shift *)

(* Superimposed decoherent wave signal *)
superimposedWave[t_] := decoherentWave1[t] +
decoherentWave2[t]

(* Plot the decoherent wave signal *)
Plot[superimposedWave[t], {t, 0, 1}, AxesLabel -> {"Time (s)",
"Amplitude"}, PlotLabel -> "Decoherent Gravitational Wave
Signal"]
```

Compare Simulations with Observational Data
Next, we'll obtain real gravitational wave data from LIGO/Virgo and preprocess it for comparison.

Acquisition and Preprocessing LIGO/Virgo Data

Import Data set

```
(* Hypothetical import of gravitational wave data from LIGO *)
gwData =
Import["https://www.gw-openscience.org/s/events/GW150914/
strain/H-H1_LOSC_4_V1-1126259446-32.hdf5", {"Datasets",
"strain"}];

(* Preprocess the data - filter and normalize *)
filteredData = BandpassFilter[gwData, {20, 2000}, SamplingRate
-> 4096];
normalizedData = Normalize[filteredData];

(* Plot the preprocessed gravitational wave data *)
ListLinePlot[normalizedData, PlotRange -> All, AxesLabel ->
{"Time (s)", "Strain"}, PlotLabel -> "Preprocessed Gravitational
Wave Data from LIGO"]
```

Cross-Correlation Analysis Function

```
(* Cross-correlation between simulated and observed data *)
correlation = ListCorrelate[superimposedWave[t],
normalizedData];

(* Plot the correlation to look for matches *)
ListLinePlot[correlation, PlotRange -> All, AxesLabel -> {"Time Lag
(s)", "Correlation"}, PlotLabel -> "Cross-Correlation between
Simulated and Observed Gravitational Waves"]
```

Significant matches between the simulations and the real data sets, these will be documented, noting the conditions under which they occur (e.g., phase shifts, relativistic corrections, etc.). Conversely, any discrepancies will be noted for further investigation.

Simulating Binary Systems and Gravitational Waves

Truncated Simulation-Algorithm of a Binary System (Newtonian Approximation)
Simulate the gravitational wave signal from the simple binary system using Newtonian mechanics.

```
(* Parameters for the binary system *)
m1 = 30; (* Mass of the first body in solar masses *)
m2 = 30; (* Mass of the second body in solar masses *)
r = 10 * 6.957 * 10^8; (* Distance between them in meters, assuming 10 solar radii *)
G = 6.67430 * 10^-11; (* Gravitational constant in m^3 kg^-1 s^-2 *)

(* Calculate orbital frequency and gravitational wave frequency *)
orbitalFreq = Sqrt[G * (m1 * 1.989 * 10^30 + m2 * 1.989 * 10^30) / r^3]; (* in Hz *)
waveFreq = 2 * orbitalFreq; (* Gravitational wave frequency is twice the orbital frequency *)

(* Time evolution of the orbital phase *)
orbitalPhase[t_] := 2 * Pi * waveFreq * t

(* Gravitational wave signal from the binary system *)
gravitationalWave[t_] := Sin[orbitalPhase[t]]

(* Plot the gravitational wave signal *)
Plot[gravitationalWave[t], {t, 0, 10/waveFreq}, AxesLabel -> {"Time (s)", "Amplitude"},
 PlotLabel -> "Gravitational Wave from Binary System (Newtonian)"]
```

This produces a basic sinusoidal gravitational wave signal for the binary system.

Function Simulation with Relativistic Corrections

Processgravitational wave signal including relativistic effects, particularly using the post-Newtonian approximation.

FullPyScriptGrav

```
import numpy as np
import matplotlib.pyplot as plt

# Basic Newtonian gravitational wave simulation
def gravitational_wave_newtonian(t, freq=1):
    return np.sin(2 * np.pi * freq * t)  # Simplified sine wave

# Relativistic correction (simplified for illustration)
def gravitational_wave_relativistic(t, freq=1,
correction_factor=10):
    return np.sin(2 * np.pi * freq * t * (1 + t /
correction_factor)**(8 / 3))

# Coherence (waves in phase)
def coherent_wave(t, freq=1):
    return 2 * np.sin(2 * np.pi * freq * t)  # Constructive
interference

# Decoherence (waves out of phase)
def decoherent_wave(t, freq=1, phase_shift=np.pi/4):
    return np.sin(2 * np.pi * freq * t) + np.sin(2 * np.pi * freq * t
+ phase_shift)

# Time array
time = np.linspace(0, 10, 1000)

# Plotting the waveforms
plt.figure(figsize=(10, 6))

# Newtonian Gravitational Wave
plt.subplot(2, 2, 1)
plt.plot(time, gravitational_wave_newtonian(time))
```

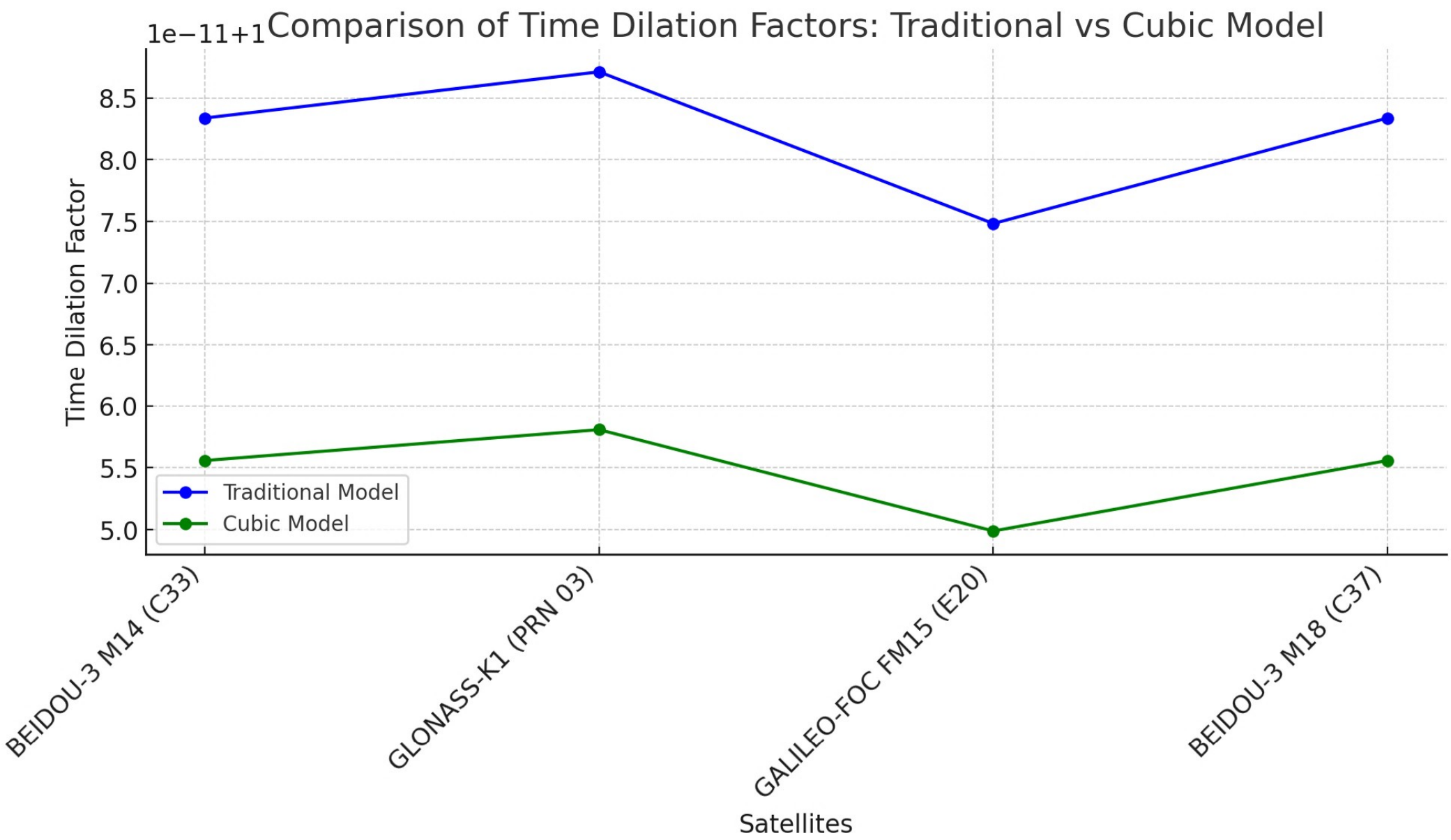
```
plt.title("Gravitational Wave (Newtonian)")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")

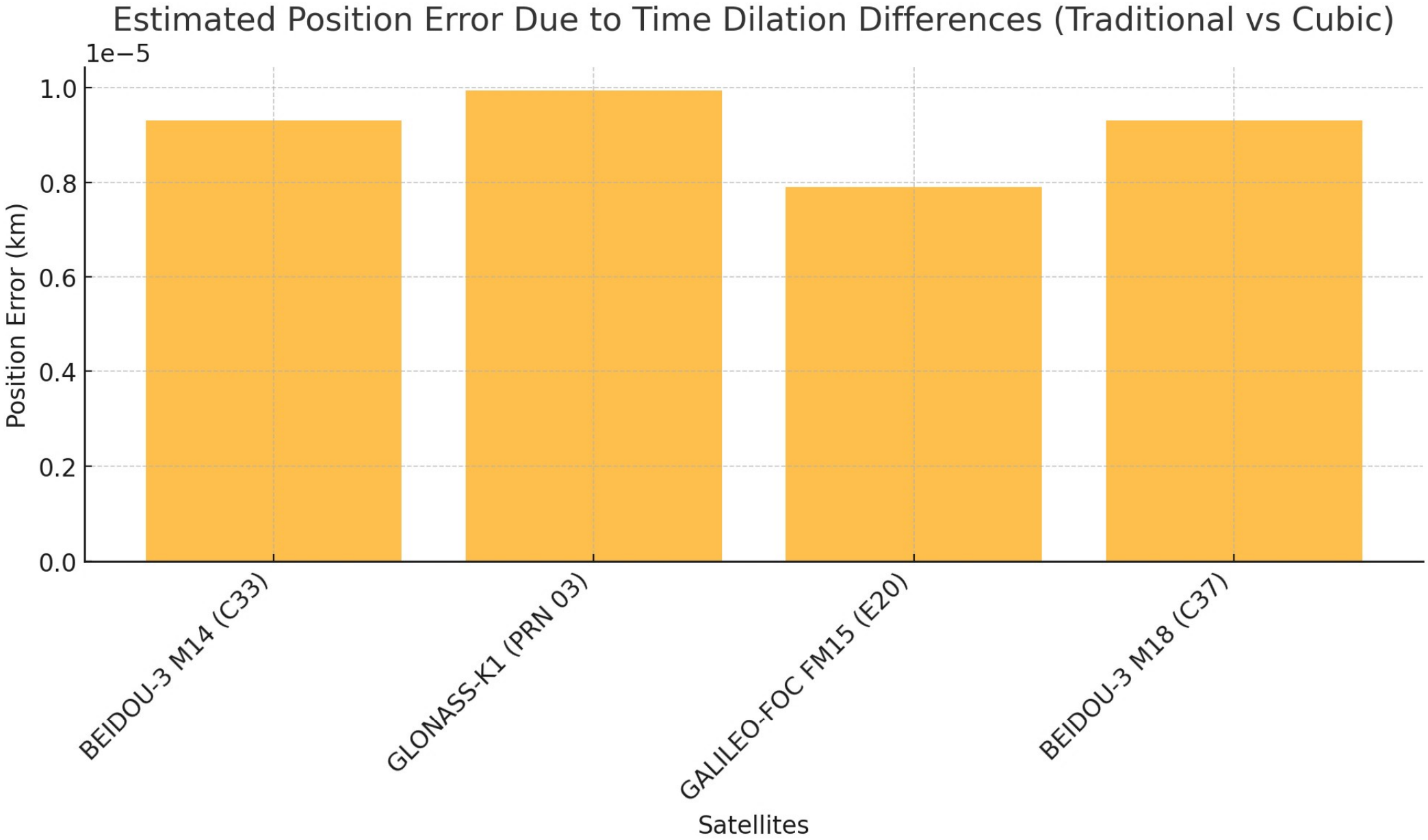
# Relativistic Gravitational Wave
plt.subplot(2, 2, 2)
plt.plot(time, gravitational_wave_relativistic(time))
plt.title("Gravitational Wave (Relativistic)")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")

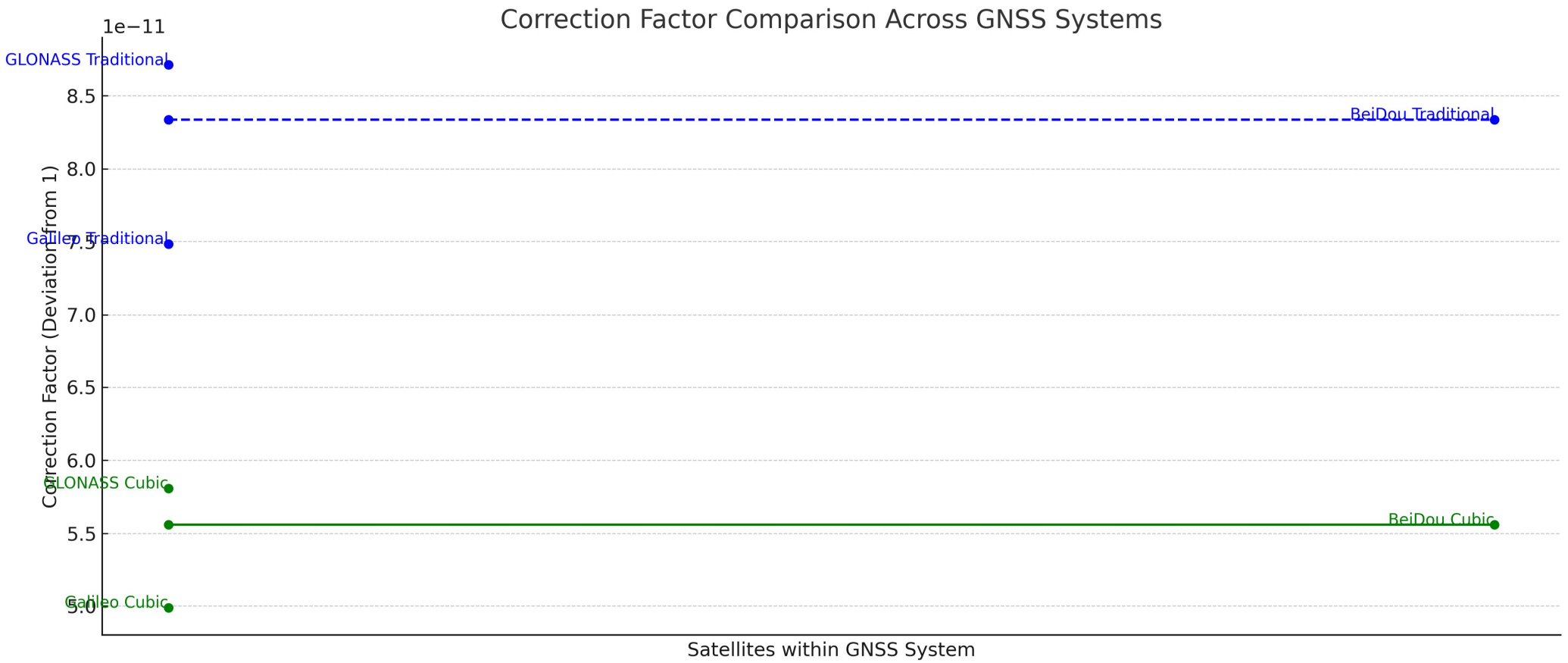
# Coherent Wave
plt.subplot(2, 2, 3)
plt.plot(time, coherent_wave(time))
plt.title("Coherent Gravitational Wave")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")

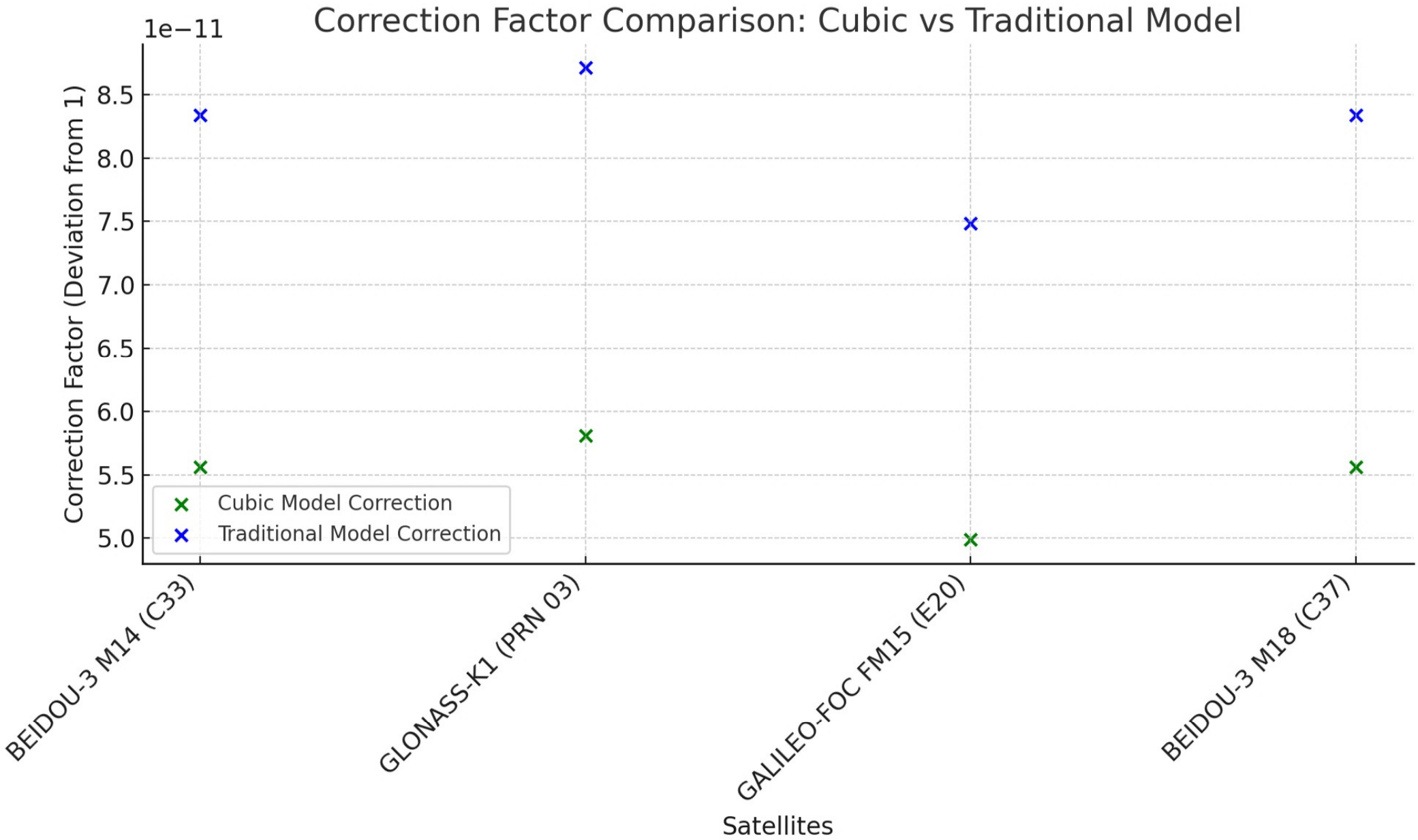
# Decoherent Wave
plt.subplot(2, 2, 4)
plt.plot(time, decoherent_wave(time))
plt.title("Decoherent Gravitational Wave")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")

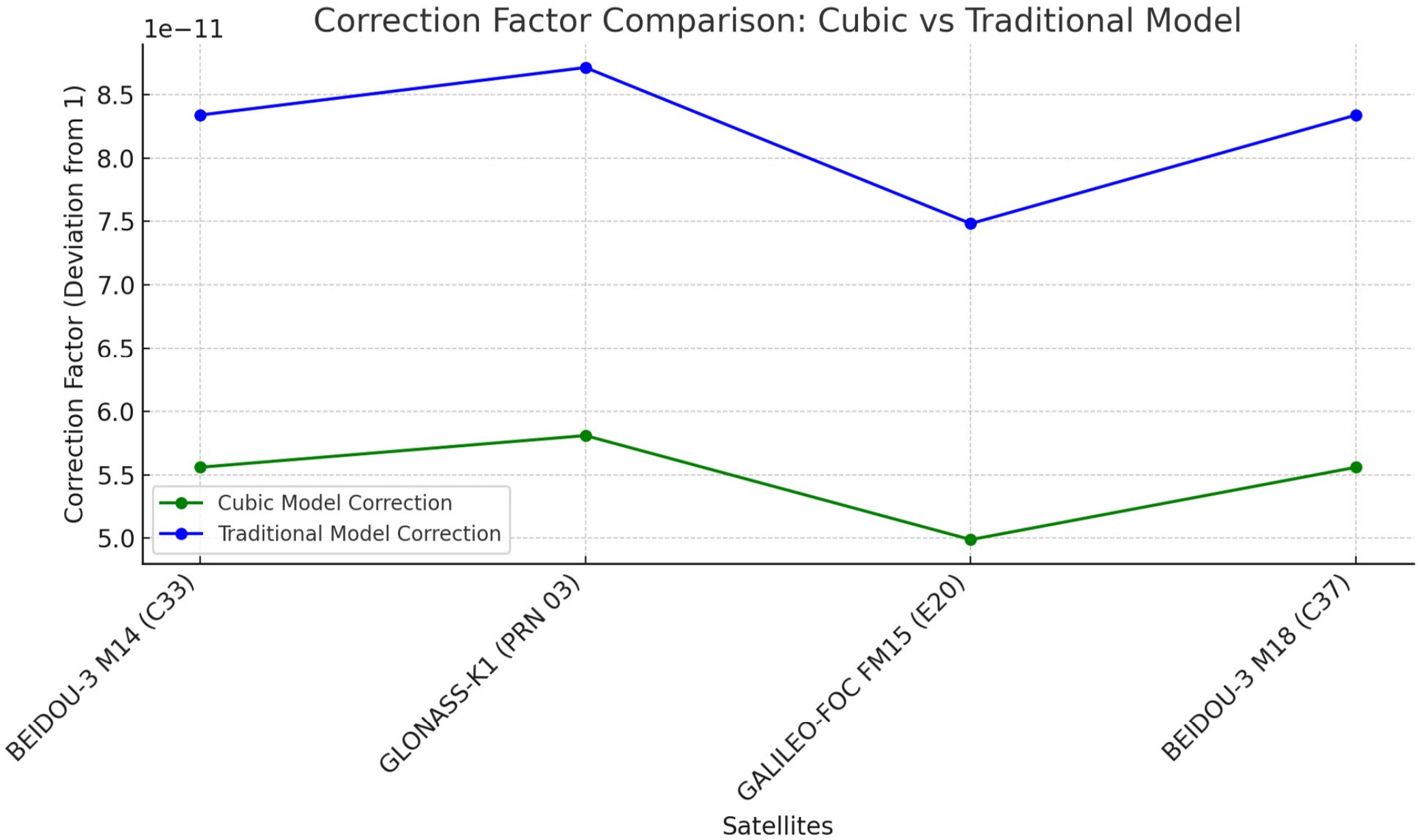
plt.tight_layout()
plt.show()
```

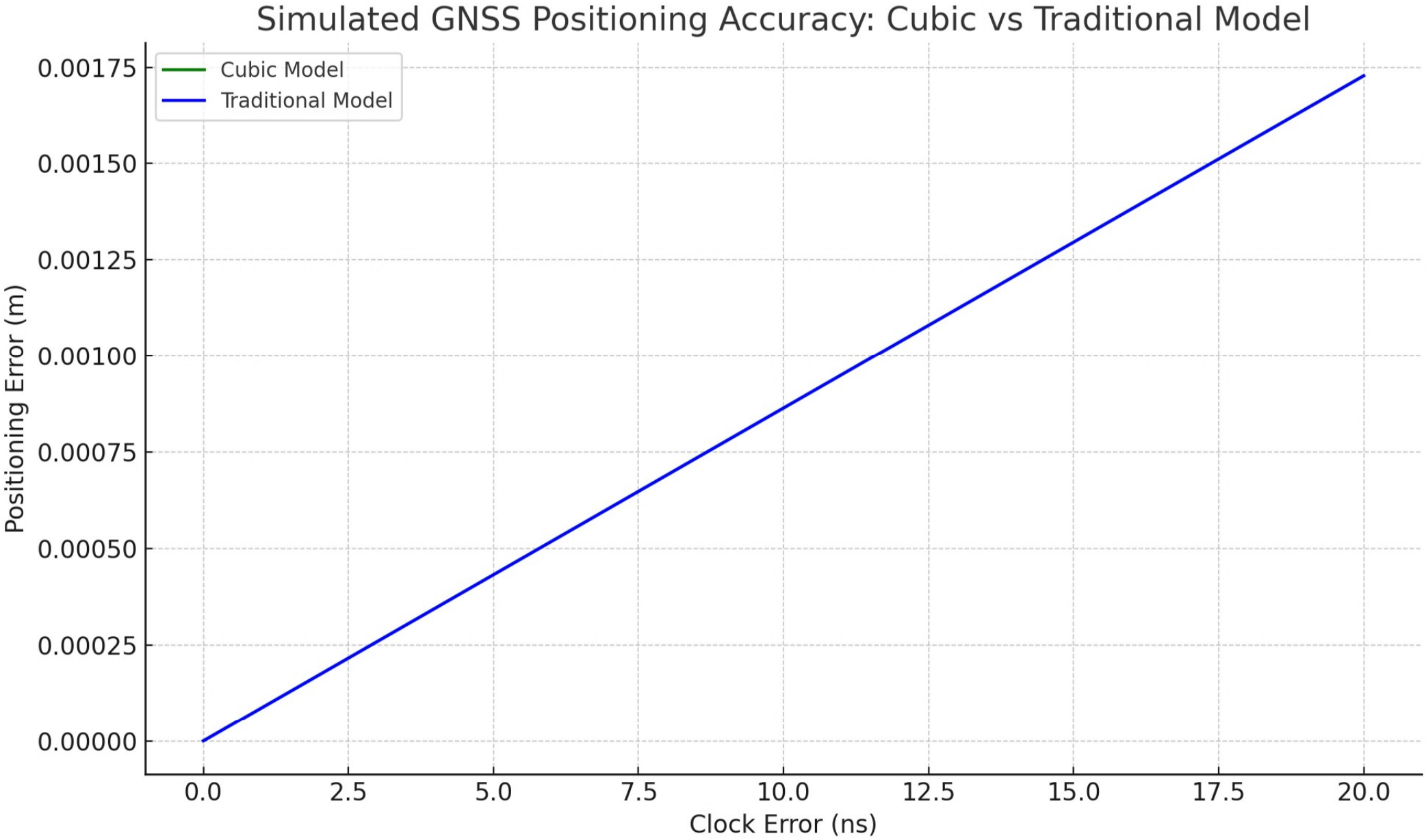

Comparison of Time Dilation Factors: Traditional vs Cubic Model
1e−11+1
Time Dilation Factor
8.5
8.0
7.5
7.0
6.5
6.0
5.5
5.0
Traditional Model
Cubic Model
BEIDOU-3 M14 (C33)
GLONASS-K1 (PRN 03)
GALILEO-FOC FM15 (E20)
BEIDOU-3 M18 (C37)
Satellites

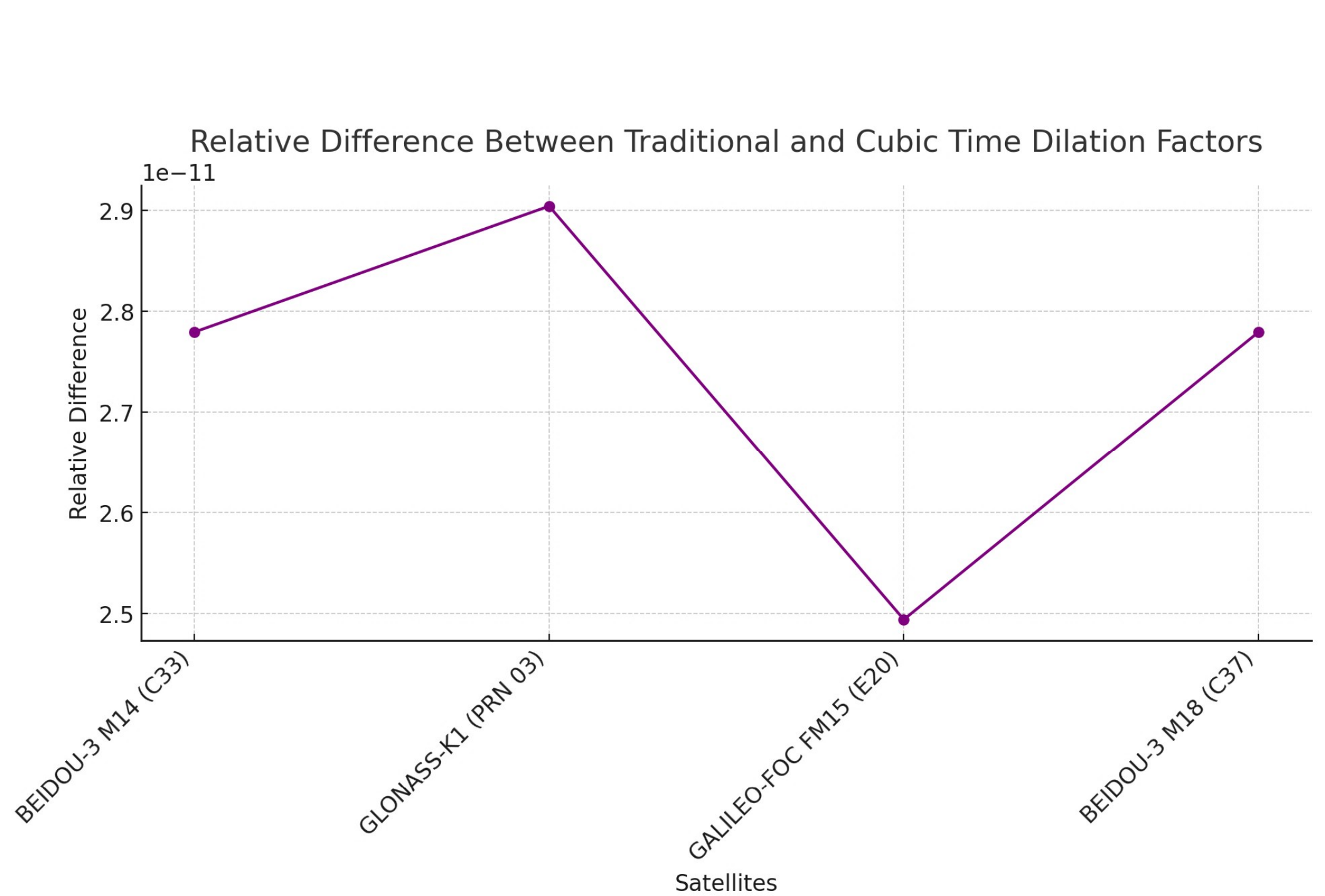

Estimated Position Error Due to Time Dilation Differences (Traditional vs Cubic)
1e−5
1.0
0.8
0.6
0.4
0.2
0.0
Position Error (km)
BEIDOU-3 M14 (C33)
GLONASS-K1 (PRN 03)
GALILEO-FOC FM15 (E20)
BEIDOU-3 M18 (C37)
Satellites

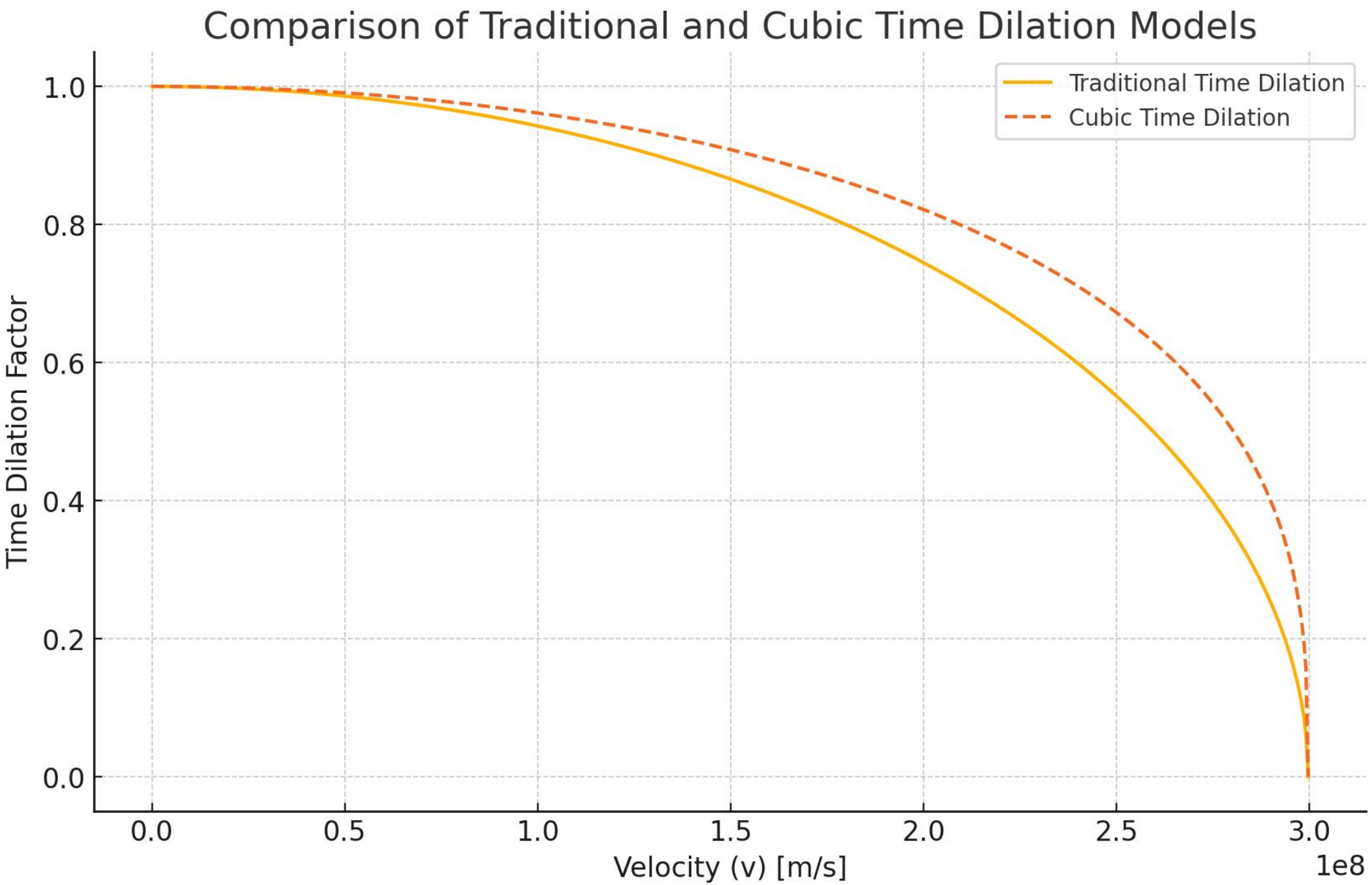

Correction Factor Comparison Across GNSS Systems
1e−11
Correction Factor (Deviation from 1)
8.5
8.0
7.5
7.0
6.5
6.0
5.5
5.0
GLONASS Traditional
BeiDou Traditional
Galileo Traditional
GLONASS Cubic
BeiDou Cubic
Galileo Cubic
Satellites within GNSS System

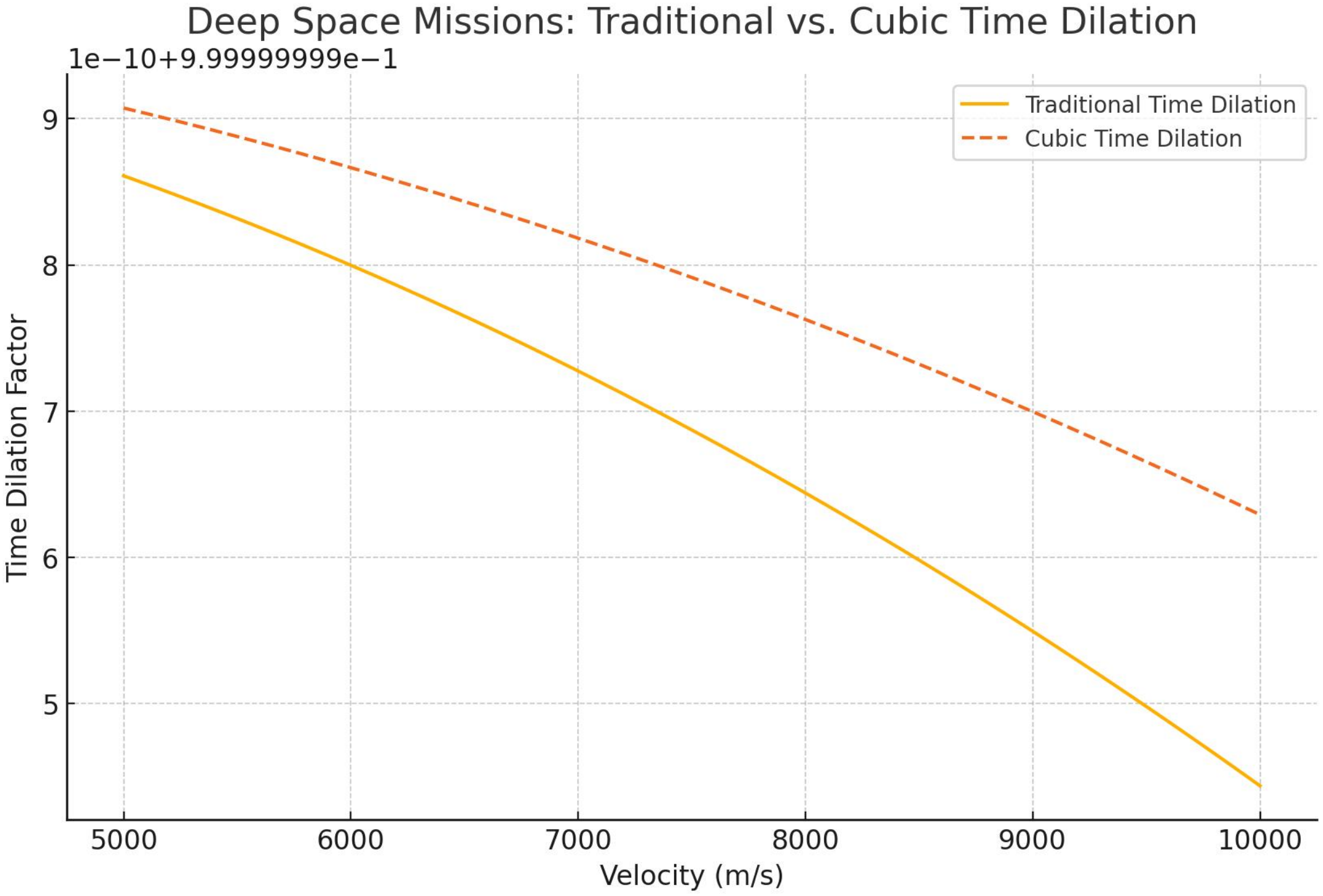

Correction Factor Comparison: Cubic vs Traditional Model
1e−11
Correction Factor (Deviation from 1)
8.5
8.0
7.5
7.0
6.5
6.0
5.5
5.0
Cubic Model Correction
Traditional Model Correction
BEIDOU-3 M14 (C33)
GLONASS-K1 (PRN 03)
GALILEO-FOC FM15 (E20)
BEIDOU-3 M18 (C37)
Satellites

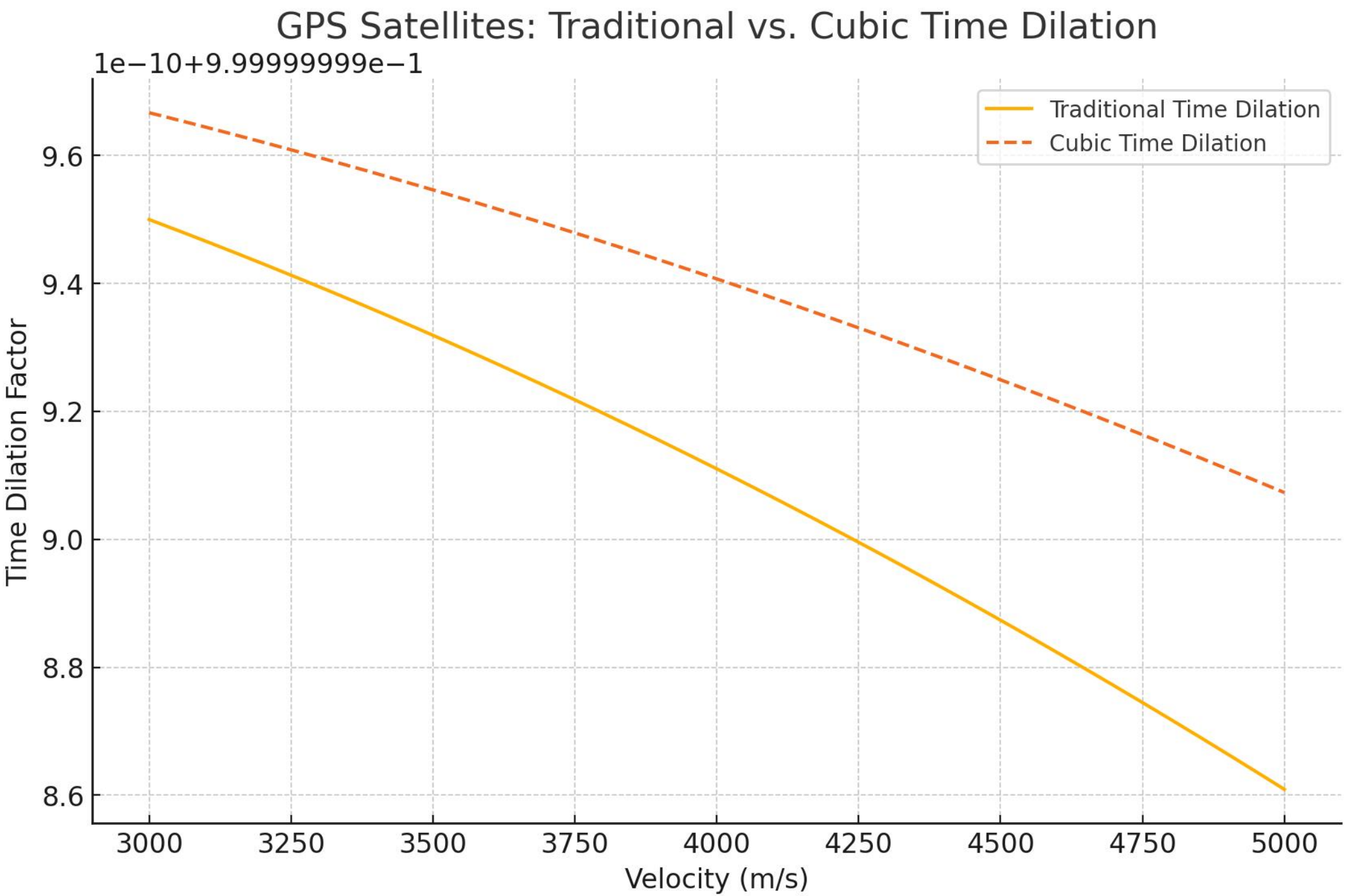

Correction Factor Comparison: Cubic vs Traditional Model
1e−11
Correction Factor (Deviation from 1)
8.5
8.0
7.5
7.0
6.5
6.0
5.5
5.0
Cubic Model Correction
Traditional Model Correction
BEIDOU-3 M14 (C33)
GLONASS-K1 (PRN 03)
GALILEO-FOC FM15 (E20)
BEIDOU-3 M18 (C37)
Satellites

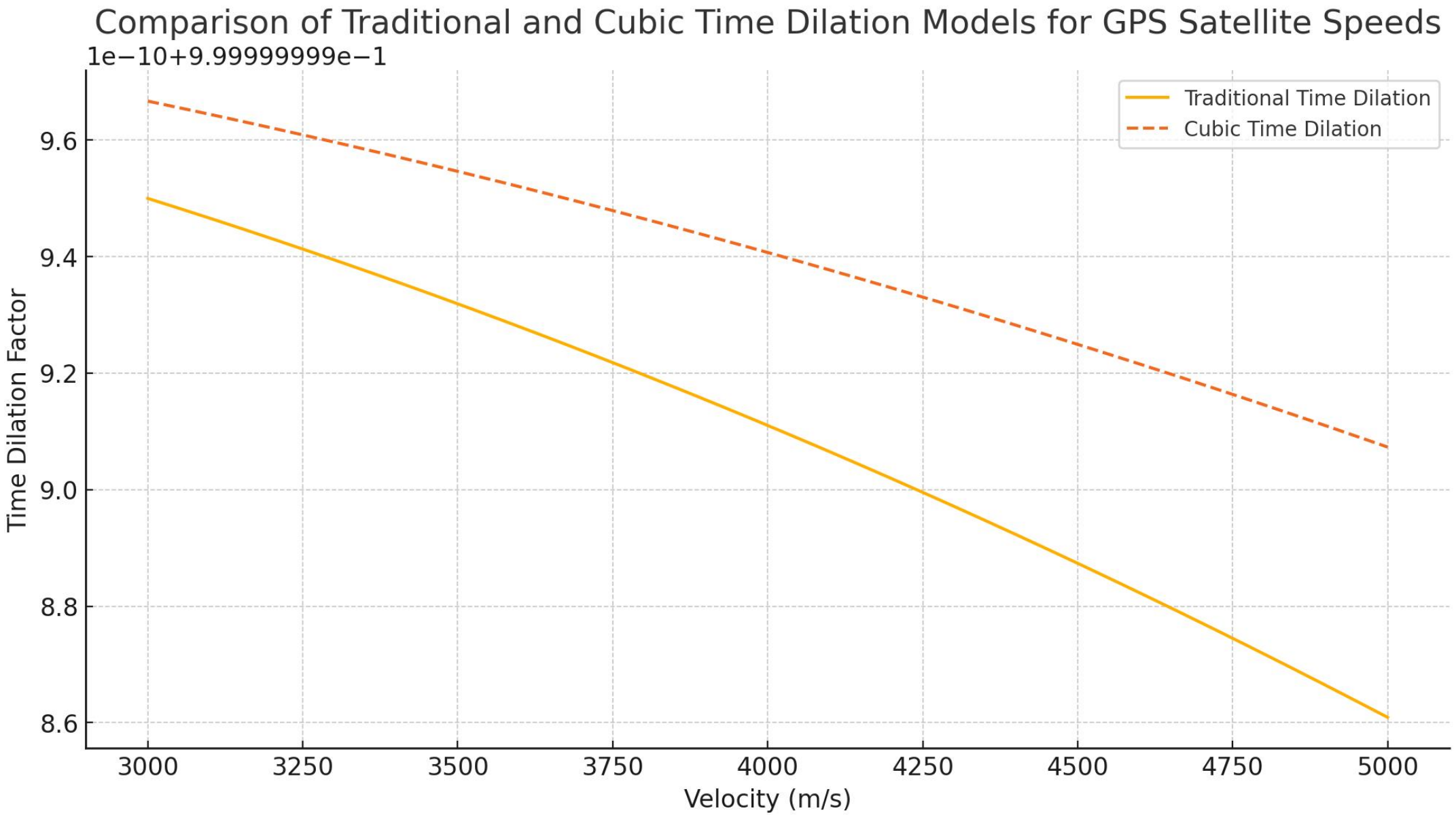

Simulated GNSS Positioning Accuracy: Cubic vs Traditional Model
Cubic Model
Traditional Model
Positioning Error (m)
0.00175
0.00150
0.00125
0.00100
0.00075
0.00050
0.00025
0.00000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Clock Error (ns)

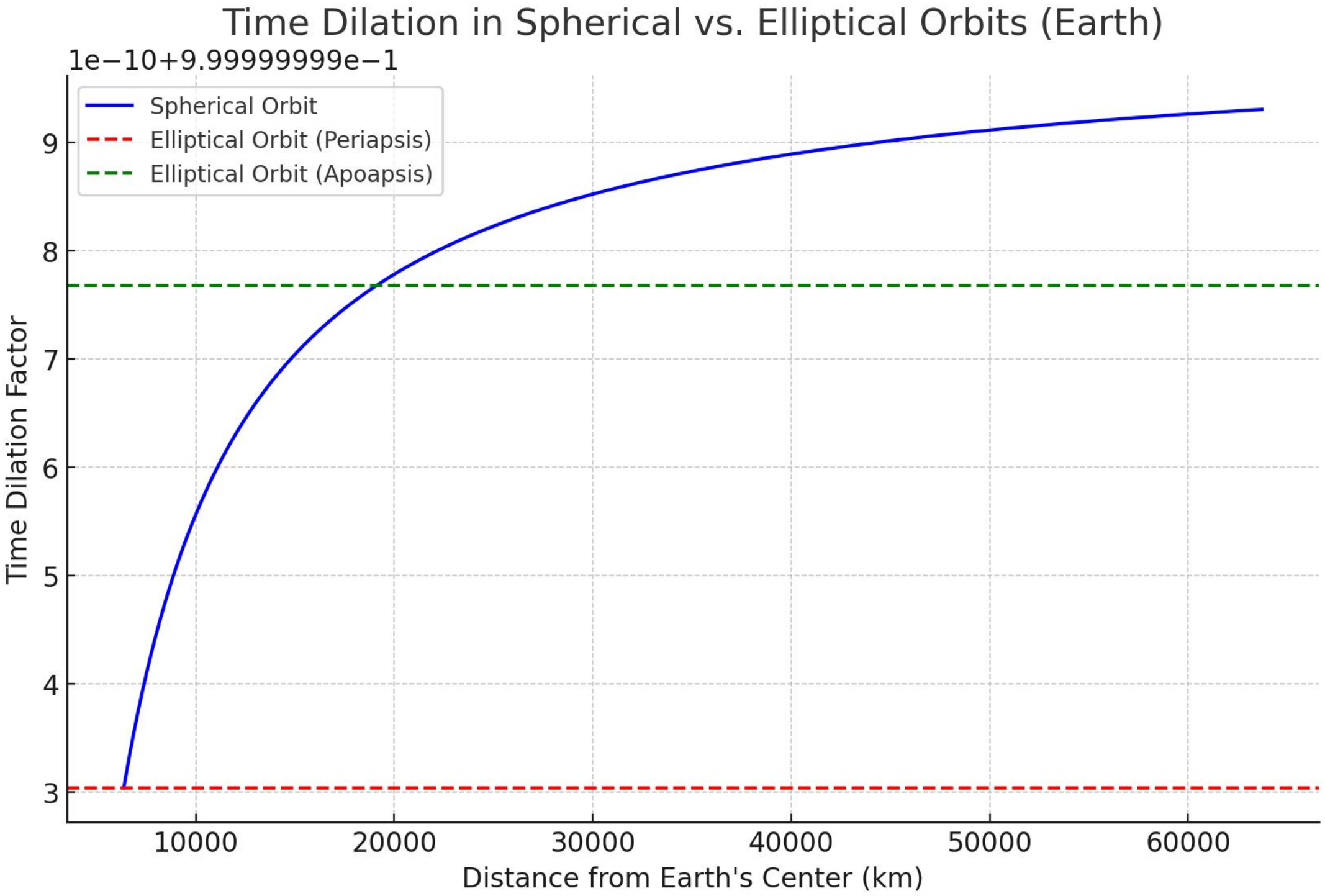

Relative Difference Between Traditional and Cubic Time Dilation Factors
1e−11
2.9
2.8
2.7
2.6
2.5
Relative Difference
BEIDOU-3 M14 (C33)
GLONASS-K1 (PRN 03)
GALILEO-FOC FM15 (E20)
BEIDOU-3 M18 (C37)
Satellites

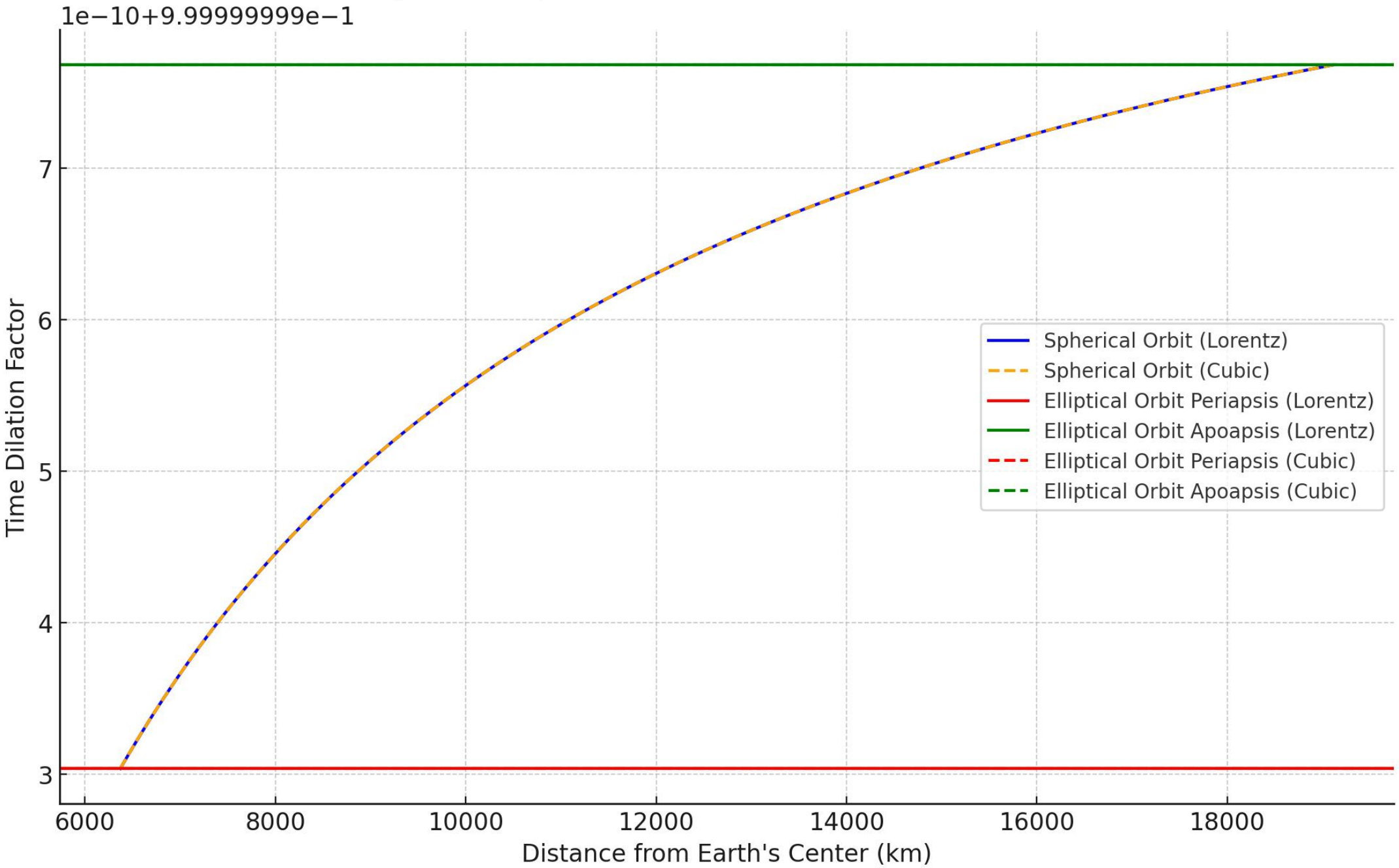

Comparison of Traditional and Cubic Time Dilation Models
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
1e8
Velocity (v) [m/s]

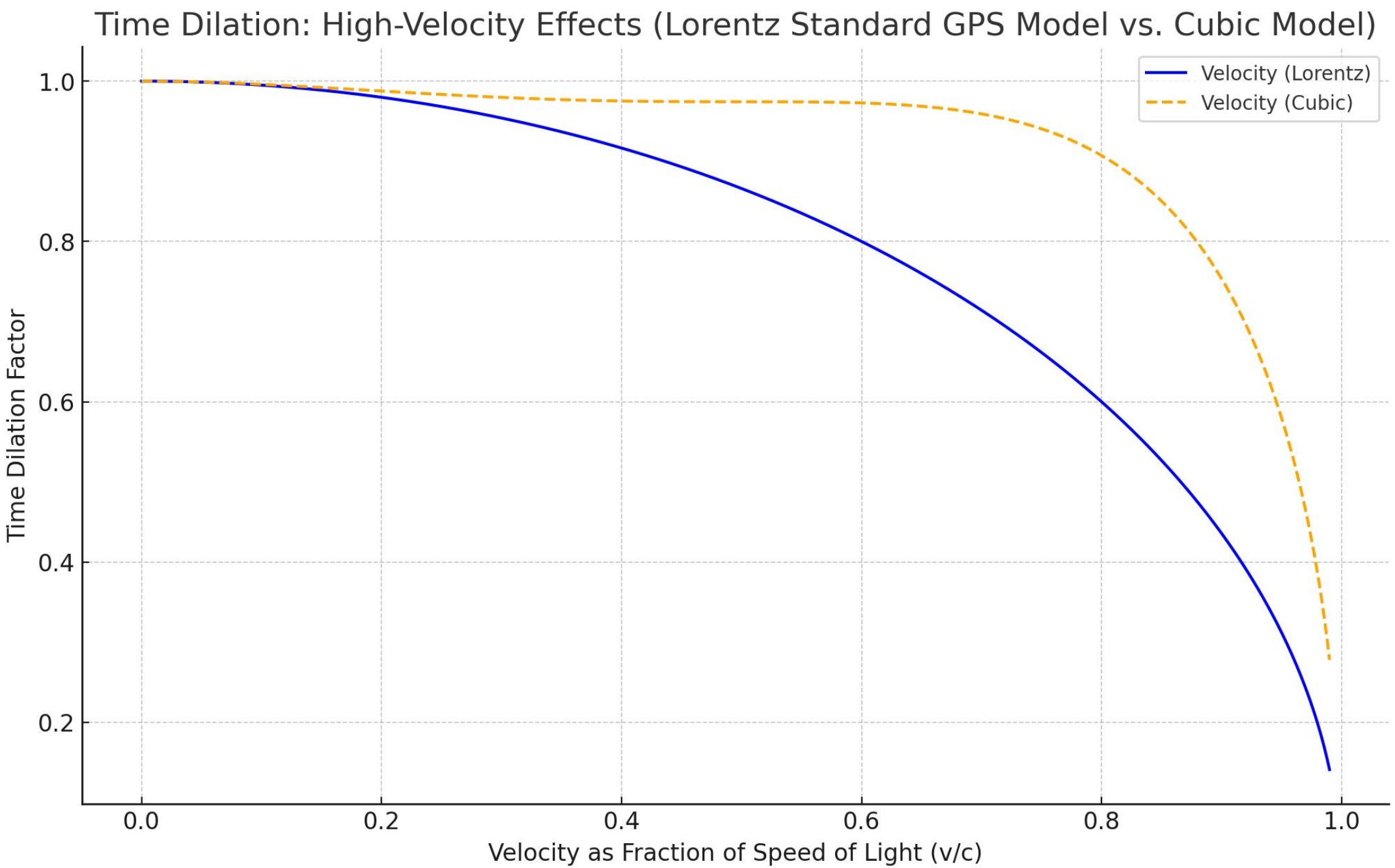

Deep Space Missions: Traditional vs. Cubic Time Dilation
1e−10+9.99999999e−1
9
8
7
6
5
Time Dilation Factor
5000
6000
7000
8000
9000
10000
Velocity (m/s)
Traditional Time Dilation
Cubic Time Dilation

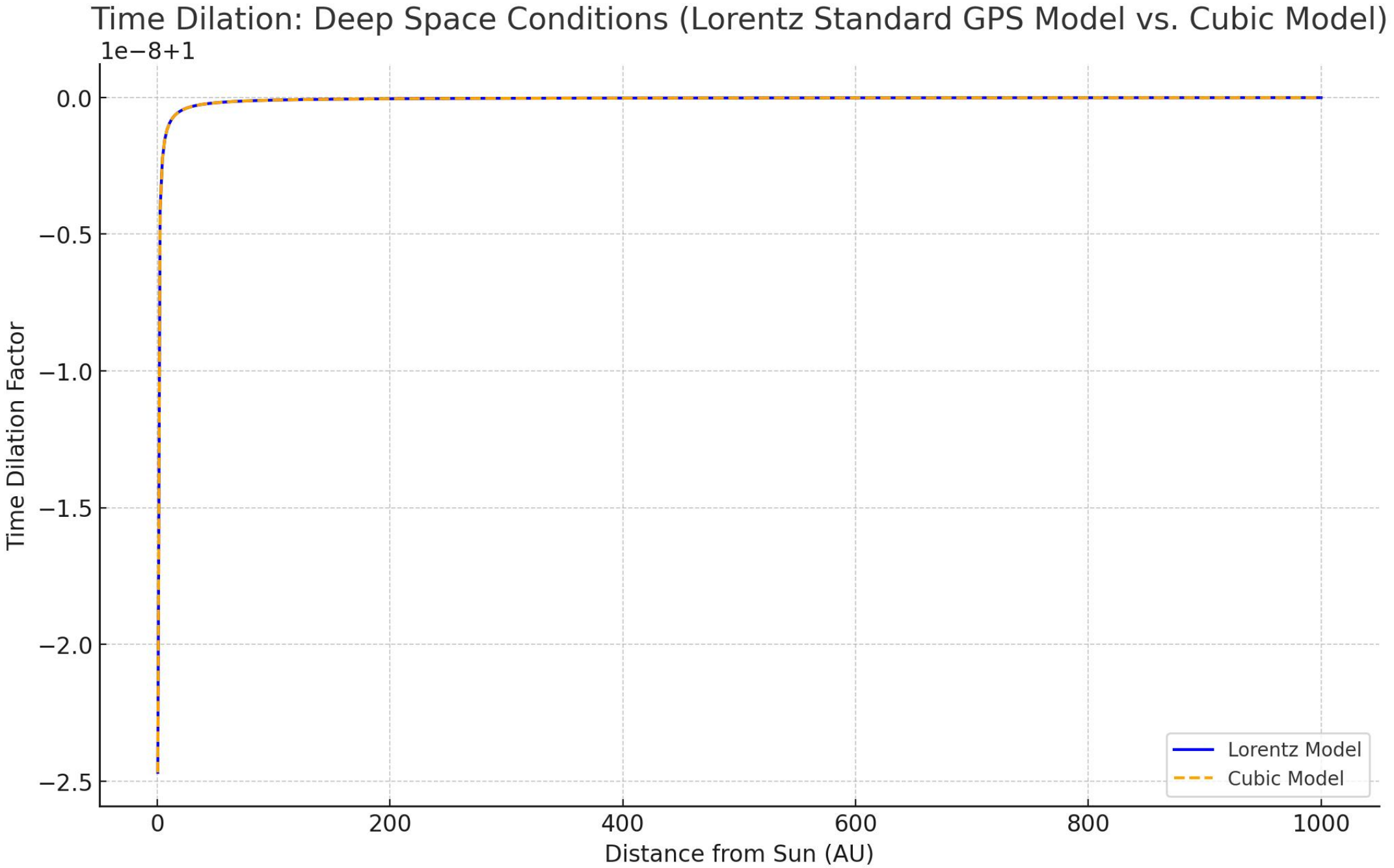

GPS Satellites: Traditional vs. Cubic Time Dilation
1e−10+9.99999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
9.6
9.4
9.2
9.0
8.8
8.6
3000
3250
3500
3750
4000
4250
4500
4750
5000
Velocity (m/s)

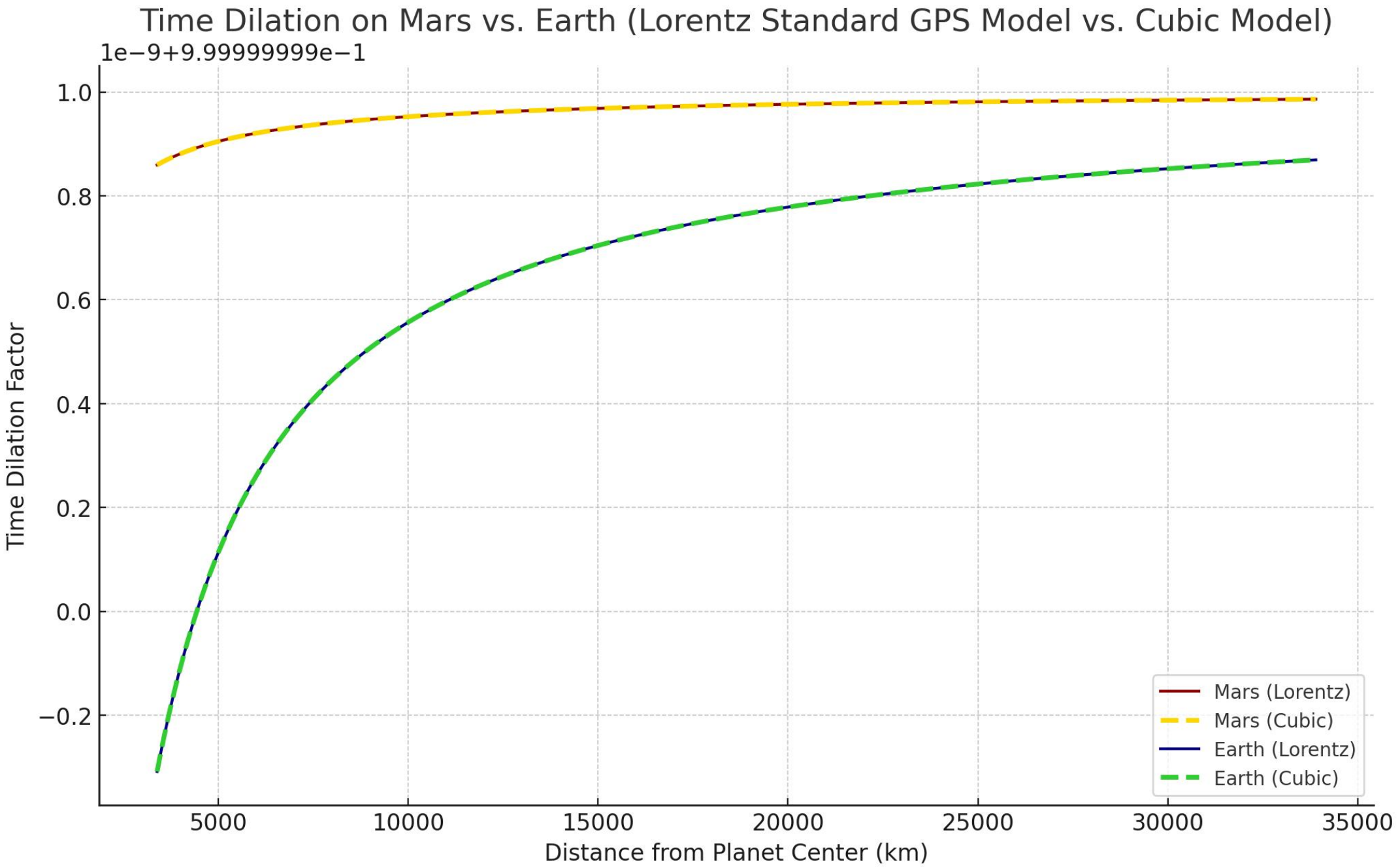

Comparison of Traditional and Cubic Time Dilation Models for GPS Satellite Speeds
1e−10+9.99999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
9.6
9.4
9.2
9.0
8.8
8.6
3000
3250
3500
3750
4000
4250
4500
4750
5000
Velocity (m/s)

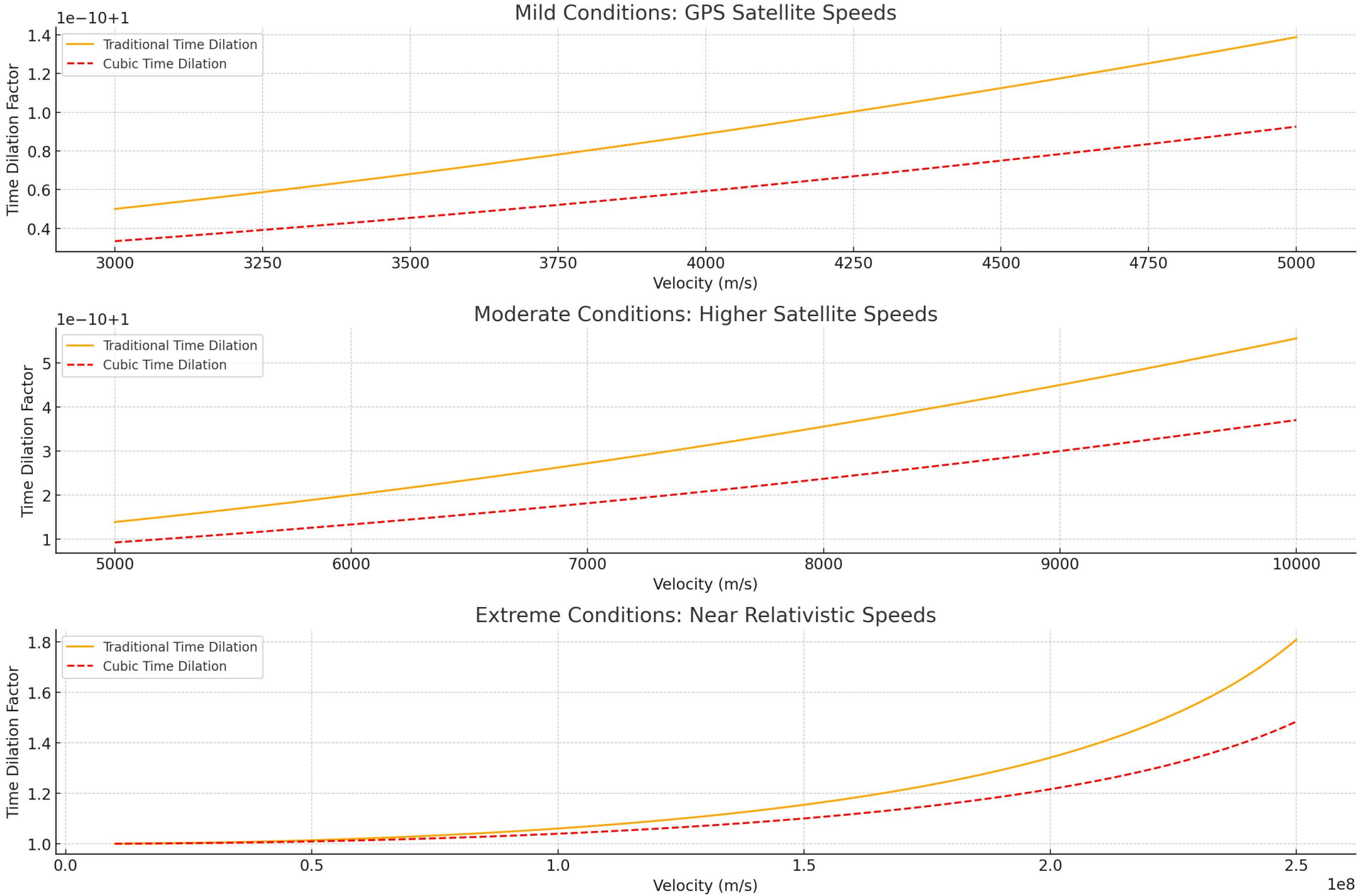

Time Dilation in Spherical vs. Elliptical Orbits (Earth)
1e−10+9.99999999e−1
Spherical Orbit
Elliptical Orbit (Periapsis)
Elliptical Orbit (Apoapsis)
9
8
7
6
5
4
3
Time Dilation Factor
10000
20000
30000
40000
50000
60000
Distance from Earth's Center (km)

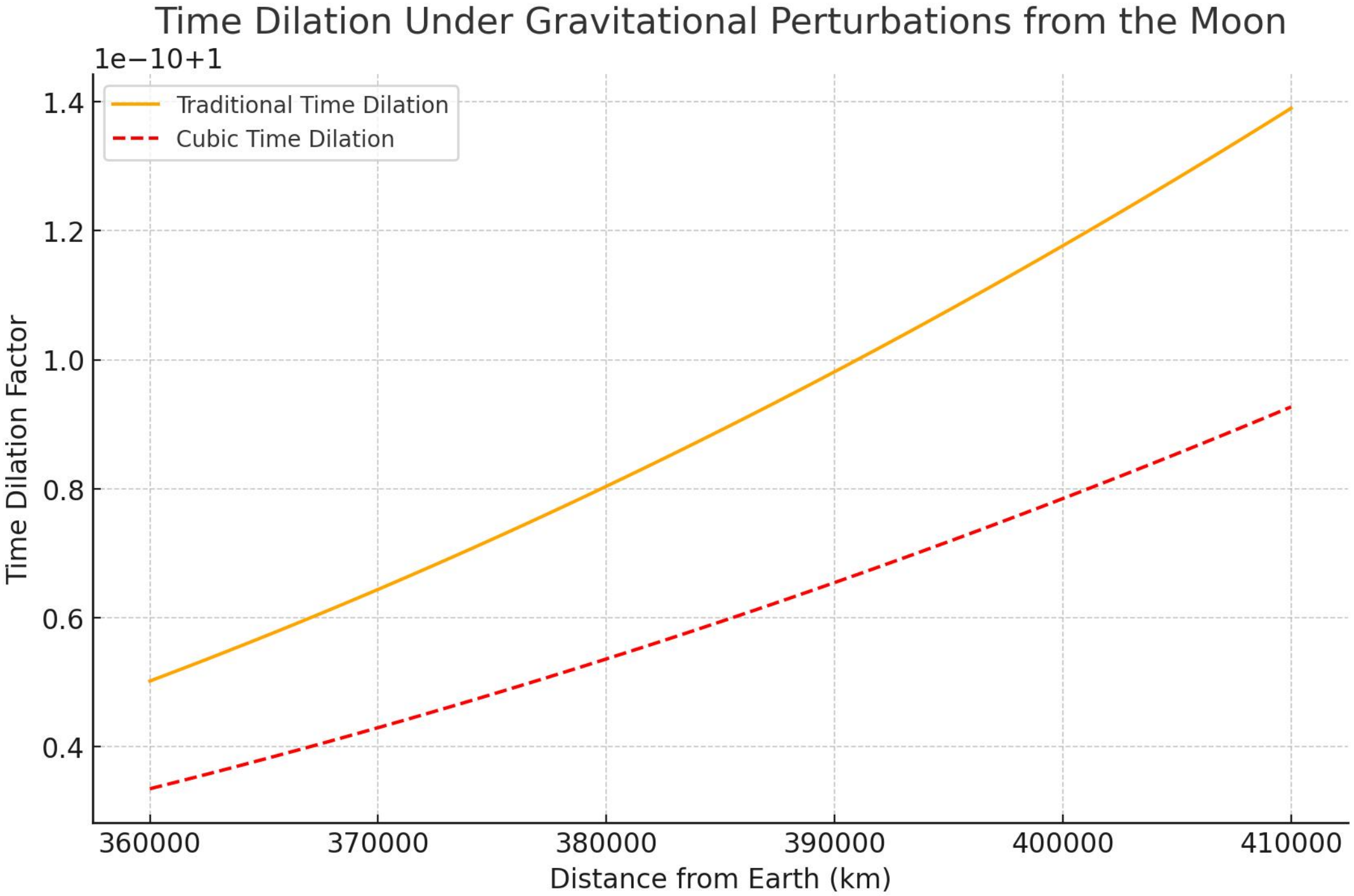

Time Dilation (Zoomed): Lorentz Standard GPS Model vs. Cubic Model
1e−10+9.99999999e−1
7
6
5
4
3
Time Dilation Factor
6000
8000
10000
12000
14000
16000
18000
Distance from Earth's Center (km)
Spherical Orbit (Lorentz)
Spherical Orbit (Cubic)
Elliptical Orbit Periapsis (Lorentz)
Elliptical Orbit Apoapsis (Lorentz)
Elliptical Orbit Periapsis (Cubic)
Elliptical Orbit Apoapsis (Cubic)

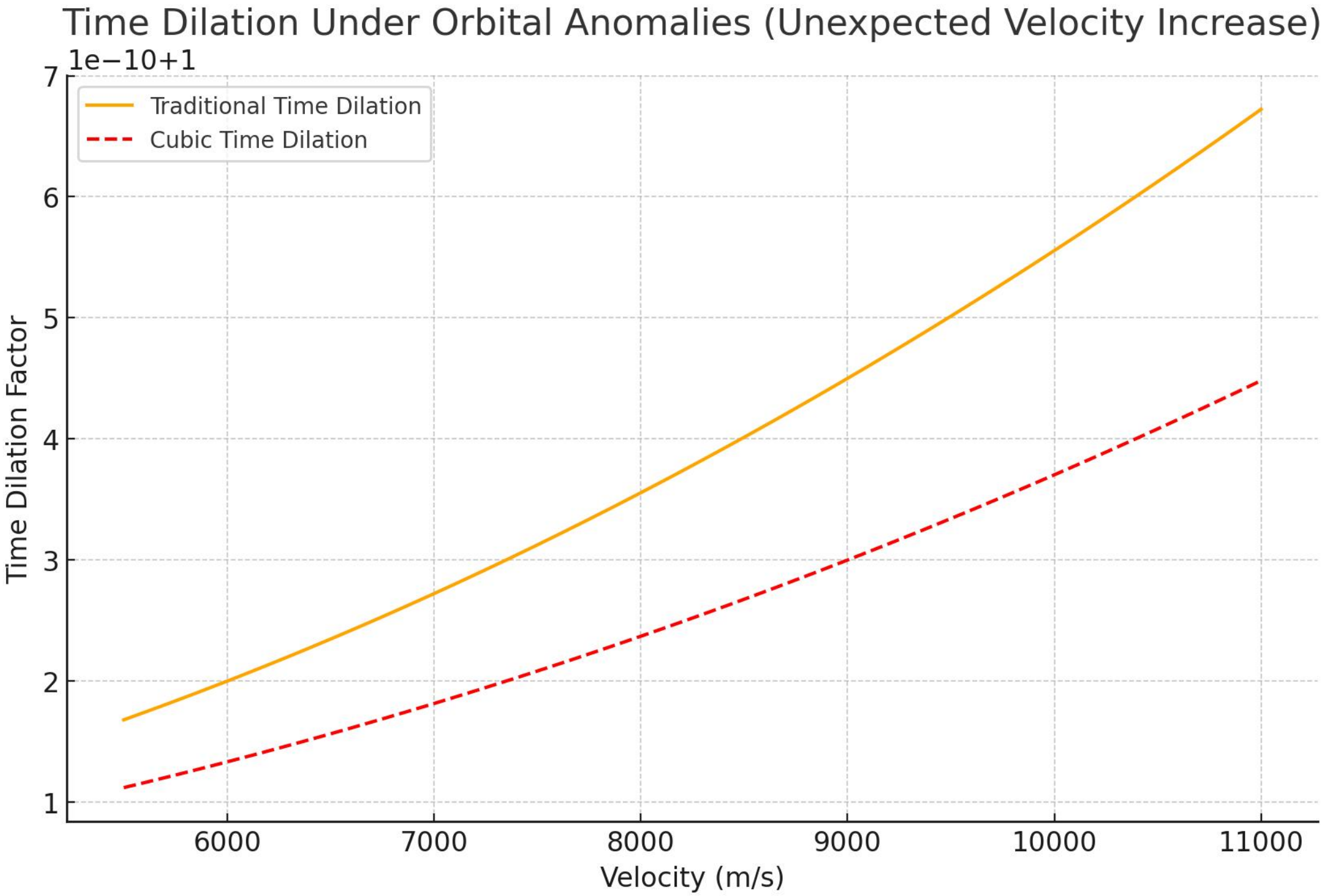

Time Dilation: High-Velocity Effects (Lorentz Standard GPS Model vs. Cubic Model)
Velocity (Lorentz)
Velocity (Cubic)
Time Dilation Factor
1.0
0.8
0.6
0.4
0.2
0.0
0.2
0.4
0.6
0.8
1.0
Velocity as Fraction of Speed of Light (v/c)

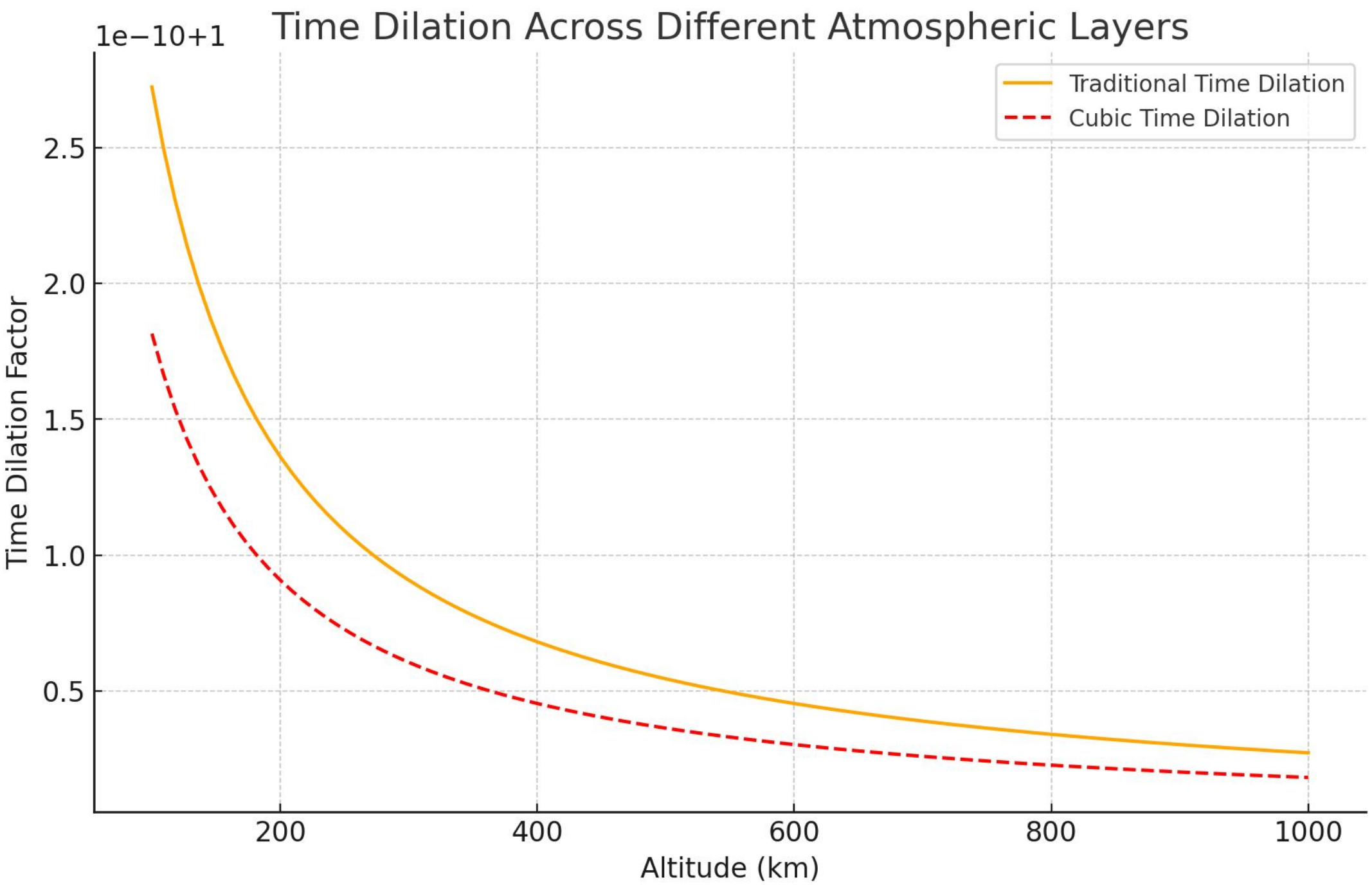

Time Dilation: Deep Space Conditions (Lorentz Standard GPS Model vs. Cubic Model)
1e−8+1
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
Time Dilation Factor
0
200
400
600
800
1000
Distance from Sun (AU)
Lorentz Model
Cubic Model

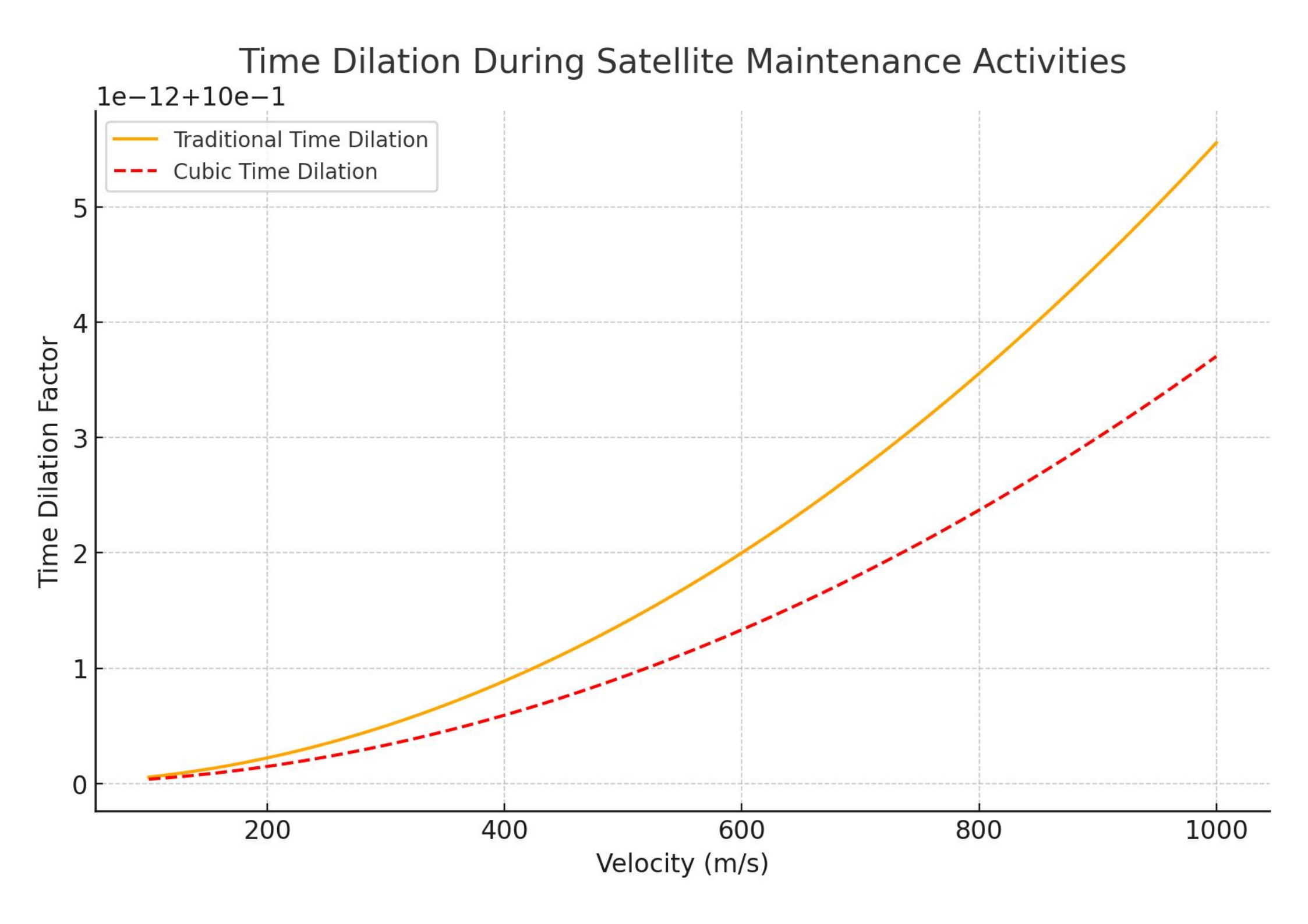

Time Dilation on Mars vs. Earth (Lorentz Standard GPS Model vs. Cubic Model)
1e−9+9.99999999e−1
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
Time Dilation Factor
5000
10000
15000
20000
25000
30000
35000
Distance from Planet Center (km)
Mars (Lorentz)
Mars (Cubic)
Earth (Lorentz)
Earth (Cubic)

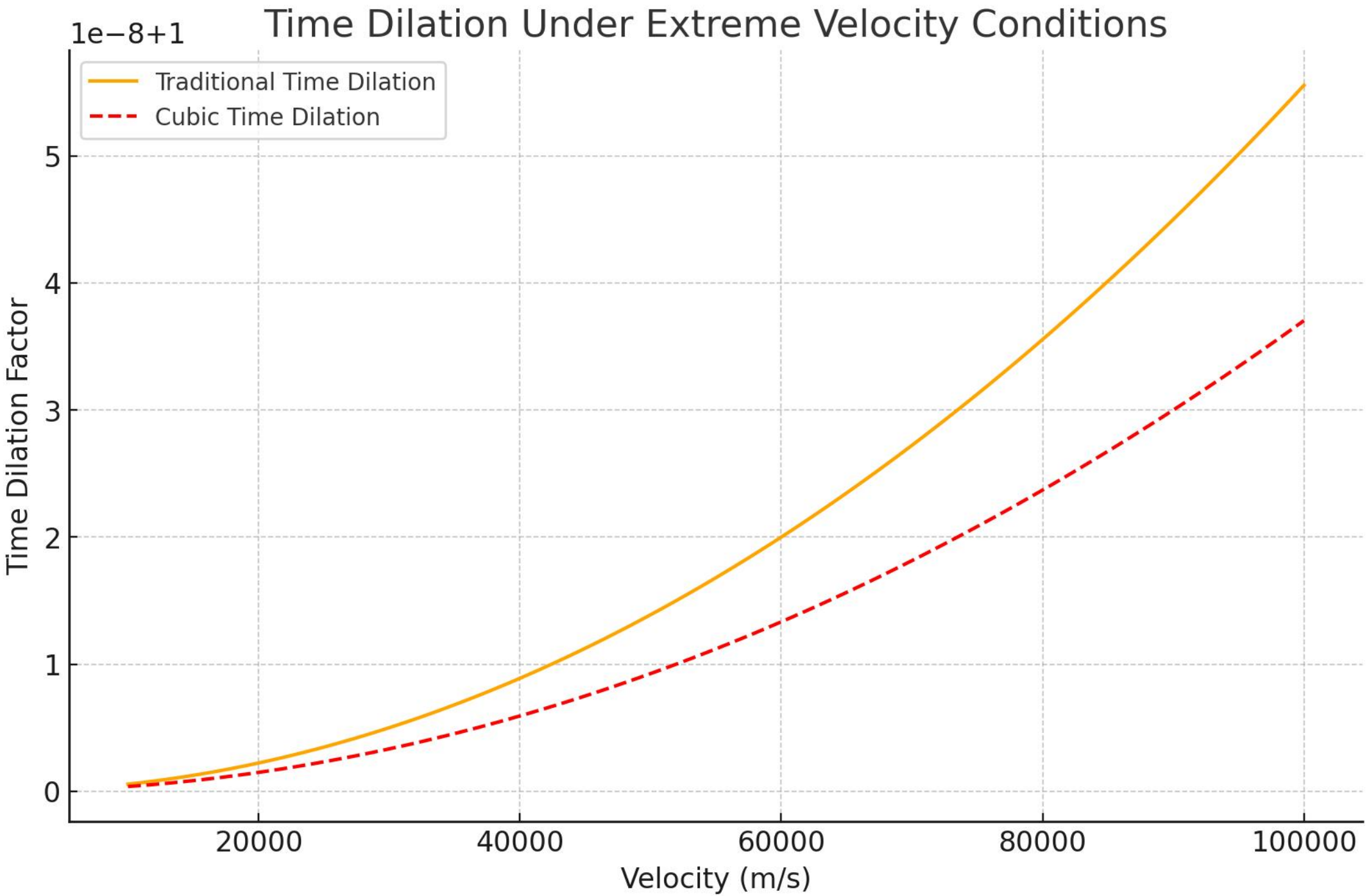

Mild Conditions: GPS Satellite Speeds
1e−10+1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
1.4
1.2
1.0
0.8
0.6
0.4
3000
3250
3500
3750
4000
4250
4500
4750
5000
Velocity (m/s)
Moderate Conditions: Higher Satellite Speeds
1e−10+1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
5
4
3
2
1
5000
6000
7000
8000
9000
10000
Velocity (m/s)
Extreme Conditions: Near Relativistic Speeds
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
1.8
1.6
1.4
1.2
1.0
0.0
0.5
1.0
1.5
2.0
2.5
1e8
Velocity (m/s)

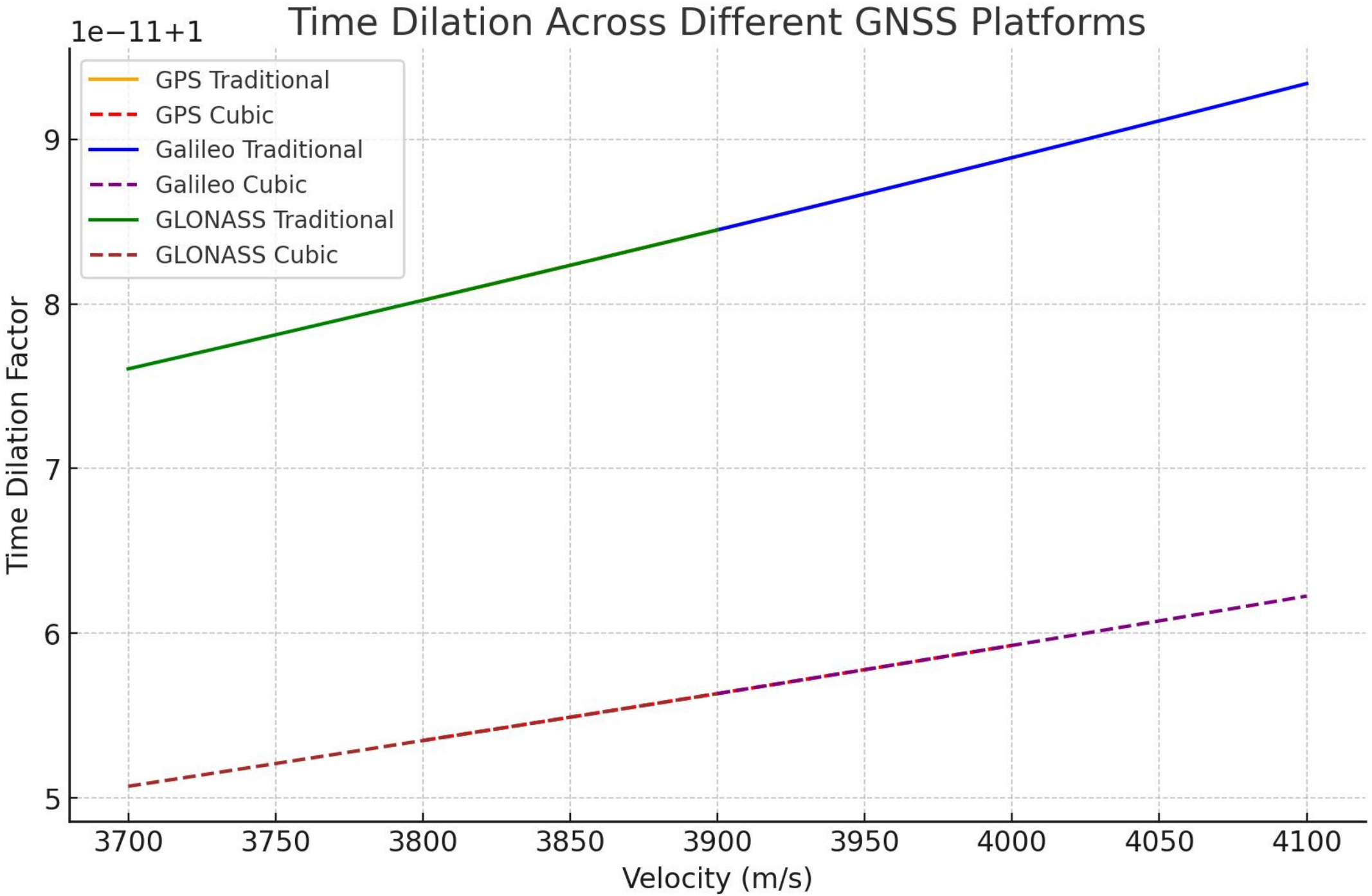

Time Dilation Under Gravitational Perturbations from the Moon
1e−10+1
Traditional Time Dilation
Cubic Time Dilation
1.4
1.2
1.0
0.8
0.6
0.4
Time Dilation Factor
360000
370000
380000
390000
400000
410000
Distance from Earth (km)

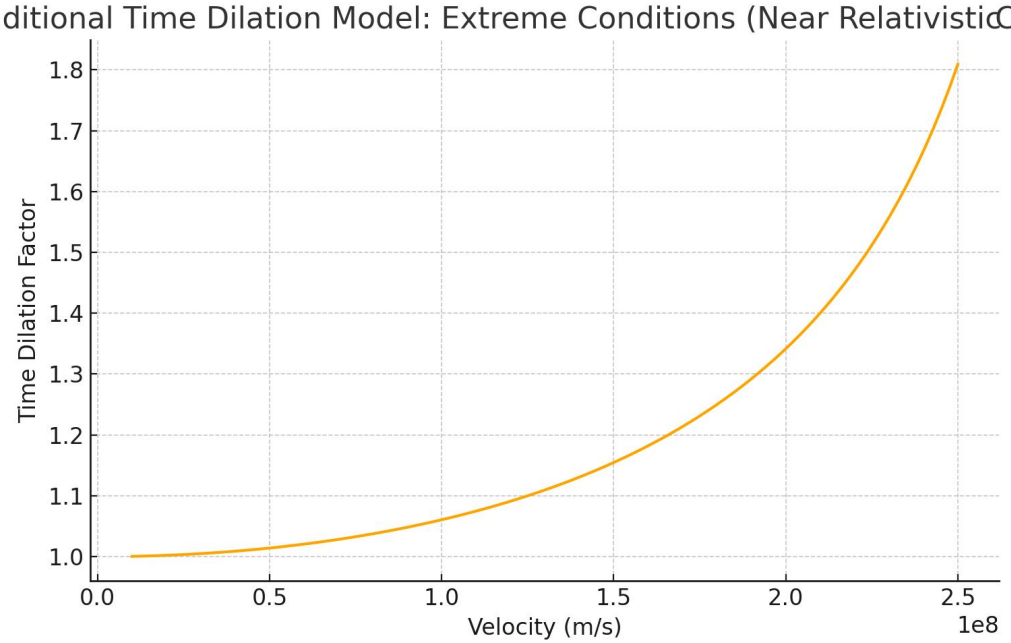

Time Dilation Under Orbital Anomalies (Unexpected Velocity Increase)
1e−10+1
7
6
5
4
3
2
1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
6000
7000
8000
9000
10000
11000
Velocity (m/s)

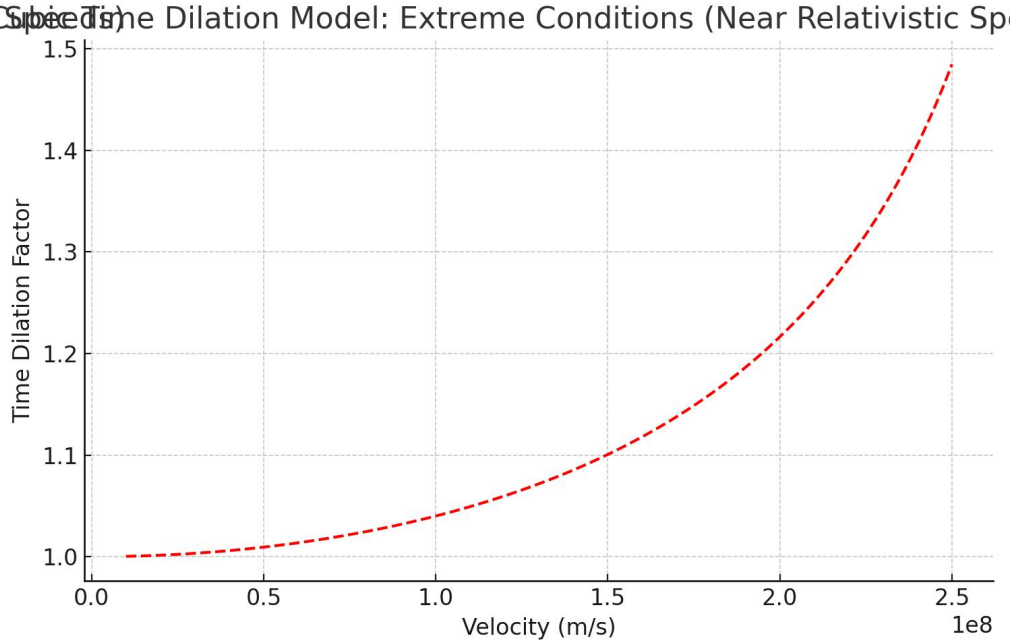

Time Dilation Across Different Atmospheric Layers
1e−10+1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
2.5
2.0
1.5
1.0
0.5
200
400
600
800
1000
Altitude (km)

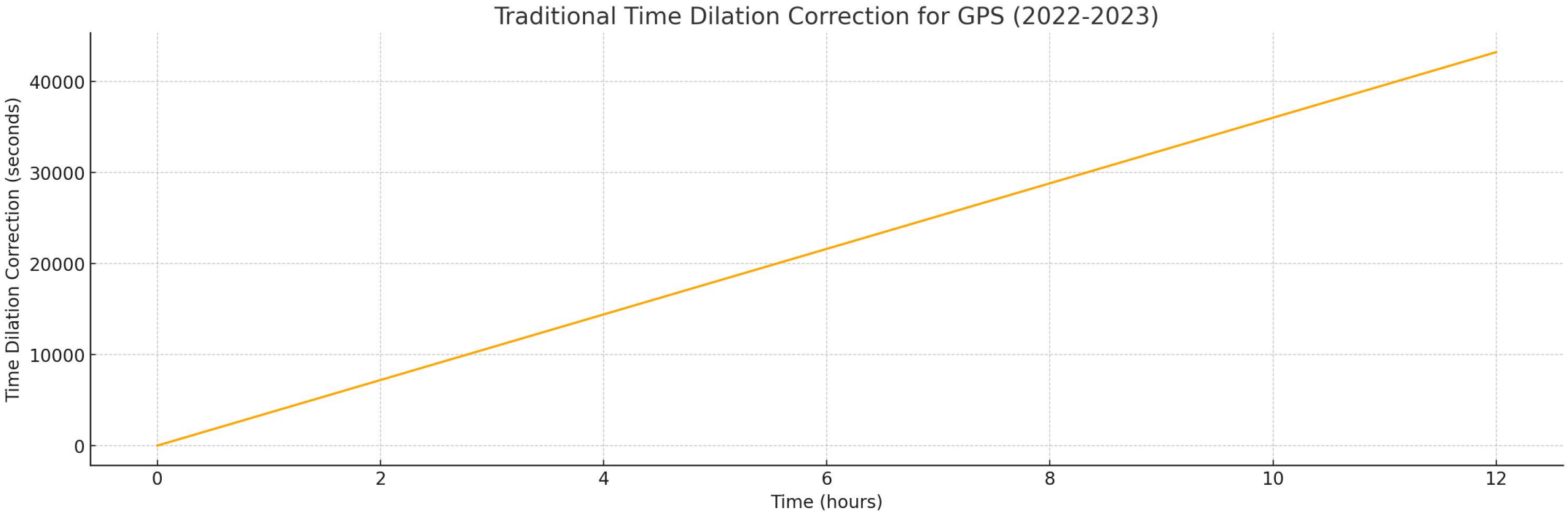

Time Dilation During Satellite Maintenance Activities
1e−12+10e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
0
1
2
3
4
5
200
400
600
800
1000
Velocity (m/s)

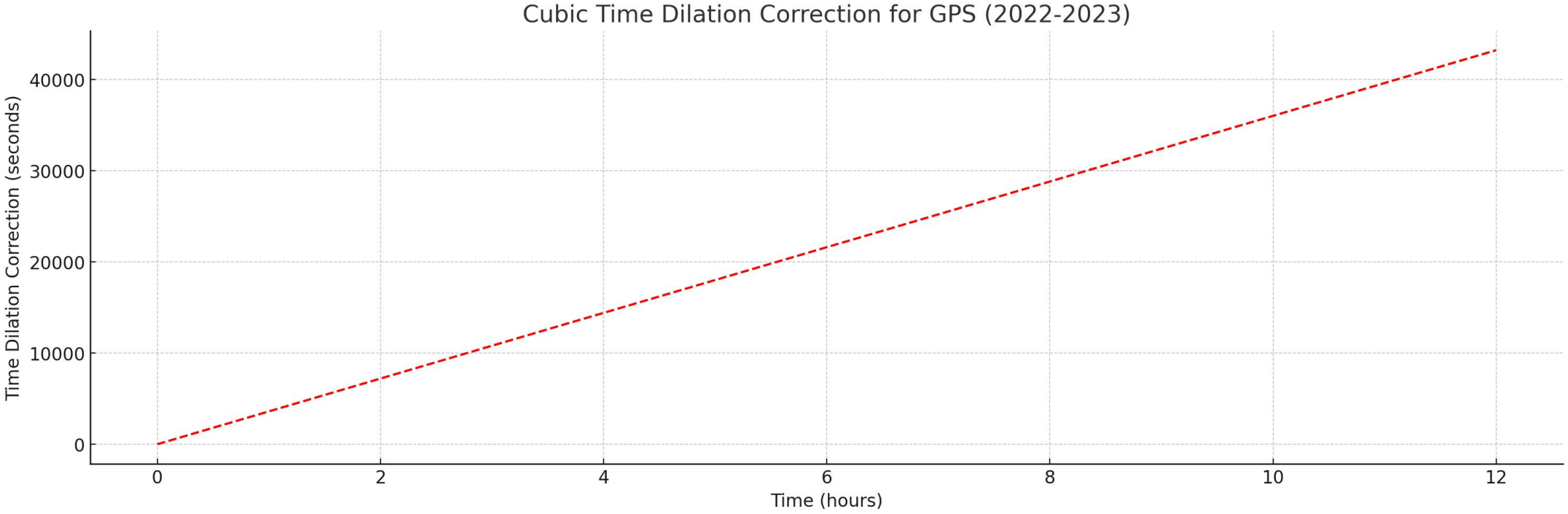

Time Dilation Under Extreme Velocity Conditions
1e−8+1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
0
1
2
3
4
5
20000
40000
60000
80000
100000
Velocity (m/s)

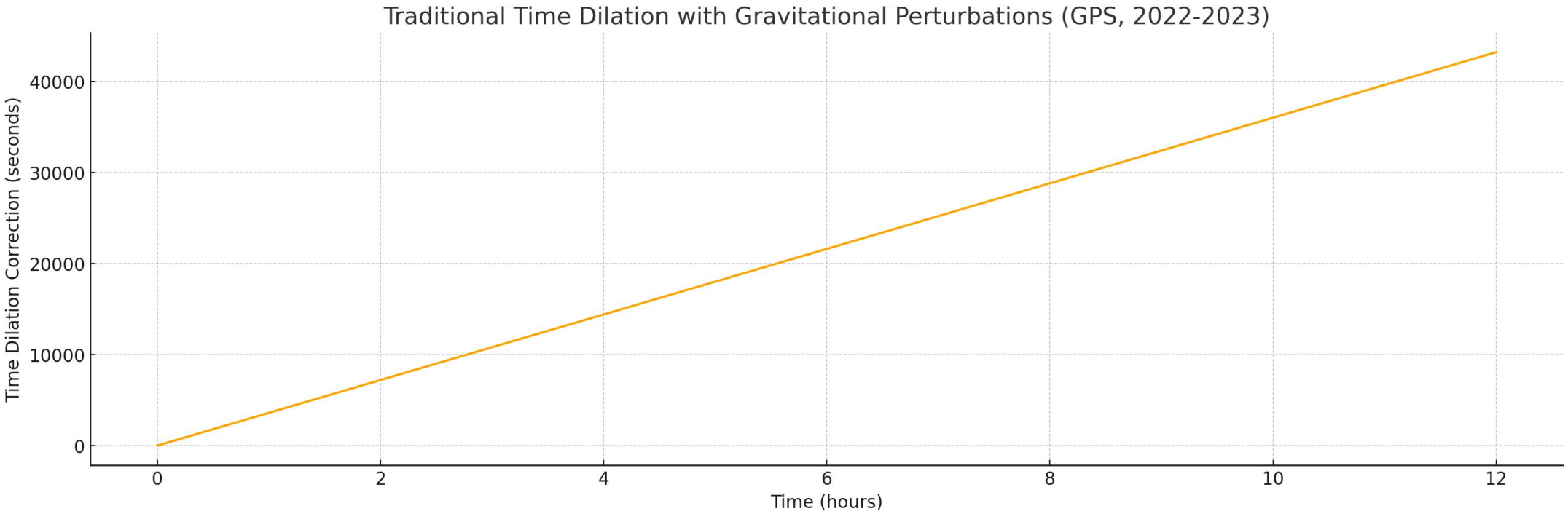

Time Dilation Across Different GNSS Platforms
1e−11+1
GPS Traditional
GPS Cubic
Galileo Traditional
Galileo Cubic
GLONASS Traditional
GLONASS Cubic
Time Dilation Factor
9
8
7
6
5
3700
3750
3800
3850
3900
3950
4000
4050
4100
Velocity (m/s)

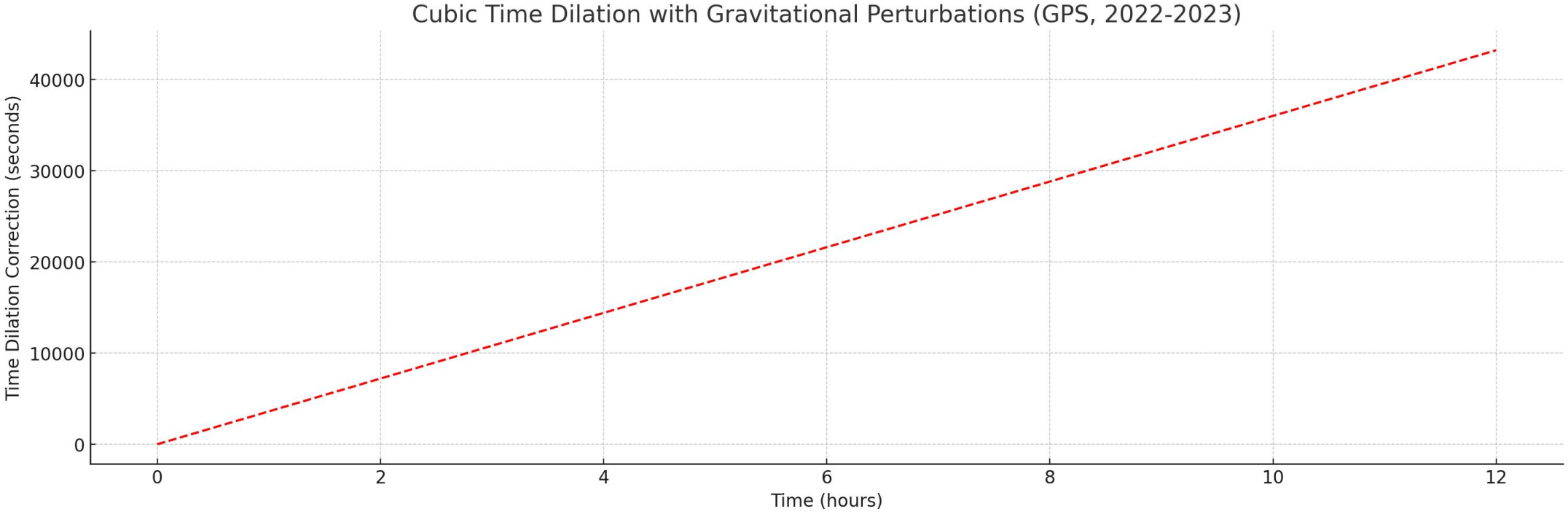

Traditional Time Dilation Model: Extreme Conditions (Near Relativistic Speeds)
Time Dilation Factor
1.0
1.1
1.2
1.3
1.4
1.5
1.6
1.7
1.8
0.0
0.5
1.0
1.5
2.0
2.5
Velocity (m/s)
1e8

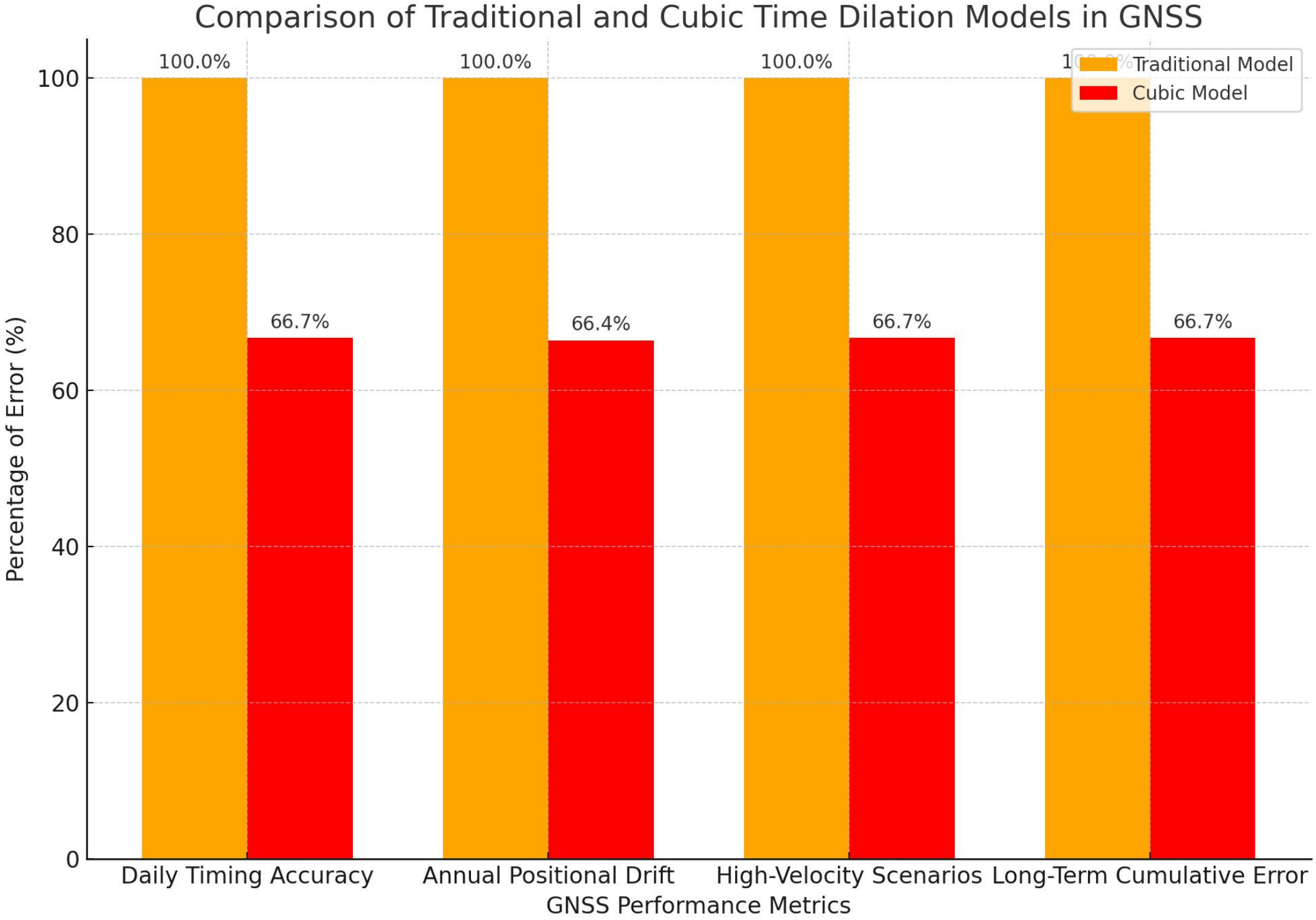

CSpace Time Dilation Model: Extreme Conditions (Near Relativistic Speeds)
Time Dilation Factor
1.0
1.1
1.2
1.3
1.4
1.5
0.0
0.5
1.0
1.5
2.0
2.5
Velocity (m/s)
1e8

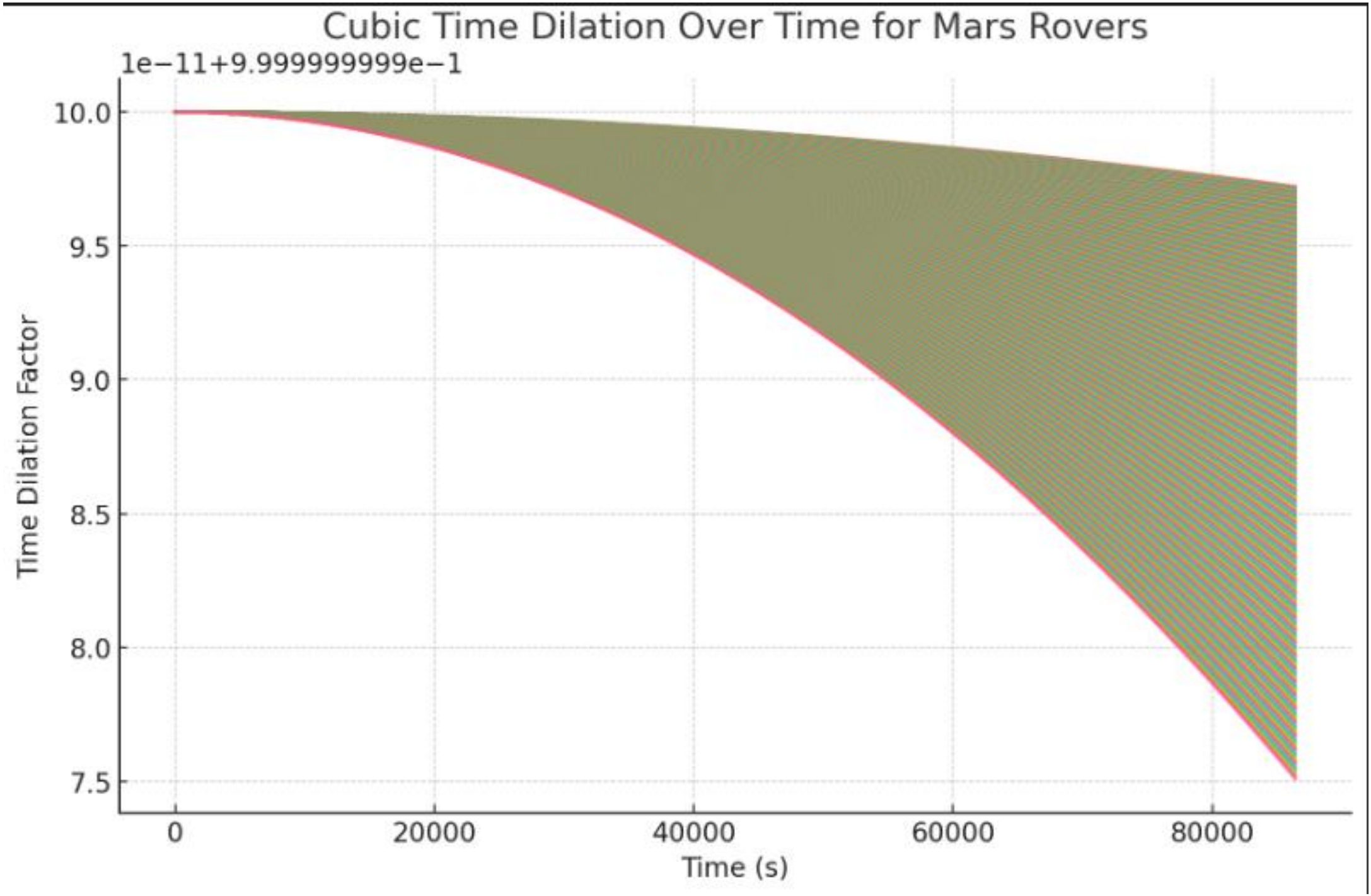

Traditional Time Dilation Correction for GPS (2022-2023)
Time Dilation Correction (seconds)
0
10000
20000
30000
40000
0
2
4
6
8
10
12
Time (hours)

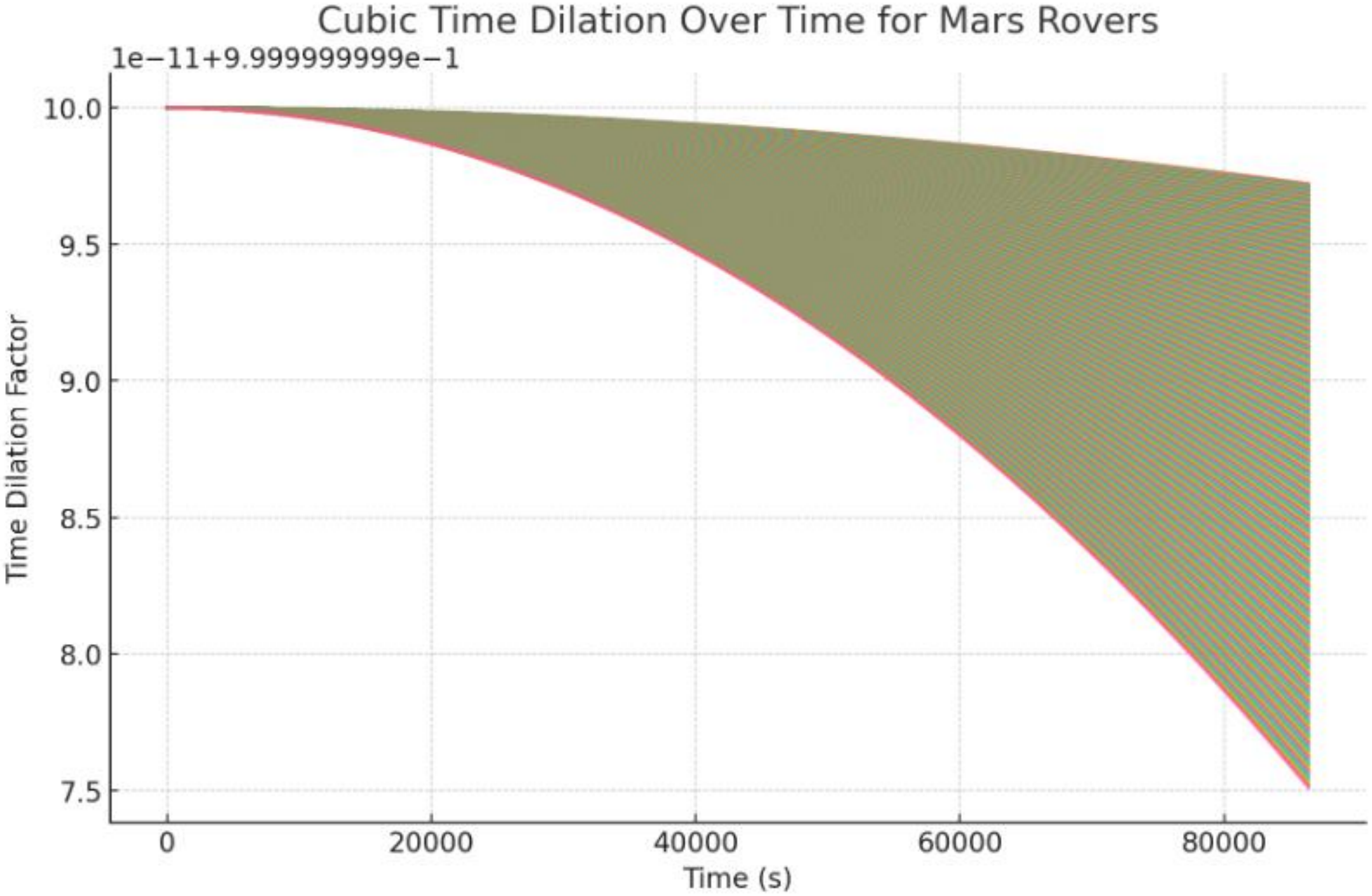

Cubic Time Dilation Correction for GPS (2022-2023)
Time Dilation Correction (seconds)
0
10000
20000
30000
40000
0
2
4
6
8
10
12
Time (hours)

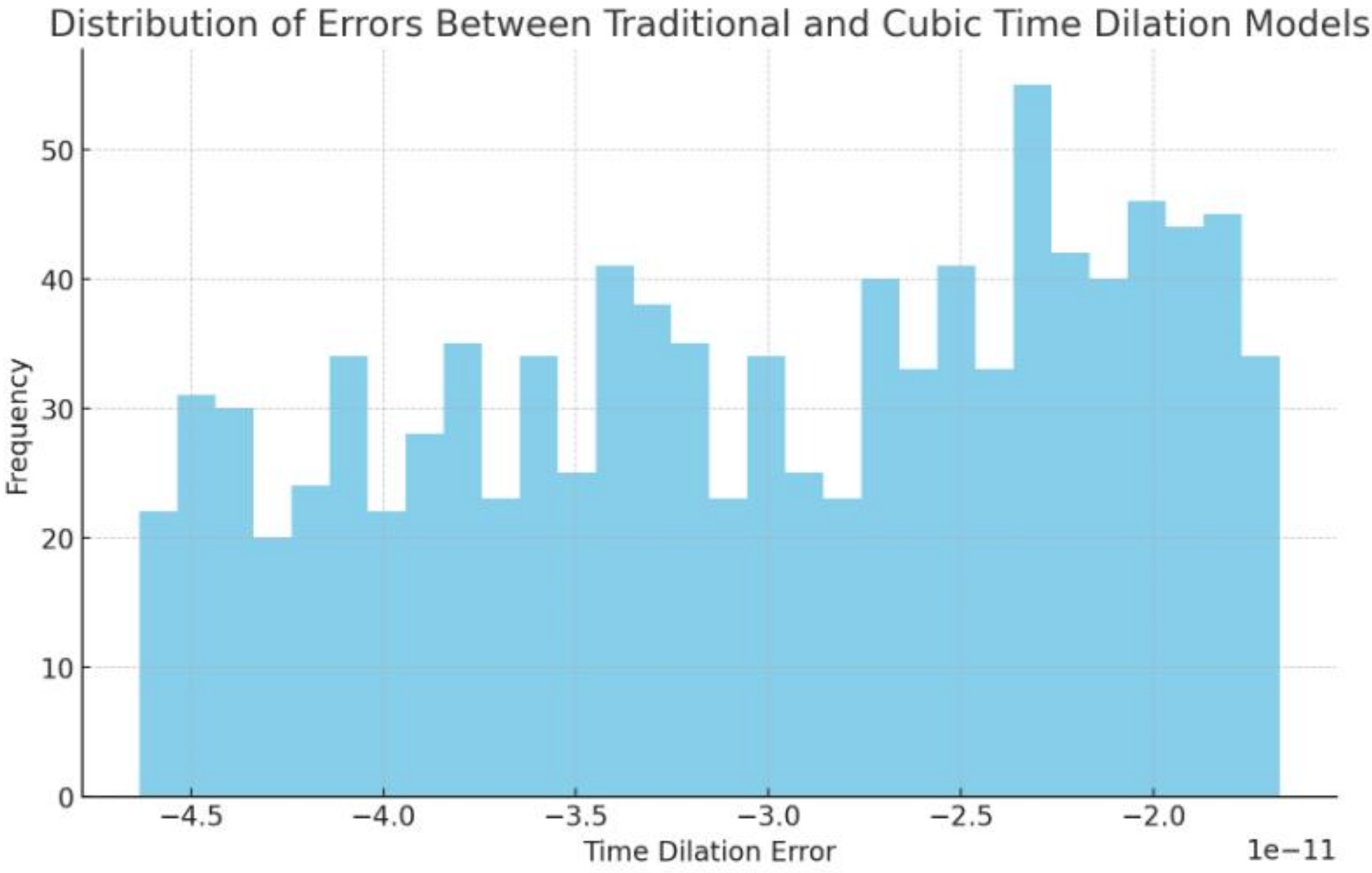

Traditional Time Dilation with Gravitational Perturbations (GPS, 2022-2023)
Time Dilation Correction (seconds)
0
10000
20000
30000
40000
0
2
4
6
8
10
12
Time (hours)

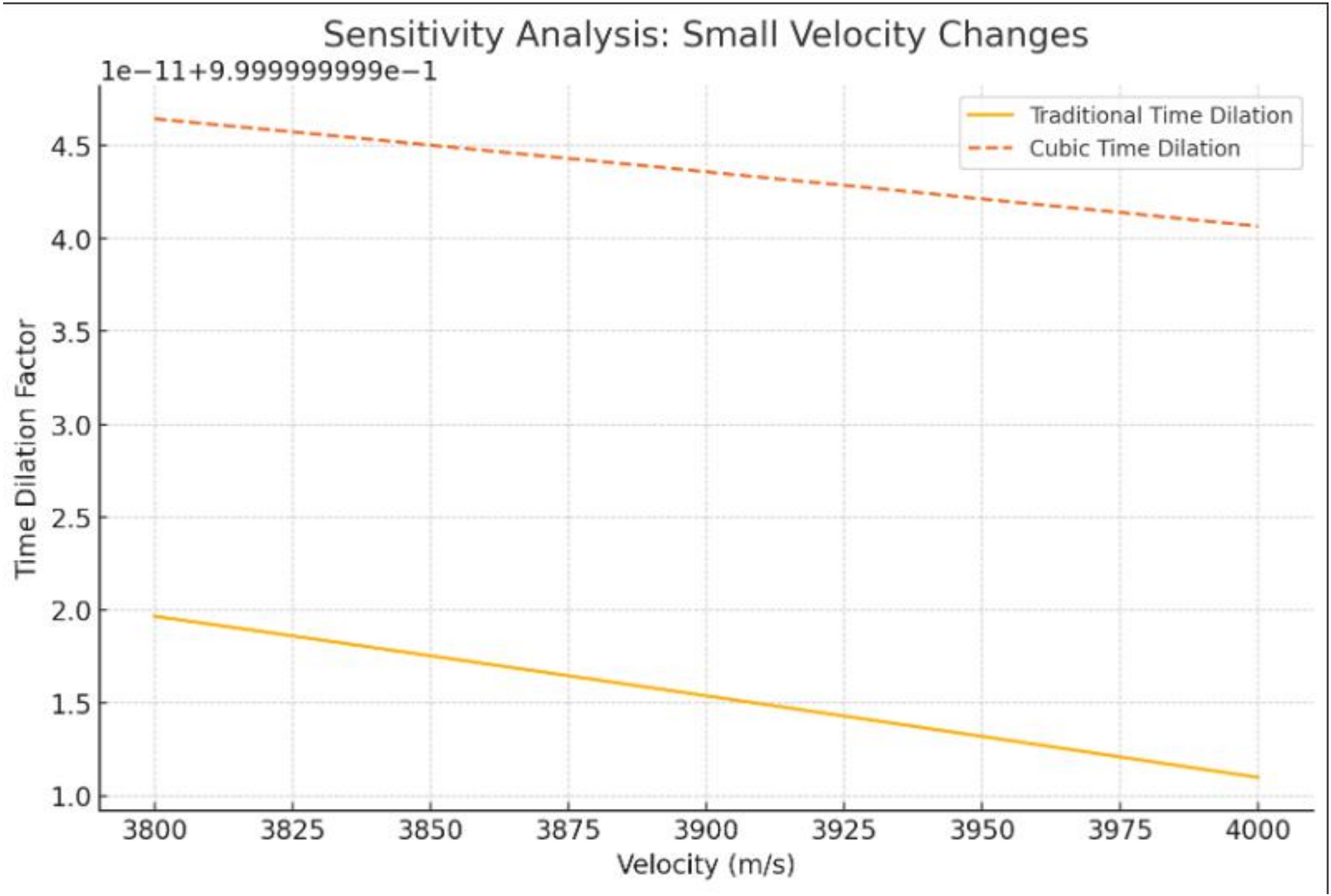

Cubic Time Dilation with Gravitational Perturbations (GPS, 2022-2023)
Time Dilation Correction (seconds)
0
10000
20000
30000
40000
0
2
4
6
8
10
12
Time (hours)

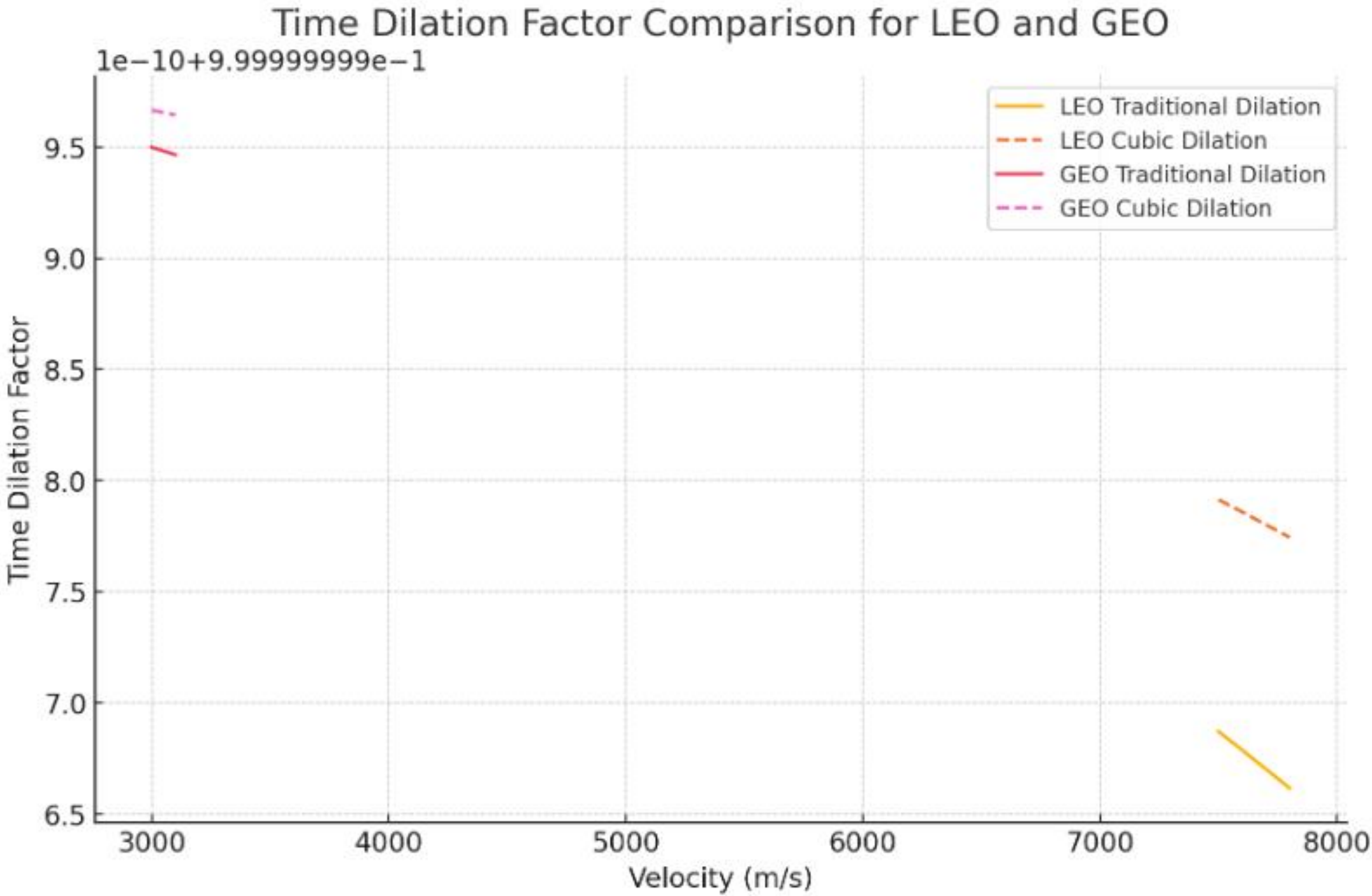

Comparison of Traditional and Cubic Time Dilation Models in GNSS
Traditional Model
Cubic Model
Percentage of Error (%)
0
20
40
60
80
100
100.0%
66.7%
100.0%
66.4%
100.0%
66.7%
66.7%
Daily Timing Accuracy
Annual Positional Drift
High-Velocity Scenarios
Long-Term Cumulative Error
GNSS Performance Metrics

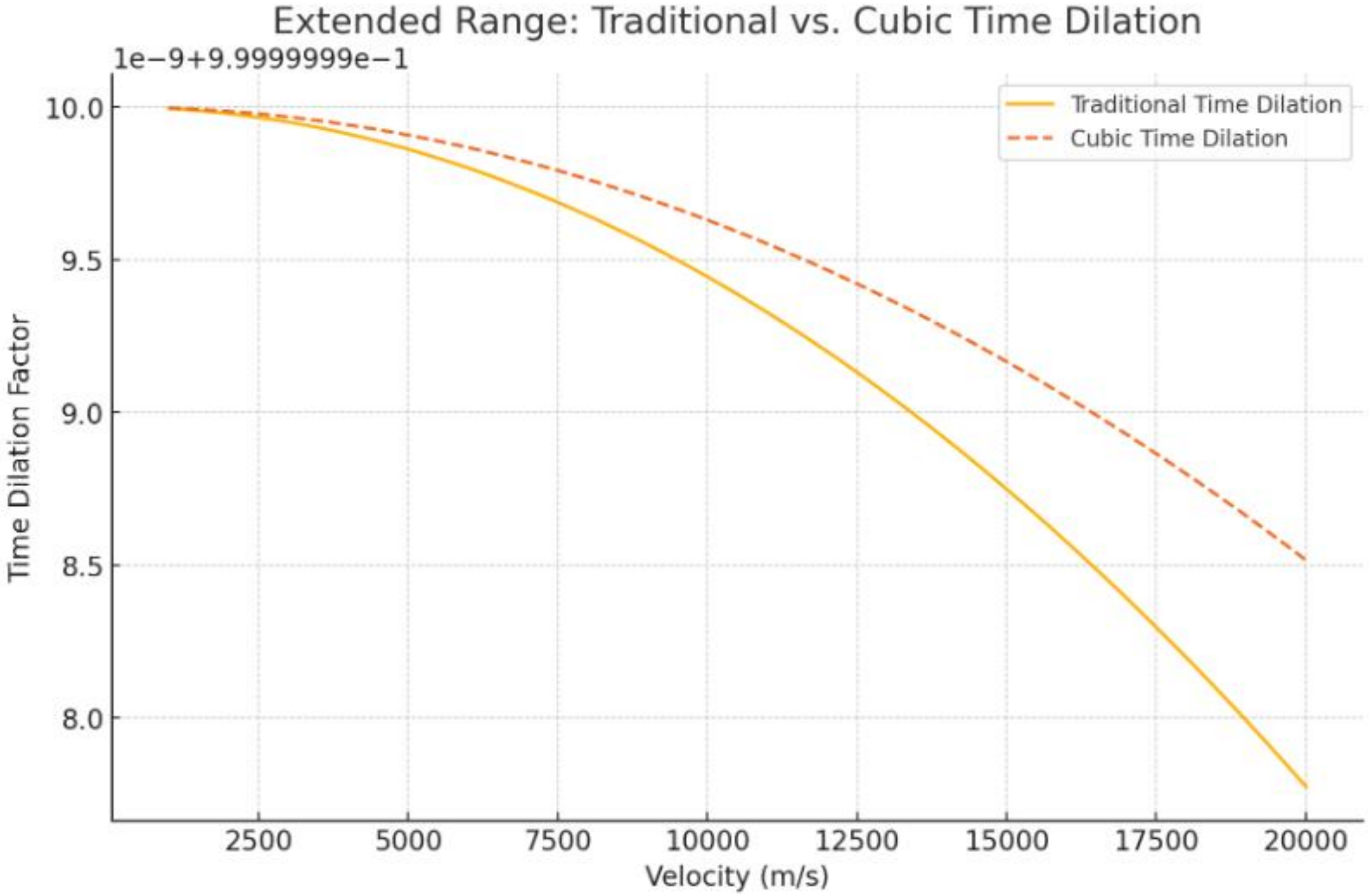

Cubic Time Dilation Over Time for Mars Rovers
1e−11+9.999999999e−1
10.0
9.5
9.0
8.5
8.0
7.5
Time Dilation Factor
0
20000
40000
60000
80000
Time (s)

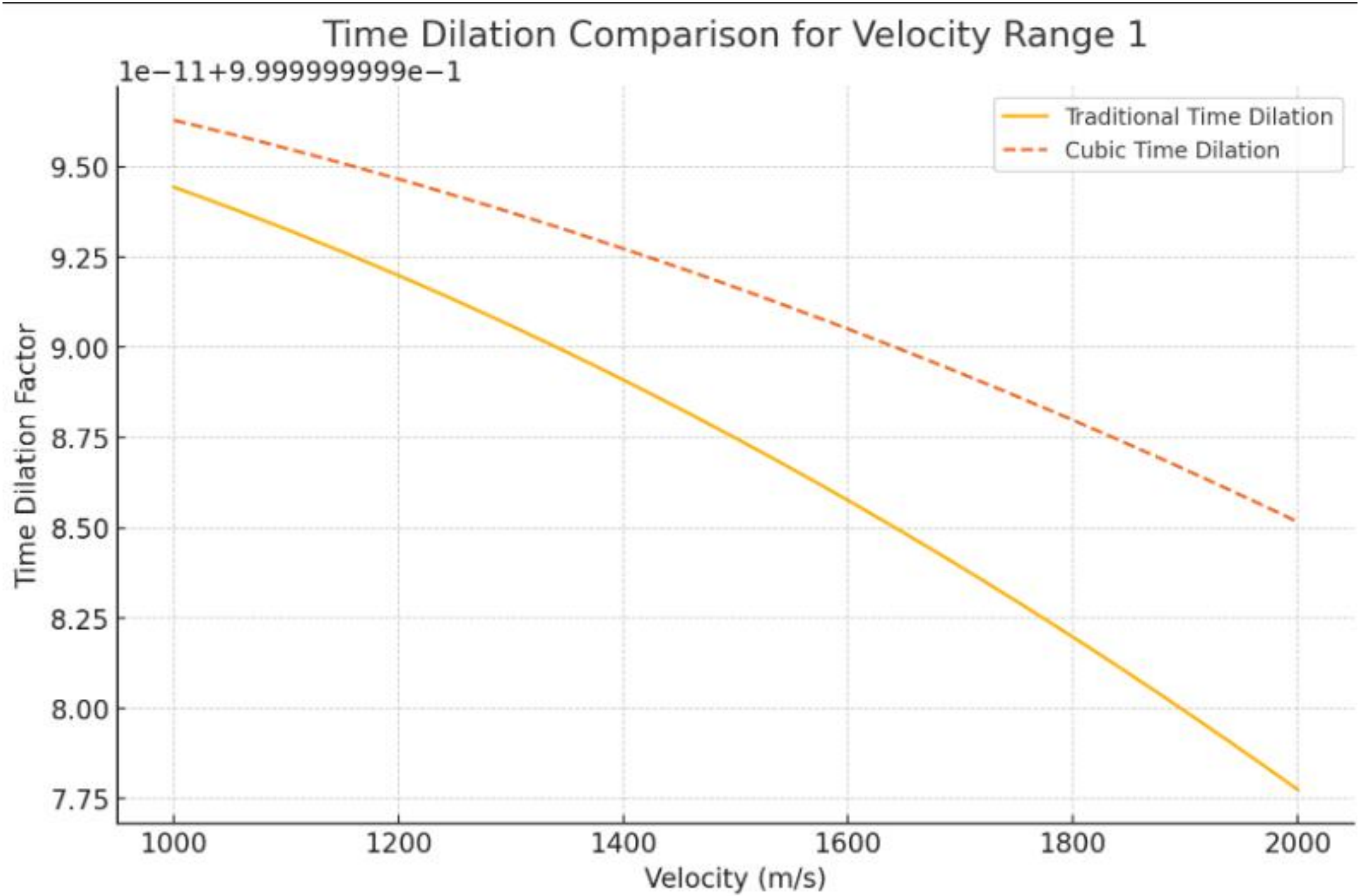

Cubic Time Dilation Over Time for Mars Rovers
1e−11+9.999999999e−1
10.0
9.5
9.0
8.5
8.0
7.5
Time Dilation Factor
0
20000
40000
60000
80000
Time (s)

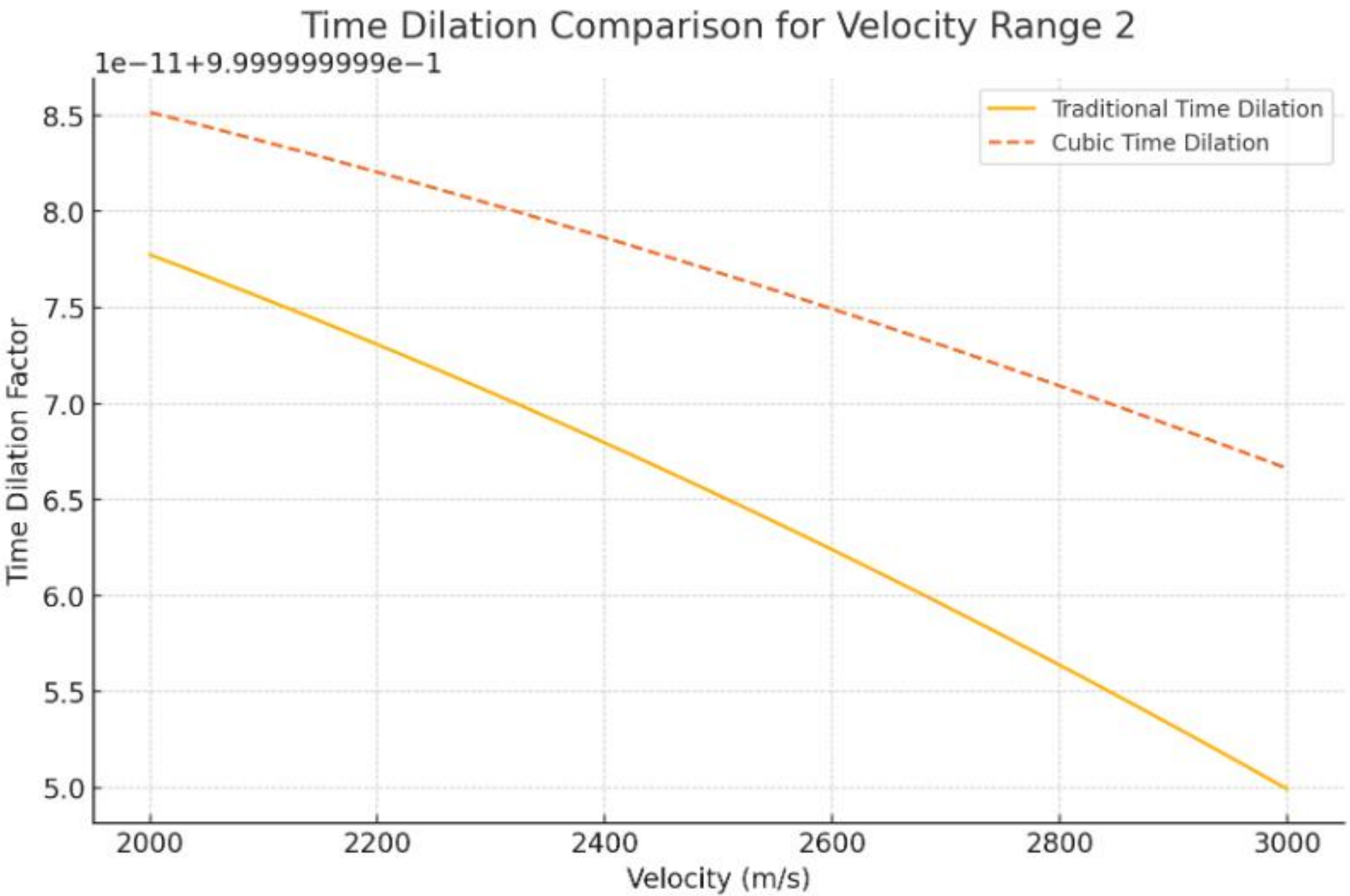

Distribution of Errors Between Traditional and Cubic Time Dilation Models
Frequency
0
10
20
30
40
50
−4.5
−4.0
−3.5
−3.0
−2.5
−2.0
Time Dilation Error
1e−11

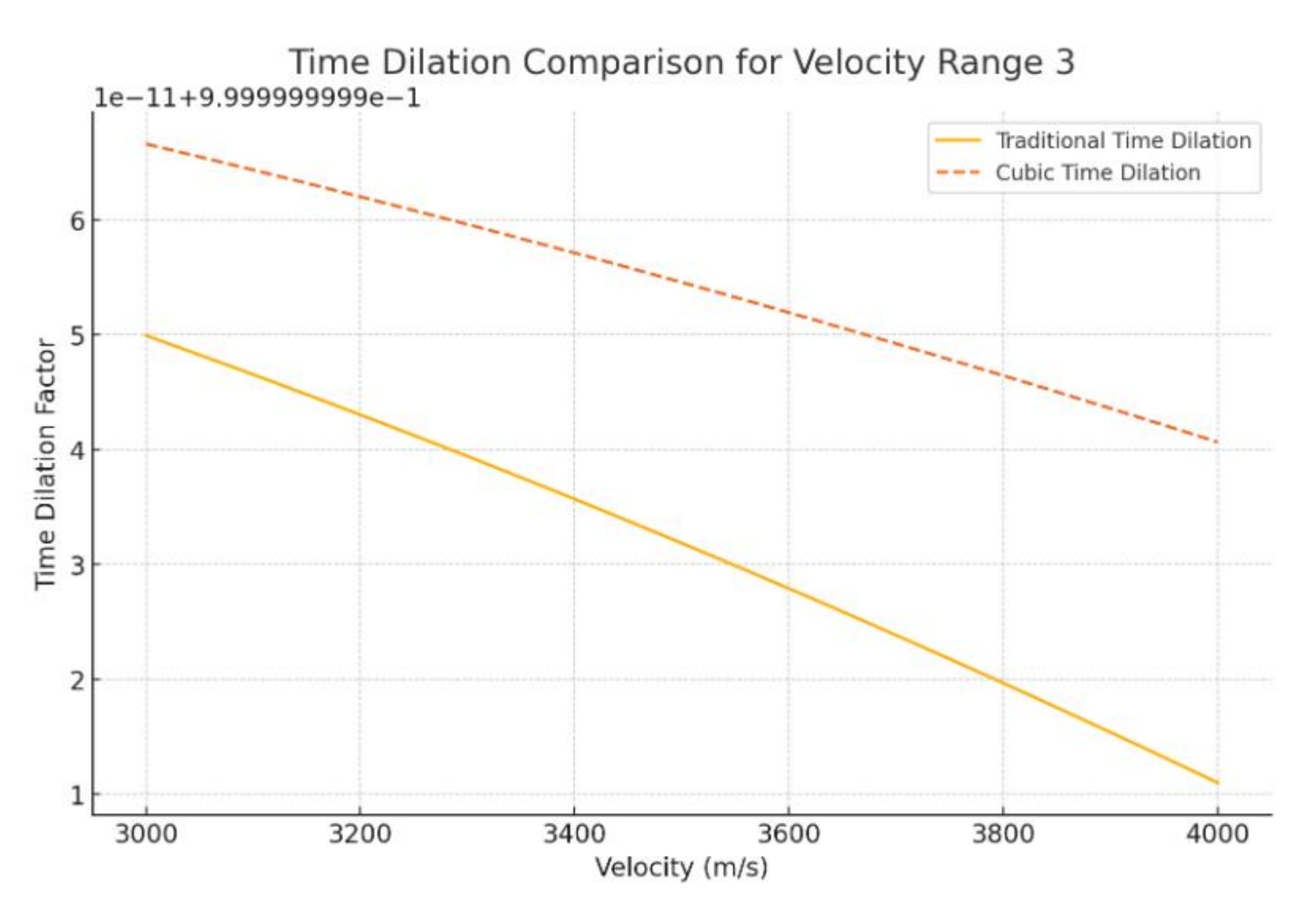

Sensitivity Analysis: Small Velocity Changes
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
3800
3825
3850
3875
3900
3925
3950
3975
4000
Velocity (m/s)

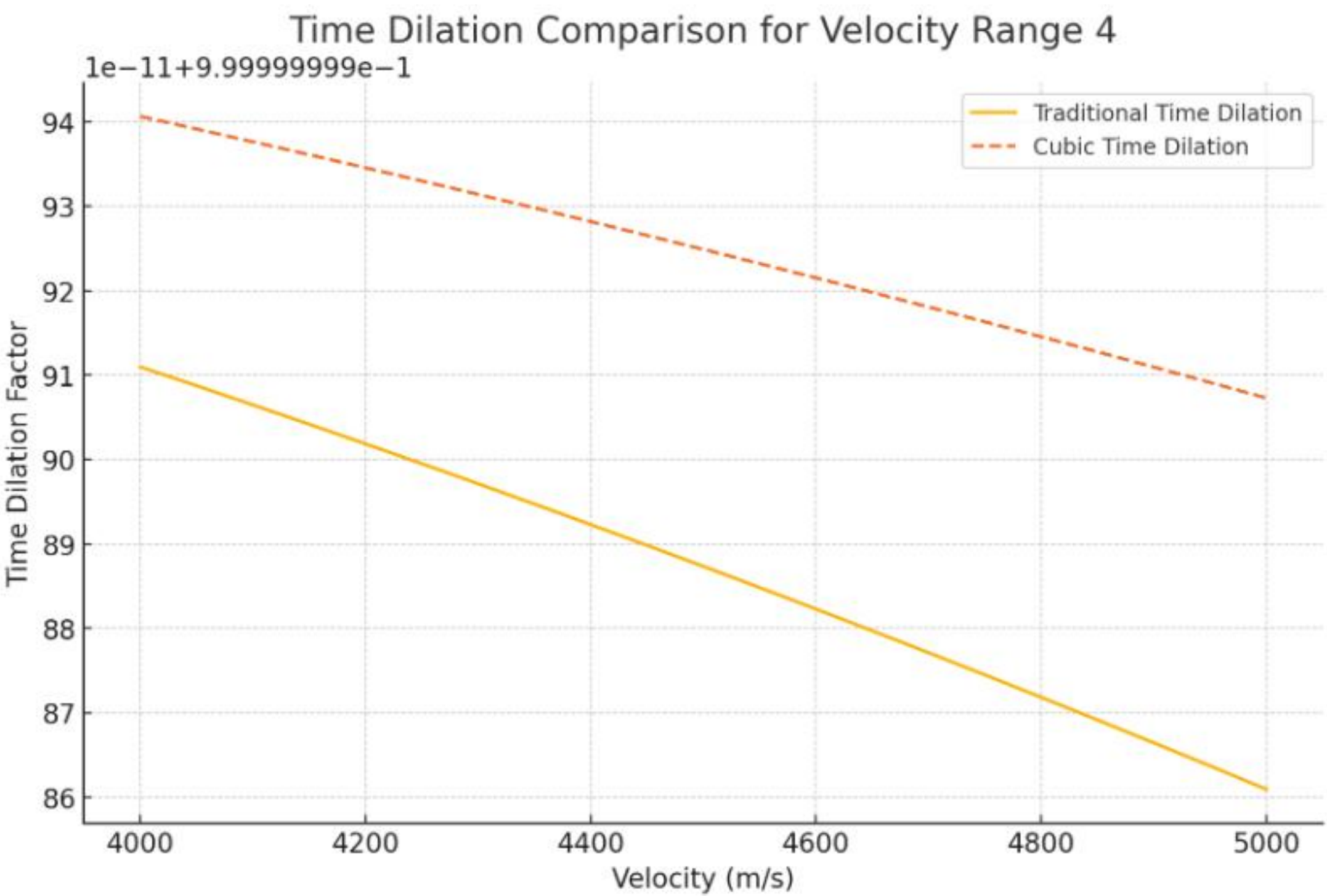

Time Dilation Factor Comparison for LEO and GEO
1e−10+9.99999999e−1
9.5
9.0
8.5
8.0
7.5
7.0
6.5
Time Dilation Factor
3000
4000
5000
6000
7000
8000
Velocity (m/s)
LEO Traditional Dilation
LEO Cubic Dilation
GEO Traditional Dilation
GEO Cubic Dilation

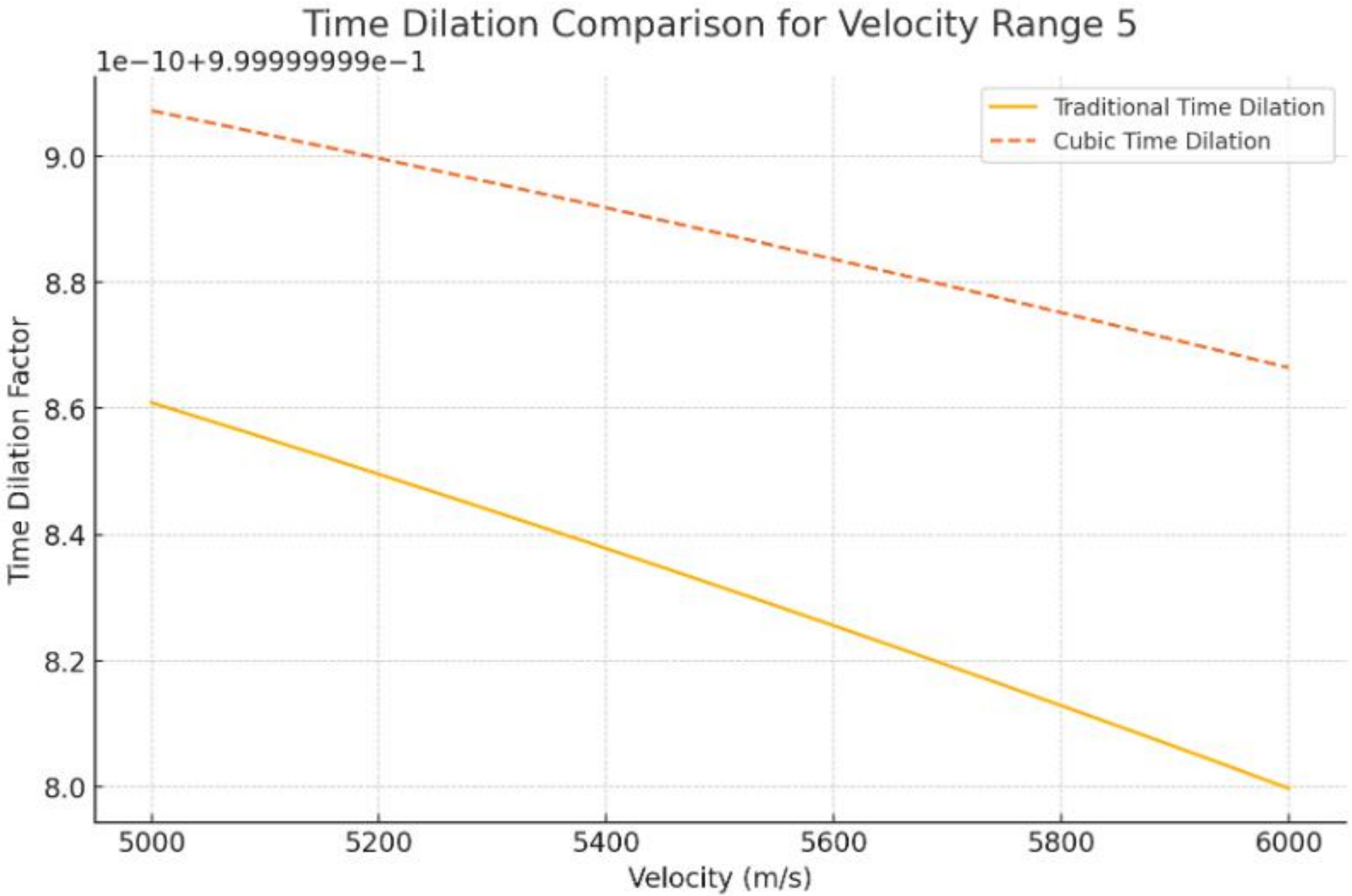

Extended Range: Traditional vs. Cubic Time Dilation
1e−9+9.9999999e−1
10.0
9.5
9.0
8.5
8.0
Time Dilation Factor
2500
5000
7500
10000
12500
15000
17500
20000
Velocity (m/s)
Traditional Time Dilation
Cubic Time Dilation

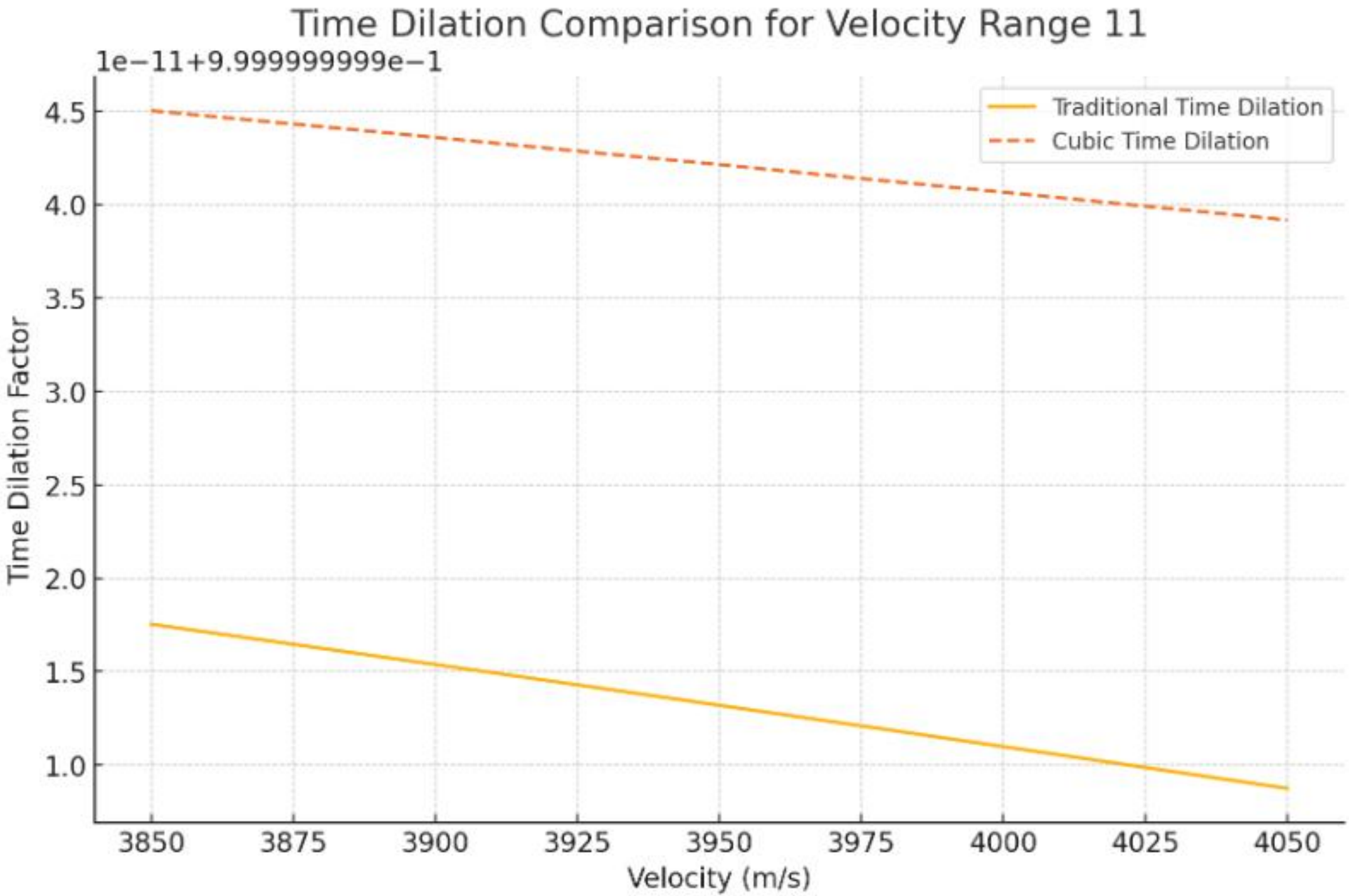

Time Dilation Comparison for Velocity Range 1
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
9.50
9.25
9.00
8.75
8.50
8.25
8.00
7.75
Time Dilation Factor
1000
1200
1400
1600
1800
2000
Velocity (m/s)

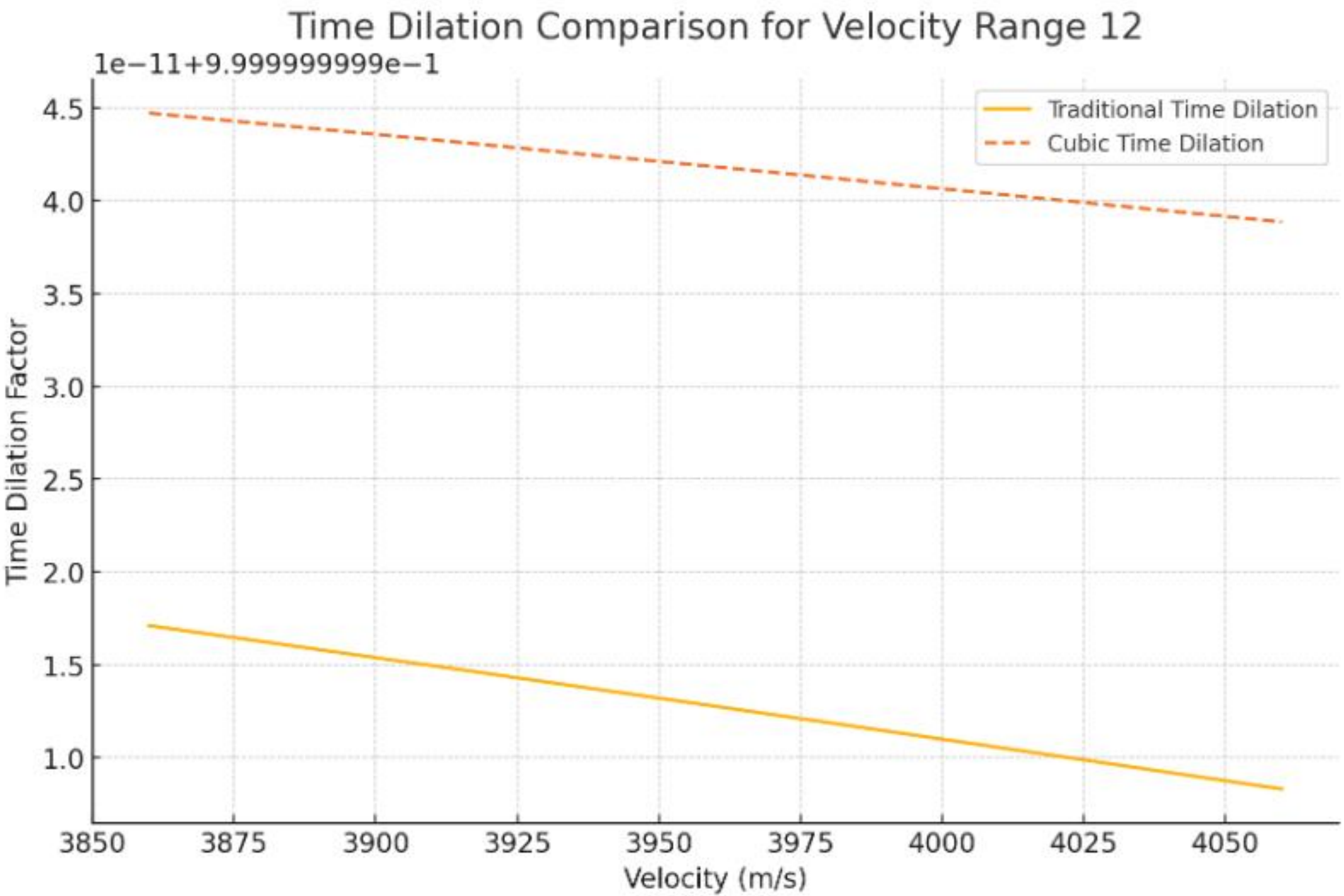

Time Dilation Comparison for Velocity Range 2
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
8.5
8.0
7.5
7.0
6.5
6.0
5.5
5.0
2000
2200
2400
2600
2800
3000
Velocity (m/s)

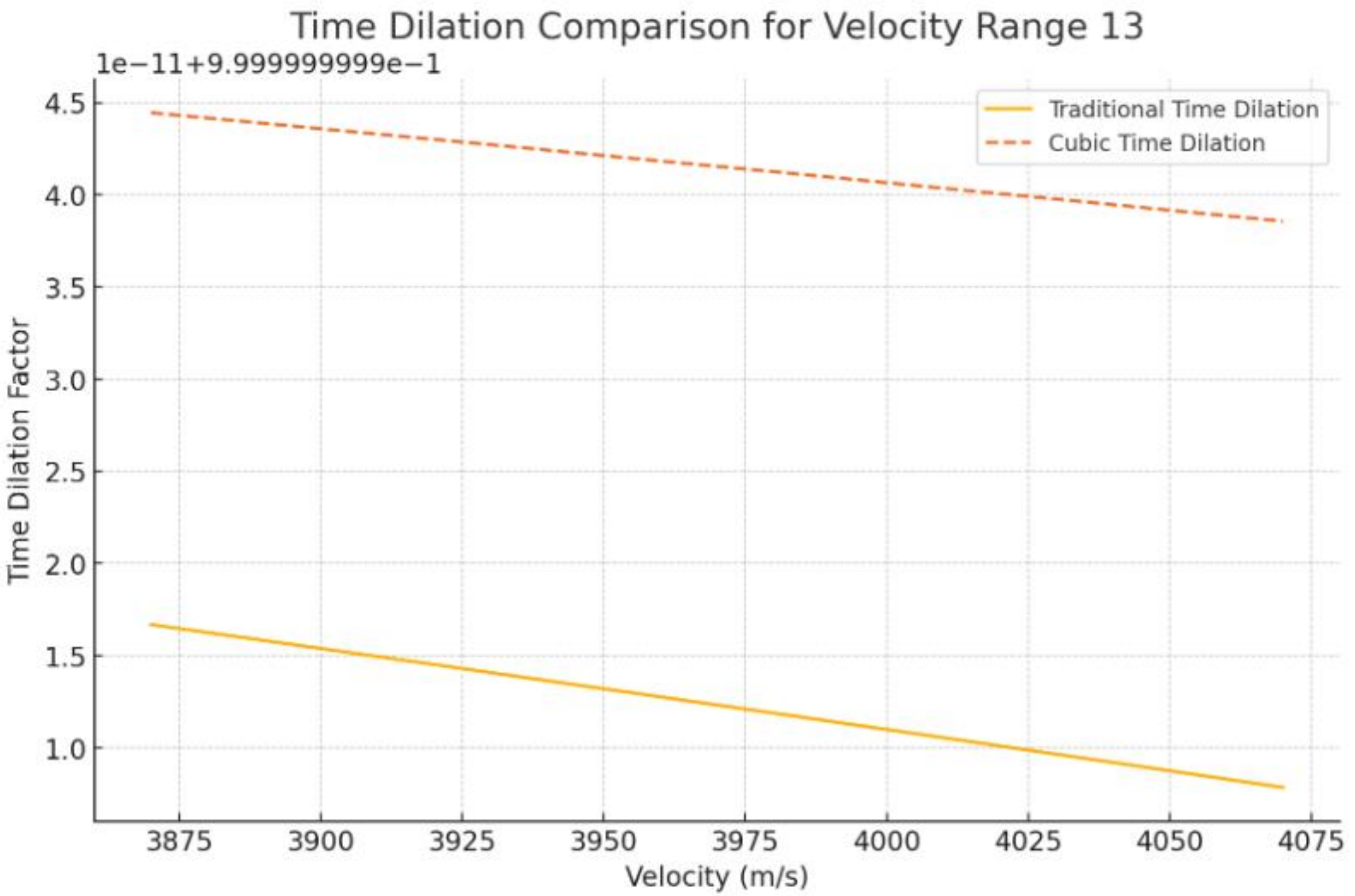

Time Dilation Comparison for Velocity Range 3
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
6
5
4
3
2
1
3000
3200
3400
3600
3800
4000
Velocity (m/s)

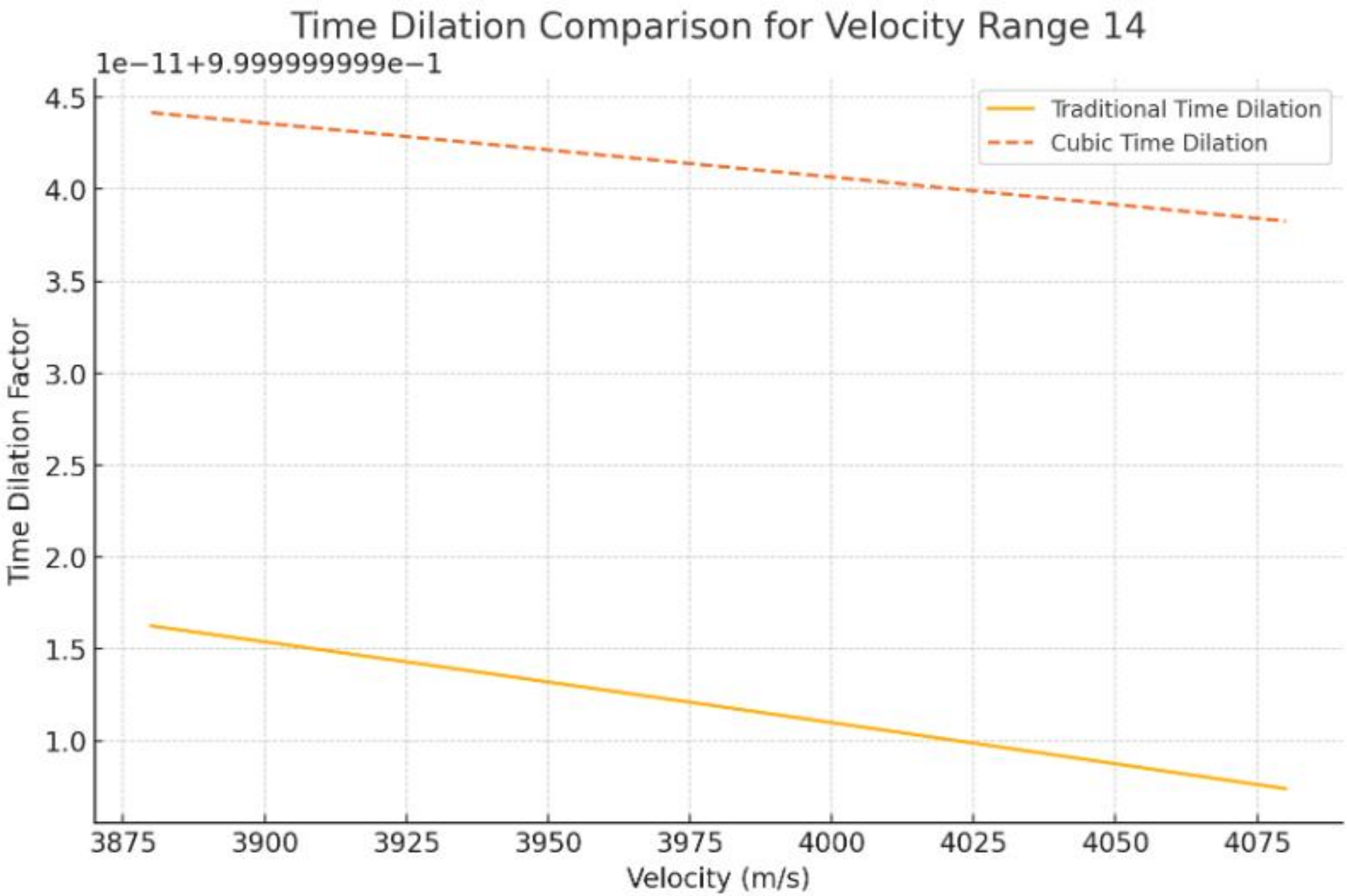

Time Dilation Comparison for Velocity Range 4
1e−11+9.99999999e−1
94
93
92
91
90
89
88
87
86
Time Dilation Factor
4000
4200
4400
4600
4800
5000
Velocity (m/s)
Traditional Time Dilation
Cubic Time Dilation

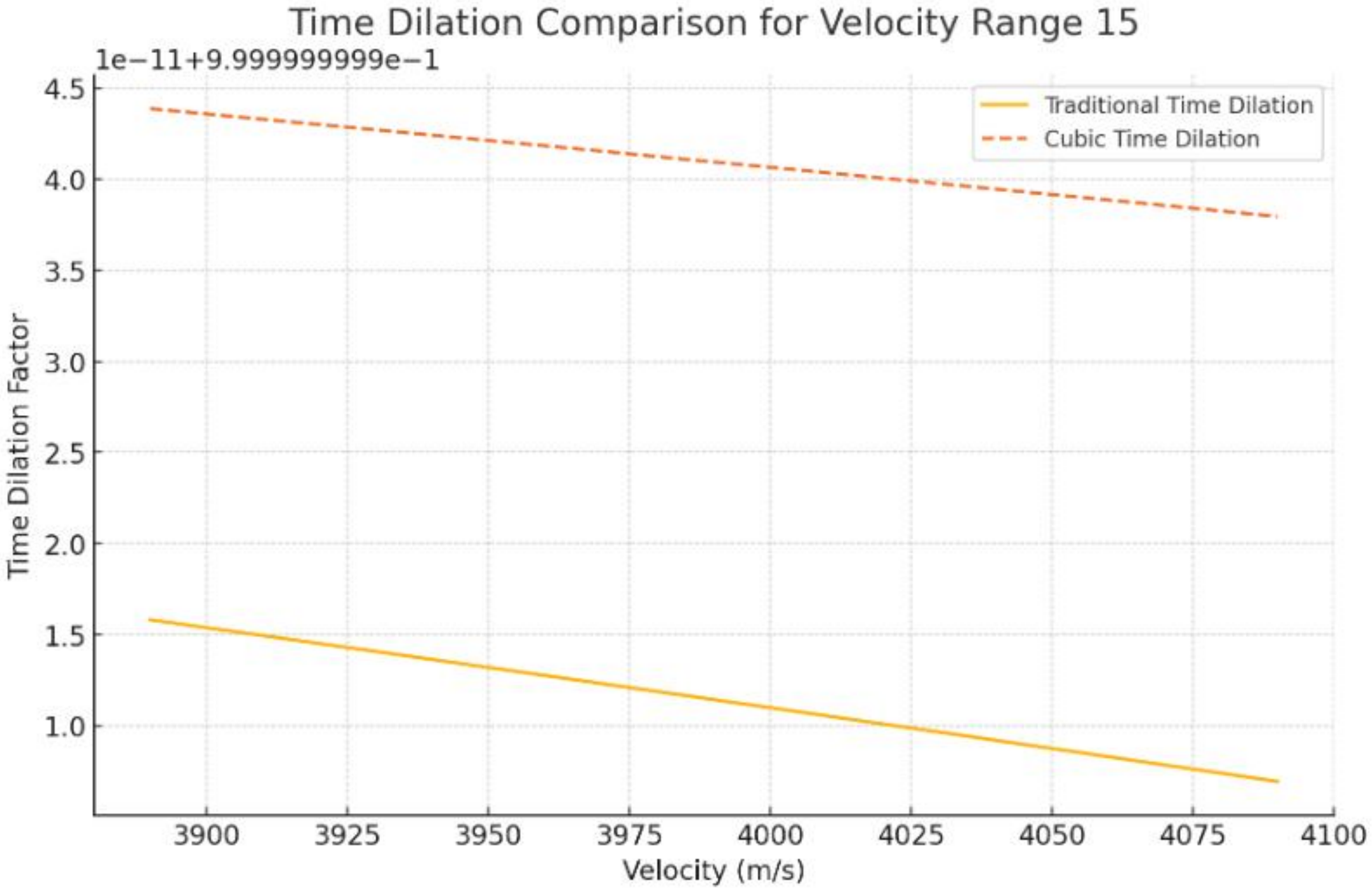

Time Dilation Comparison for Velocity Range 5
1e−10+9.99999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
9.0
8.8
8.6
8.4
8.2
8.0
5000
5200
5400
5600
5800
6000
Velocity (m/s)

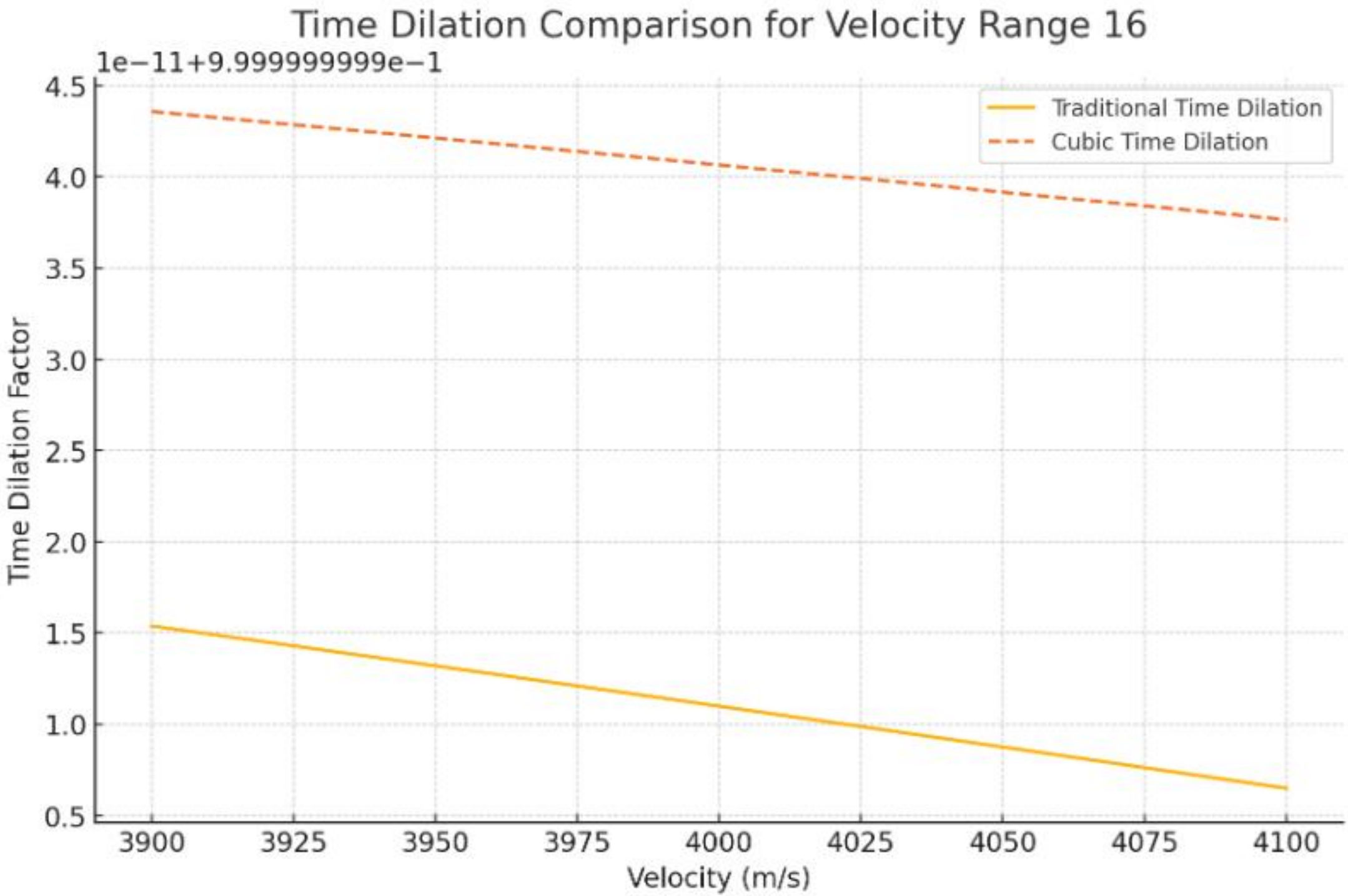

Time Dilation Comparison for Velocity Range 11
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
3850
3875
3900
3925
3950
3975
4000
4025
4050
Velocity (m/s)

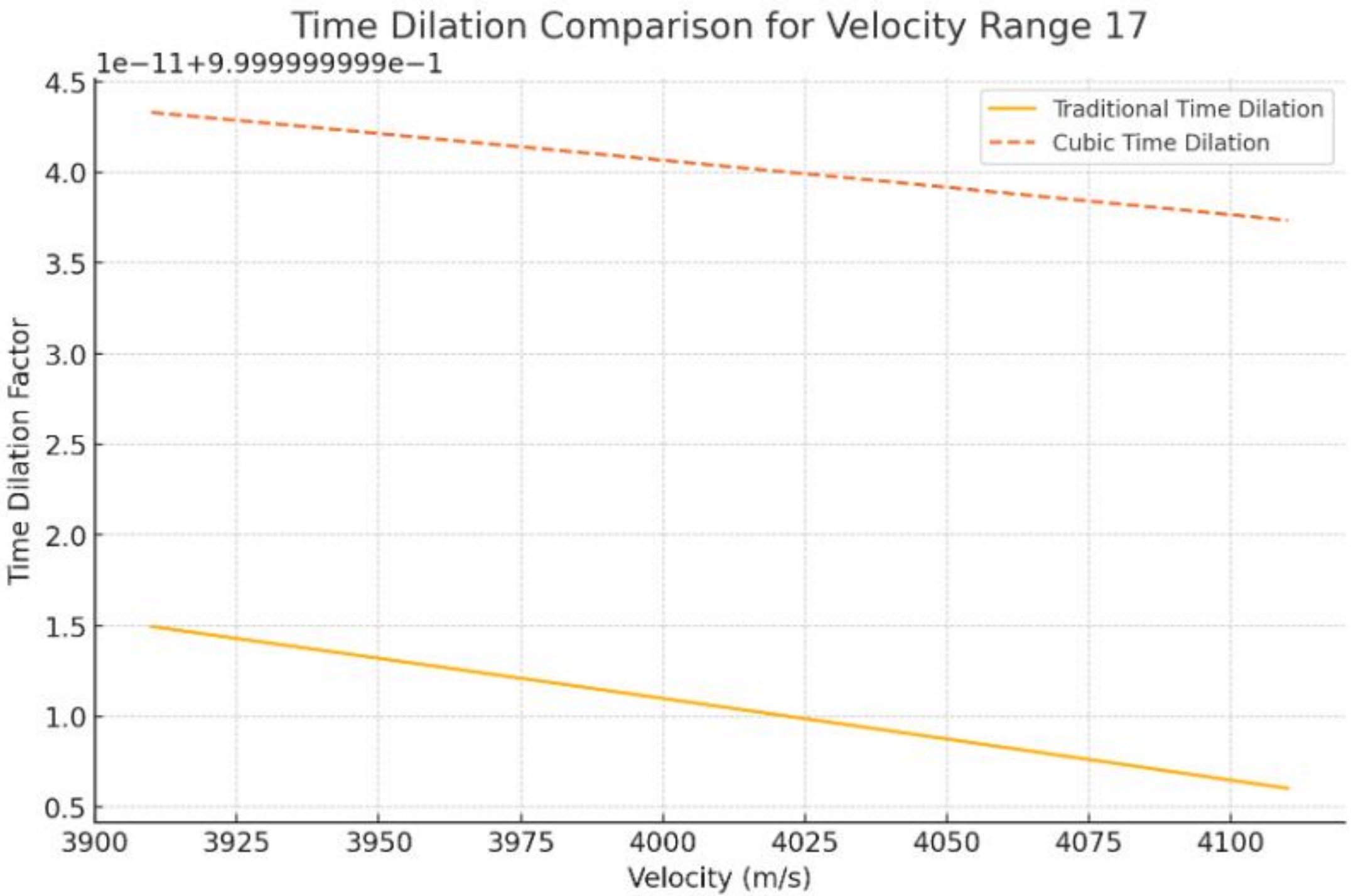

Time Dilation Comparison for Velocity Range 12
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
3850
3875
3900
3925
3950
3975
4000
4025
4050
Velocity (m/s)

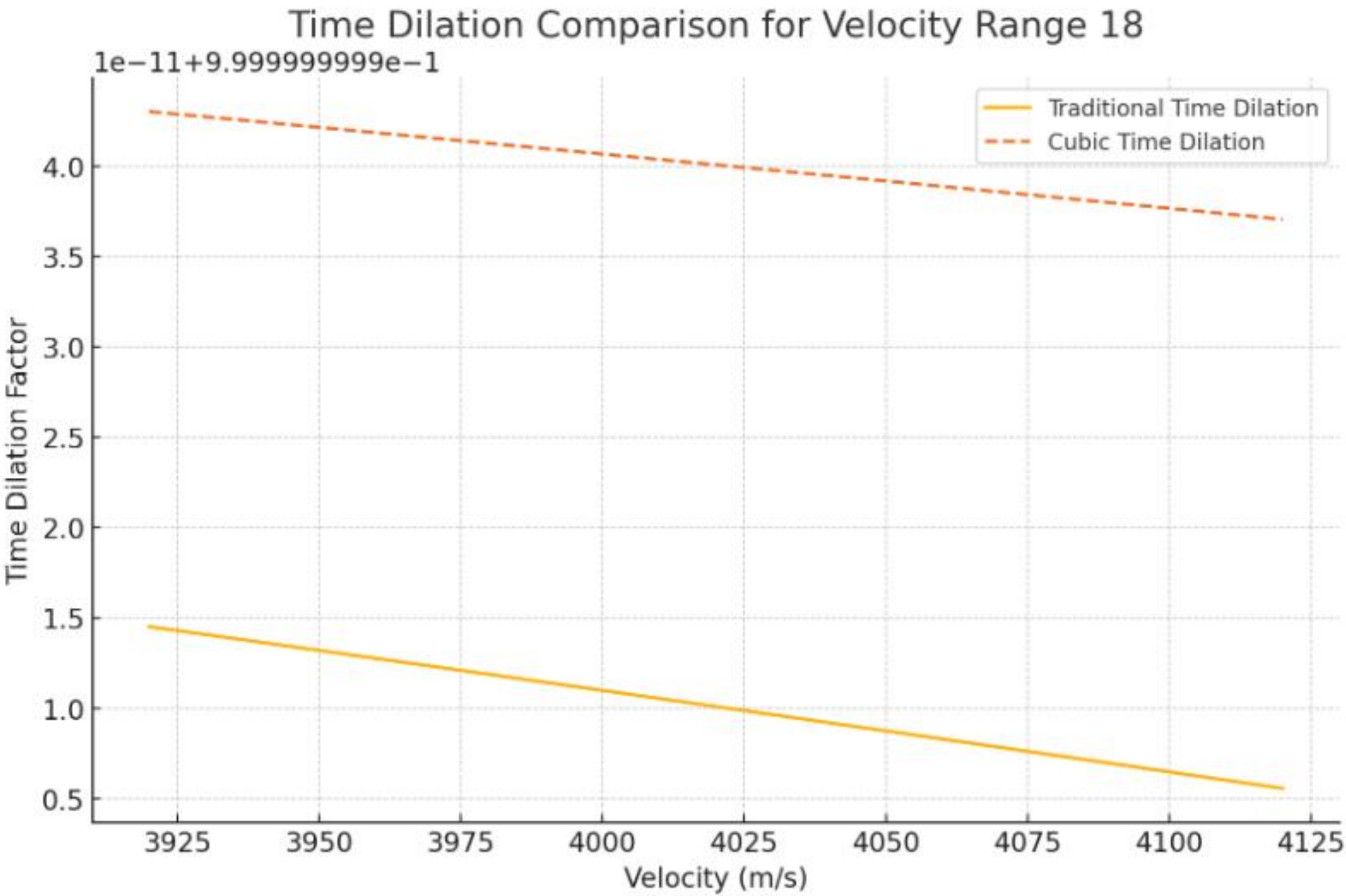

Time Dilation Comparison for Velocity Range 13
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
3875
3900
3925
3950
3975
4000
4025
4050
4075
Velocity (m/s)

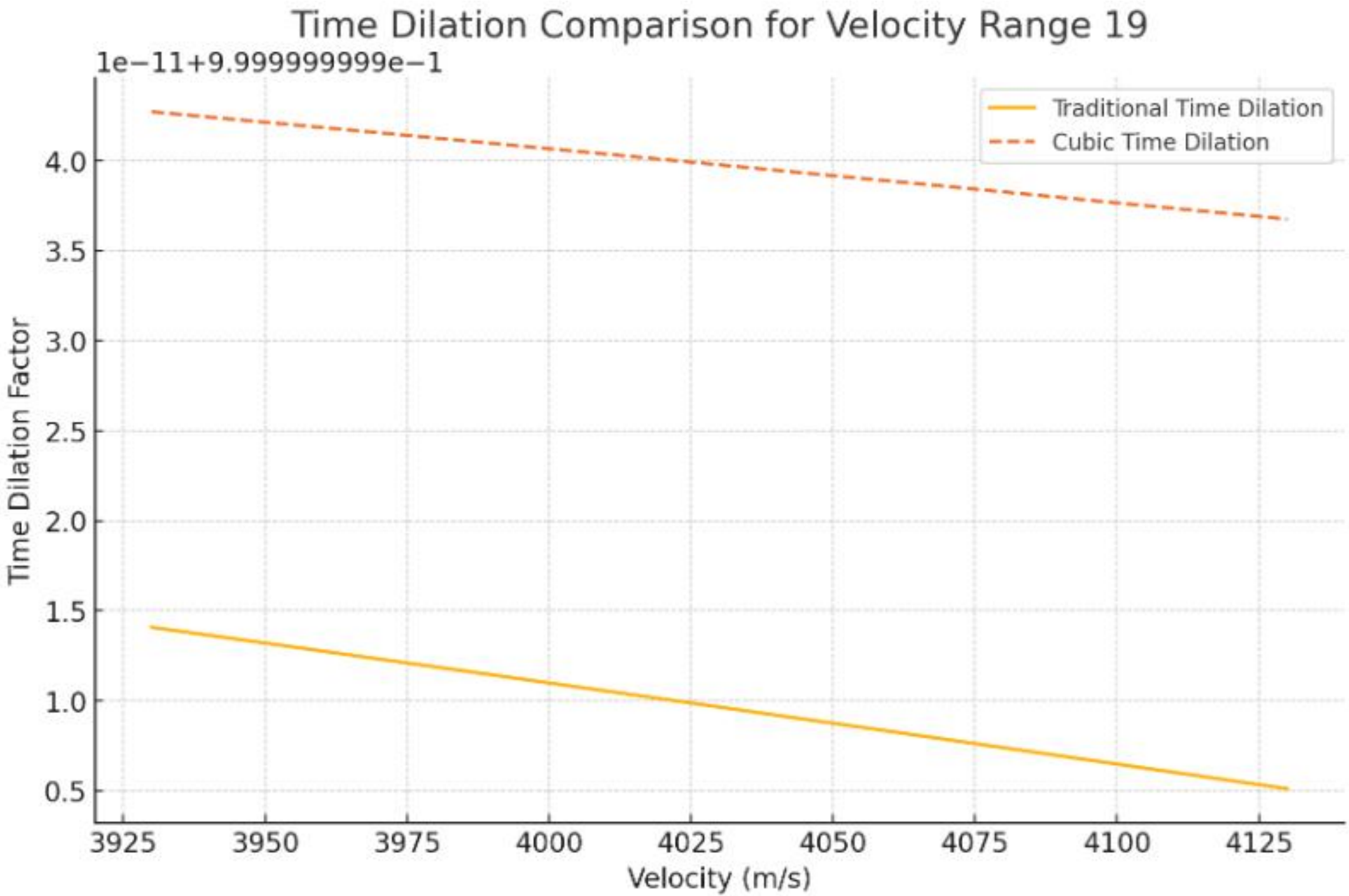

Time Dilation Comparison for Velocity Range 14
1e−11+9.999999999e−1
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
Time Dilation Factor
3875
3900
3925
3950
3975
4000
4025
4050
4075
Velocity (m/s)
Traditional Time Dilation
Cubic Time Dilation

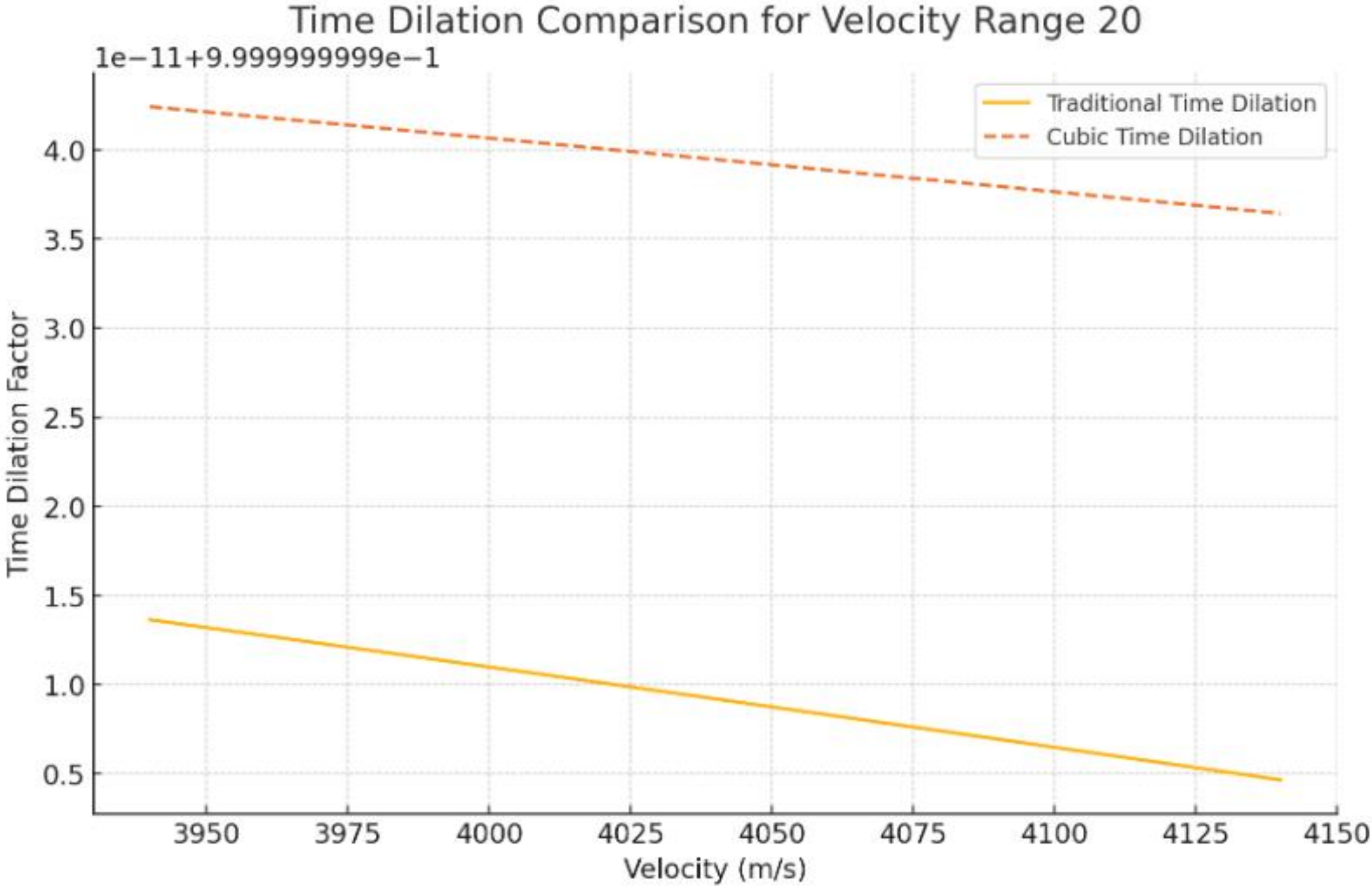

Time Dilation Comparison for Velocity Range 15
1e−11+9.999999999e−1
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
Time Dilation Factor
Traditional Time Dilation
Cubic Time Dilation
3900
3925
3950
3975
4000
4025
4050
4075
4100
Velocity (m/s)

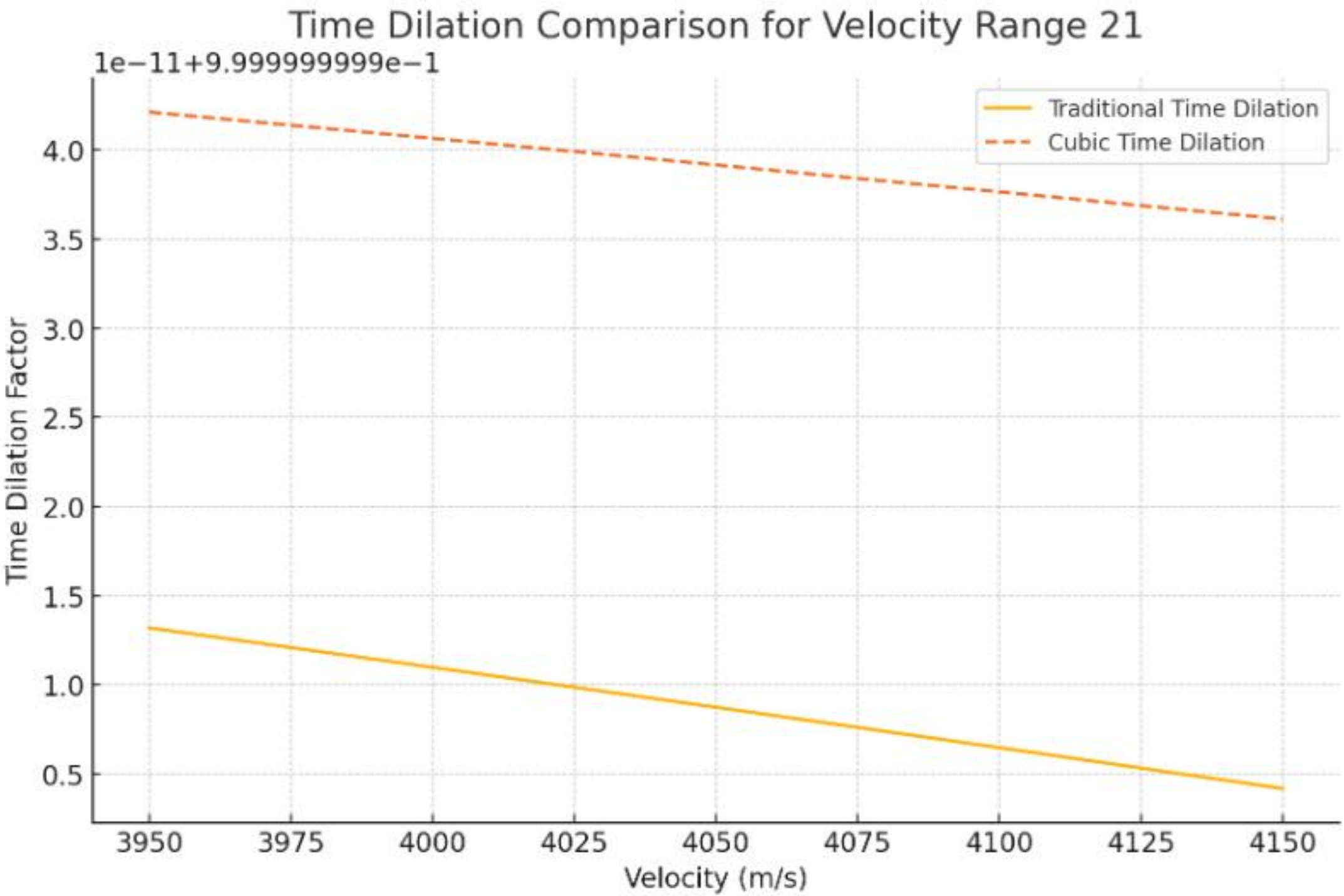

Time Dilation Comparison for Velocity Range 16
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
3900
3925
3950
3975
4000
4025
4050
4075
4100
Velocity (m/s)

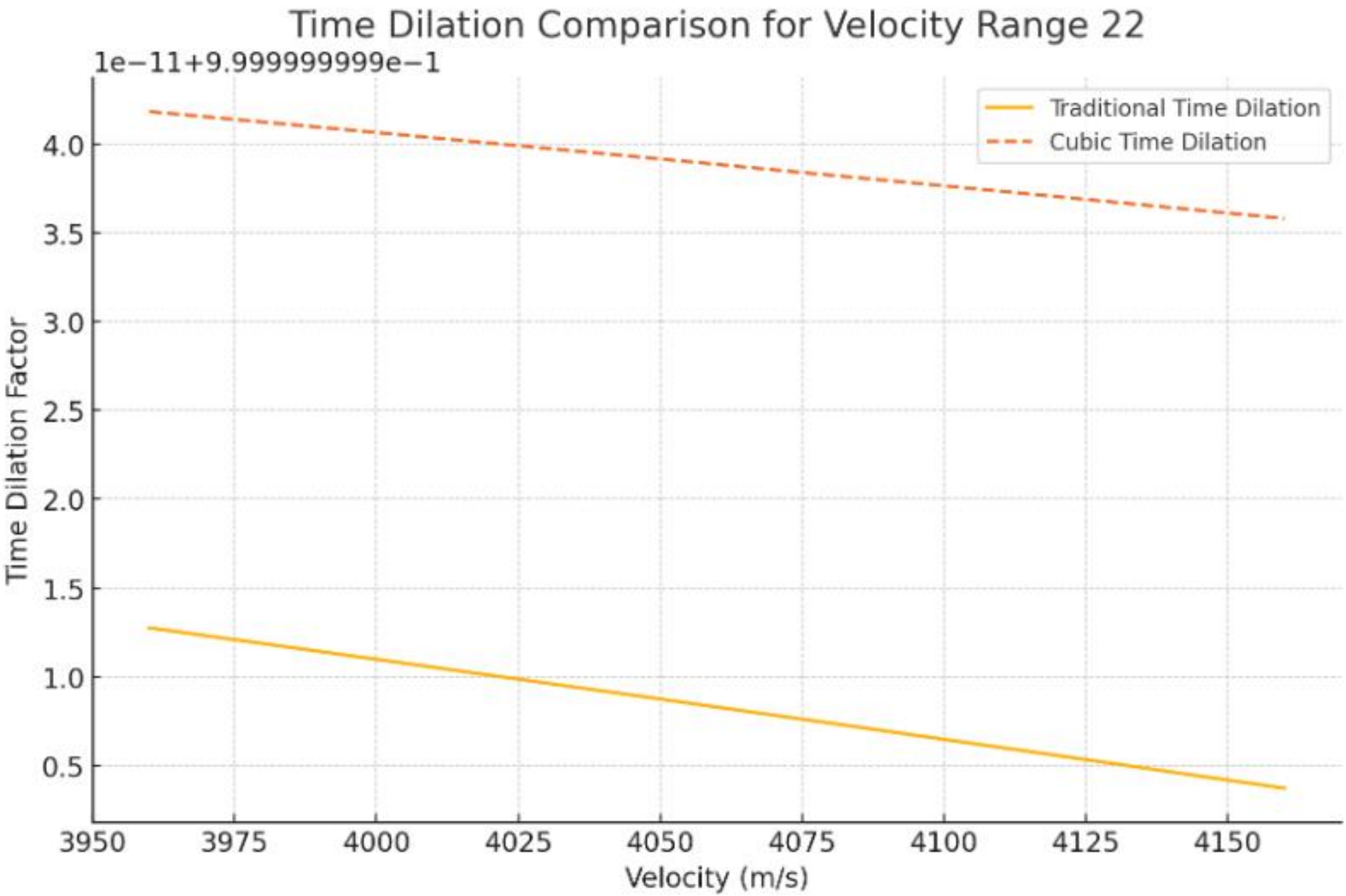

Time Dilation Comparison for Velocity Range 17
1e−11+9.999999999e−1
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
Time Dilation Factor
3900
3925
3950
3975
4000
4025
4050
4075
4100
Velocity (m/s)
Traditional Time Dilation
Cubic Time Dilation

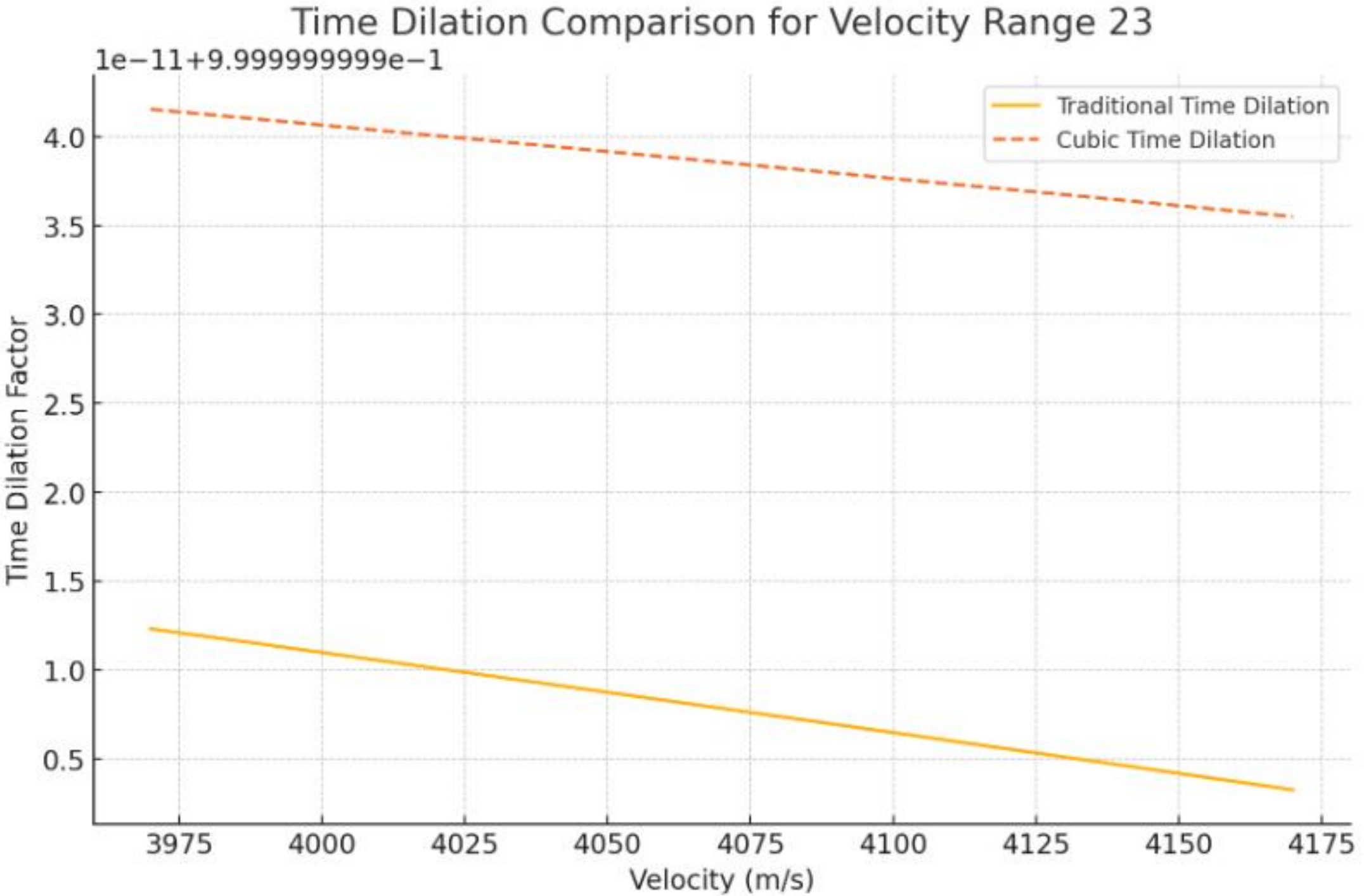

Time Dilation Comparison for Velocity Range 18
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
3925
3950
3975
4000
4025
4050
4075
4100
4125
Velocity (m/s)

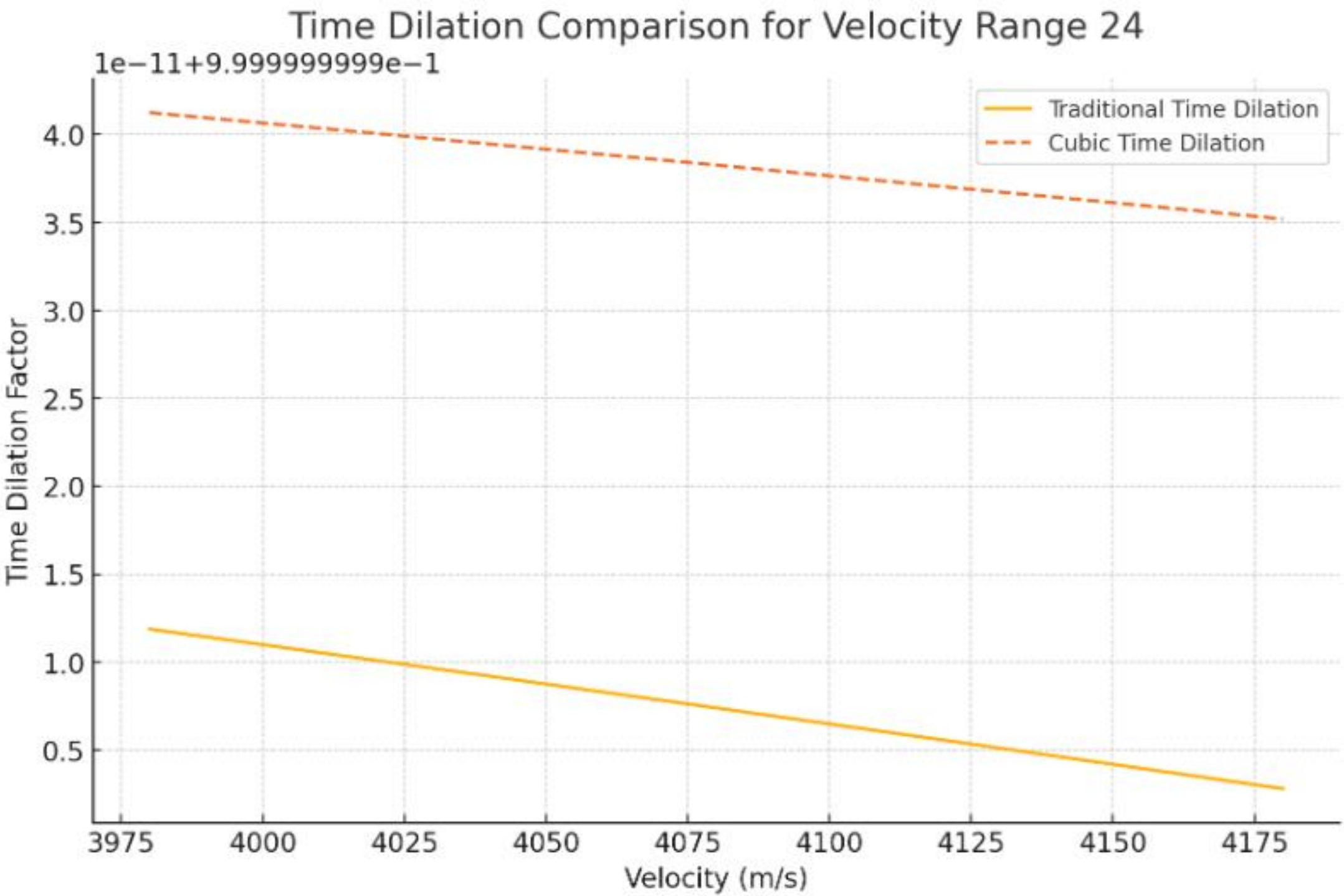

Time Dilation Comparison for Velocity Range 19
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
3925
3950
3975
4000
4025
4050
4075
4100
4125
Velocity (m/s)

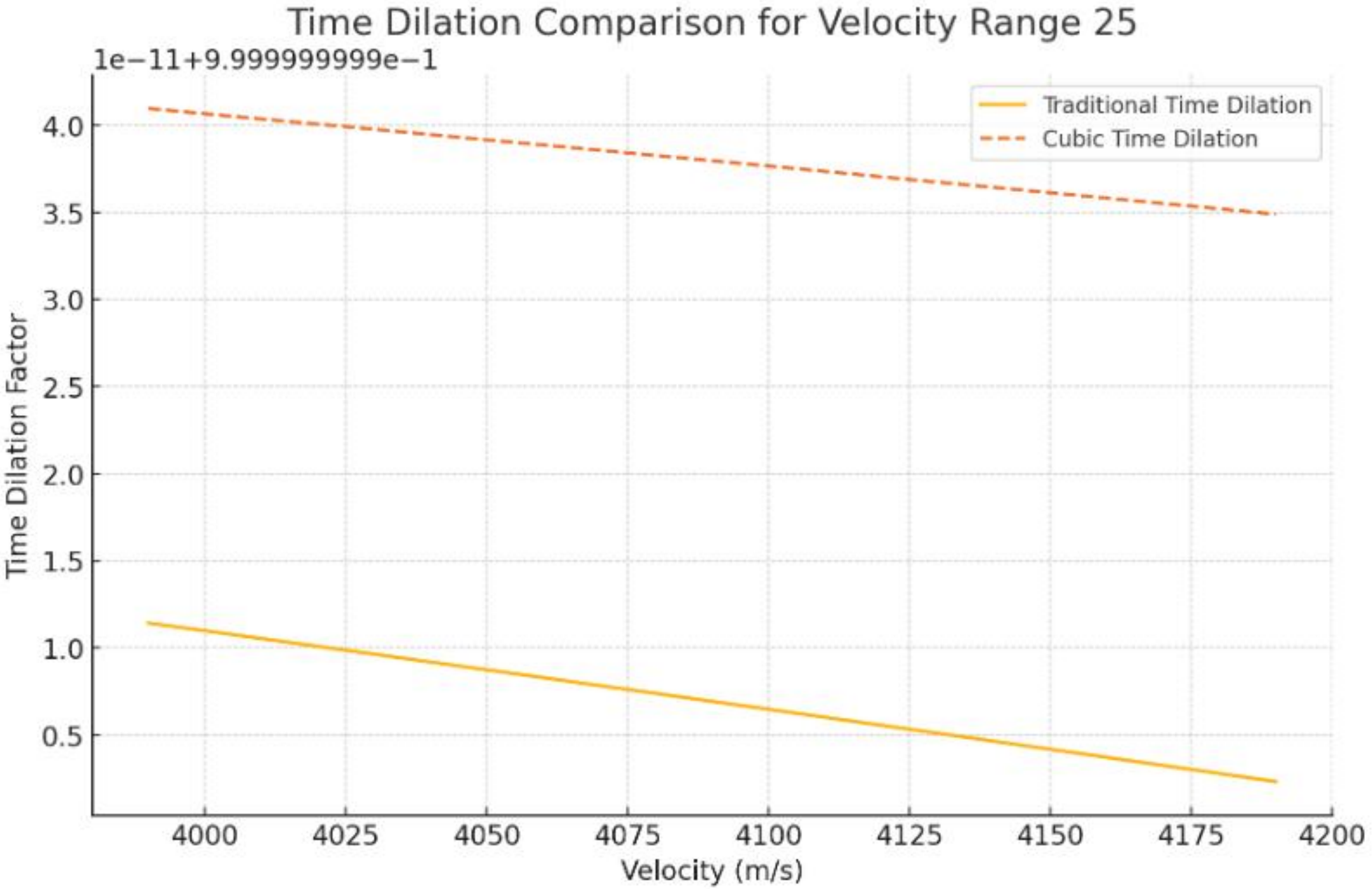

Time Dilation Comparison for Velocity Range 20
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
3950
3975
4000
4025
4050
4075
4100
4125
4150
Velocity (m/s)

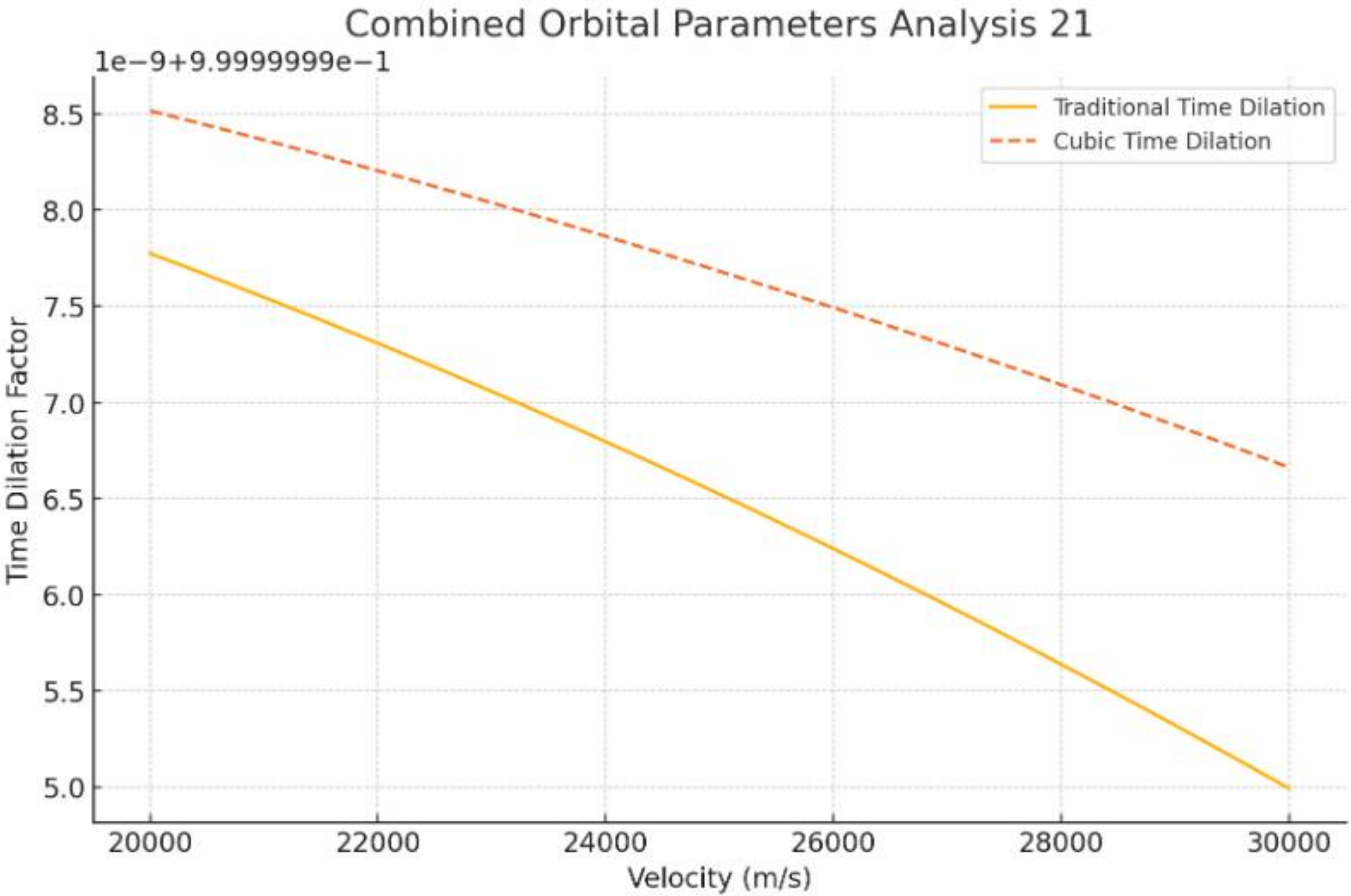

Time Dilation Comparison for Velocity Range 21
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
3950
3975
4000
4025
4050
4075
4100
4125
4150
Velocity (m/s)

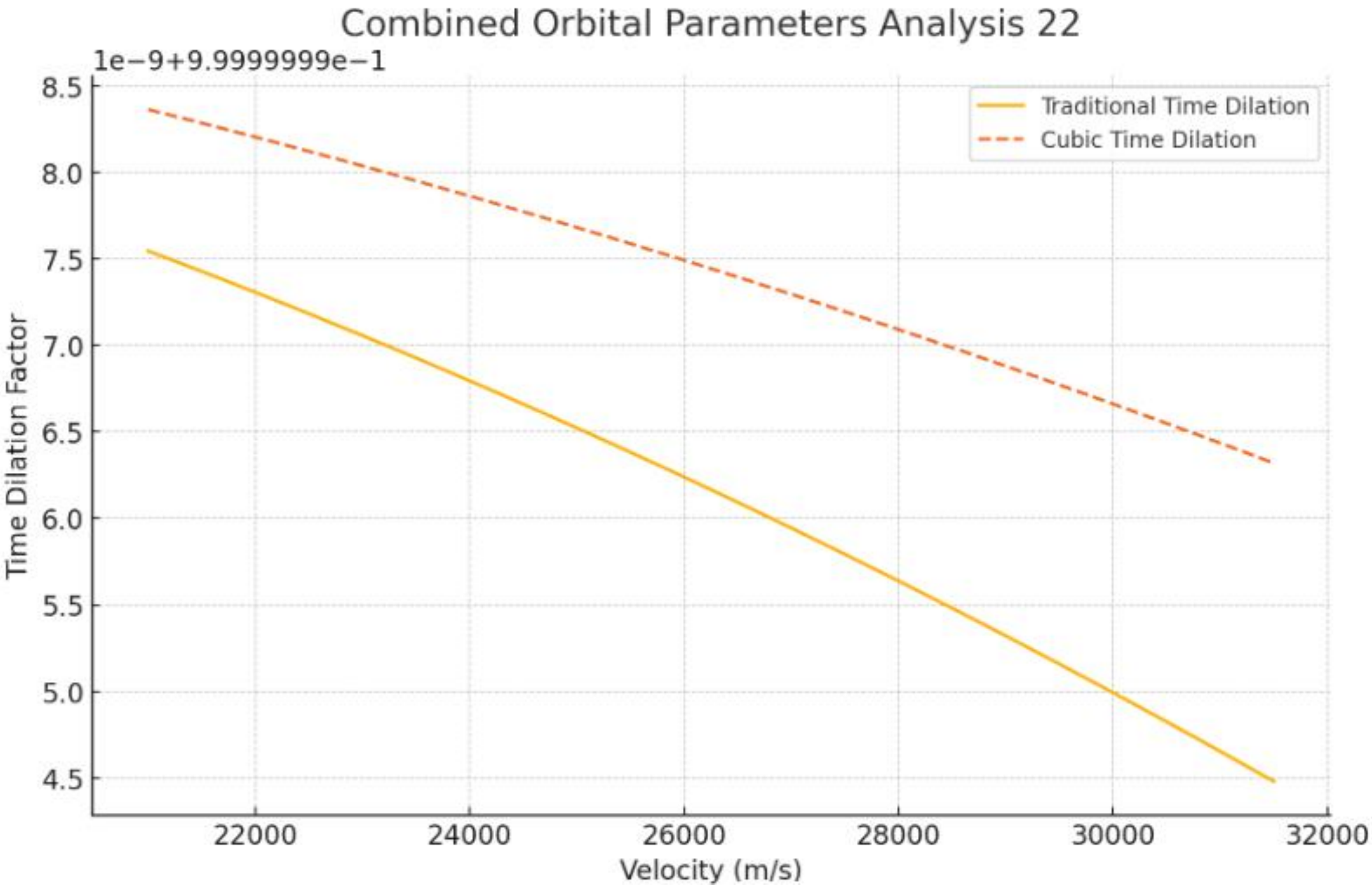

Time Dilation Comparison for Velocity Range 22
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
3950
3975
4000
4025
4050
4075
4100
4125
4150
Velocity (m/s)

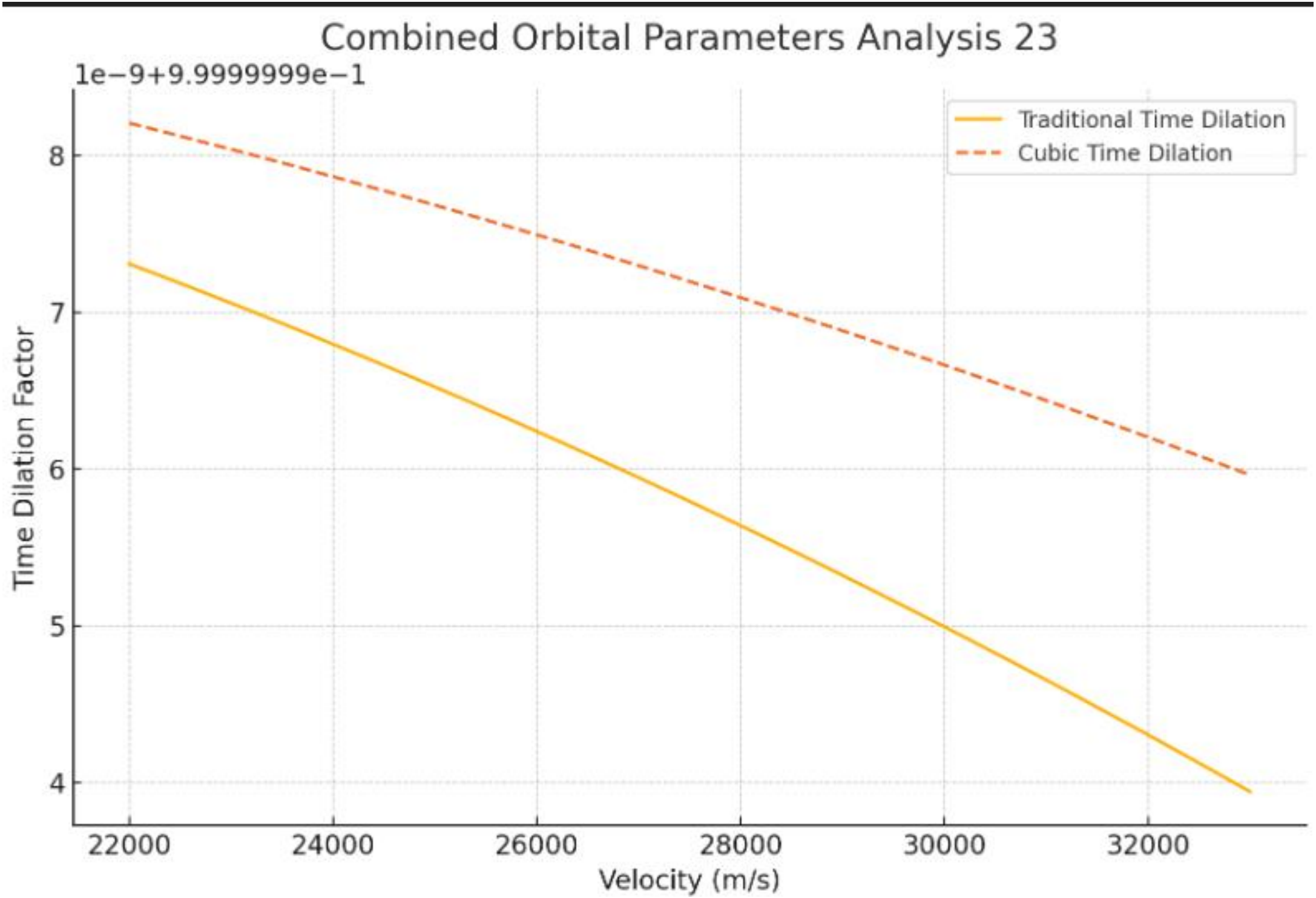

Time Dilation Comparison for Velocity Range 23
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
3975
4000
4025
4050
4075
4100
4125
4150
4175
Velocity (m/s)

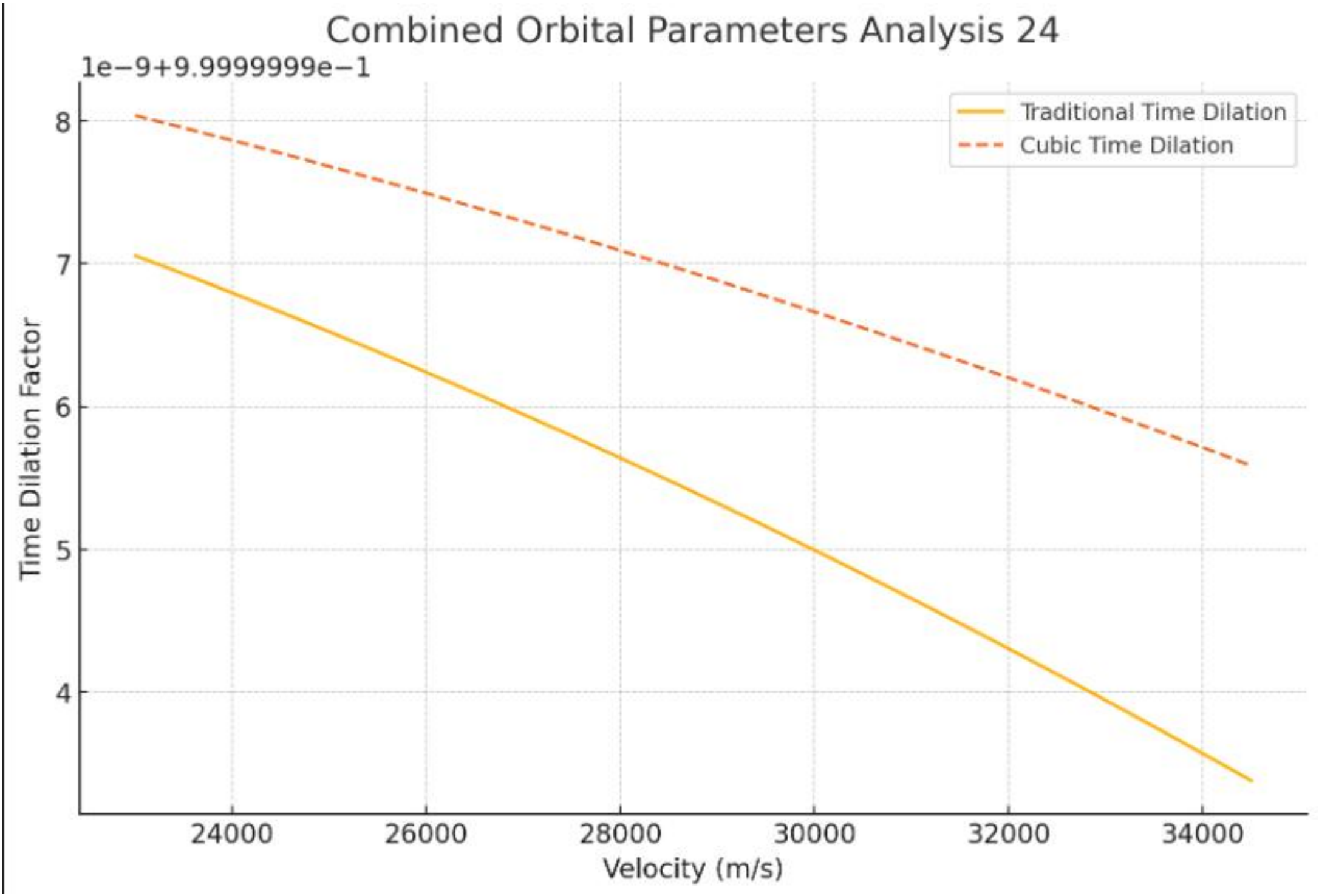

Time Dilation Comparison for Velocity Range 24
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
3975
4000
4025
4050
4075
4100
4125
4150
4175
Velocity (m/s)

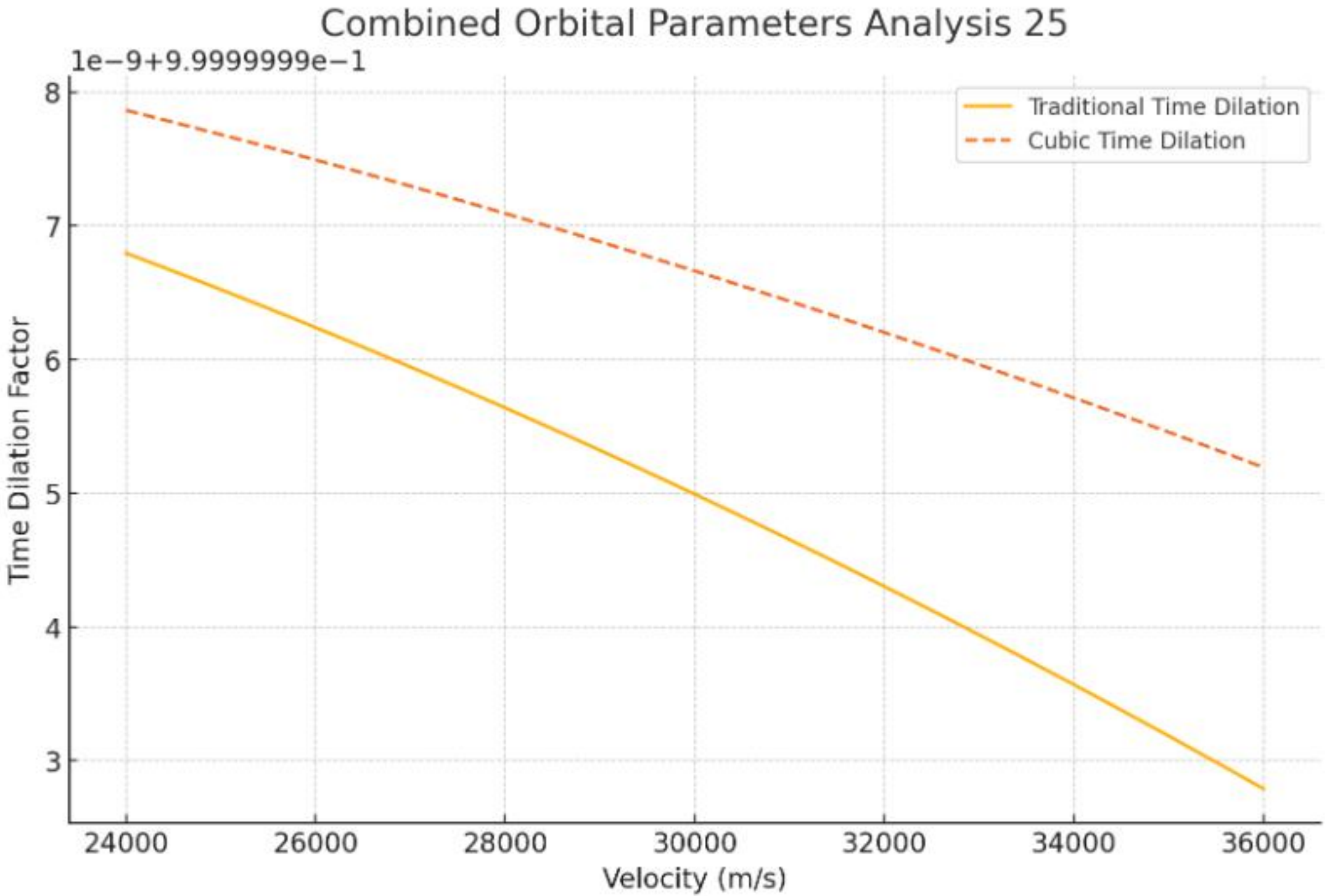

Time Dilation Comparison for Velocity Range 25
1e−11+9.999999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
4000
4025
4050
4075
4100
4125
4150
4175
4200
Velocity (m/s)

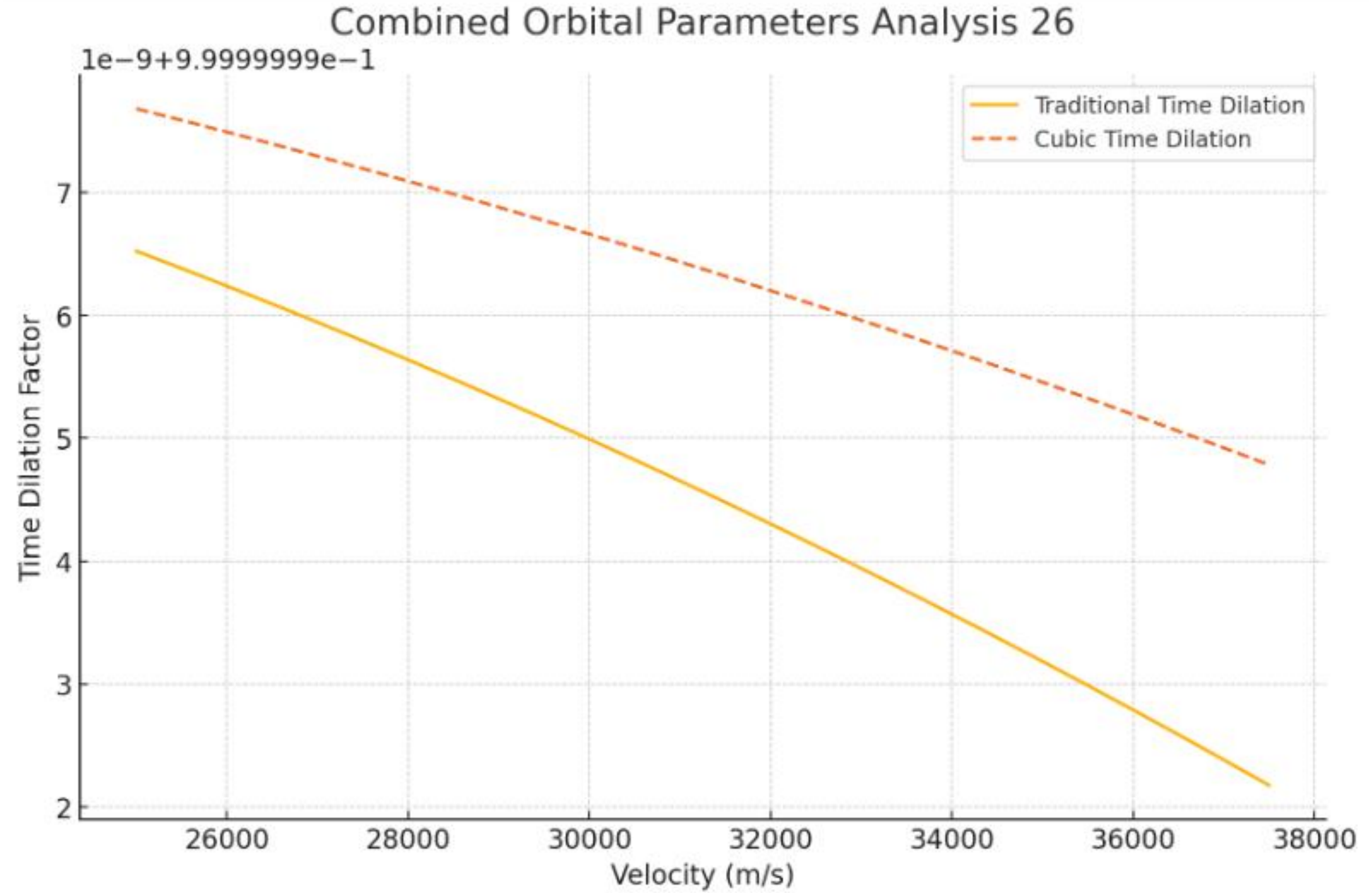

Combined Orbital Parameters Analysis 21
1e−9+9.9999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
8.5
8.0
7.5
7.0
6.5
6.0
5.5
5.0
20000
22000
24000
26000
28000
30000
Velocity (m/s)

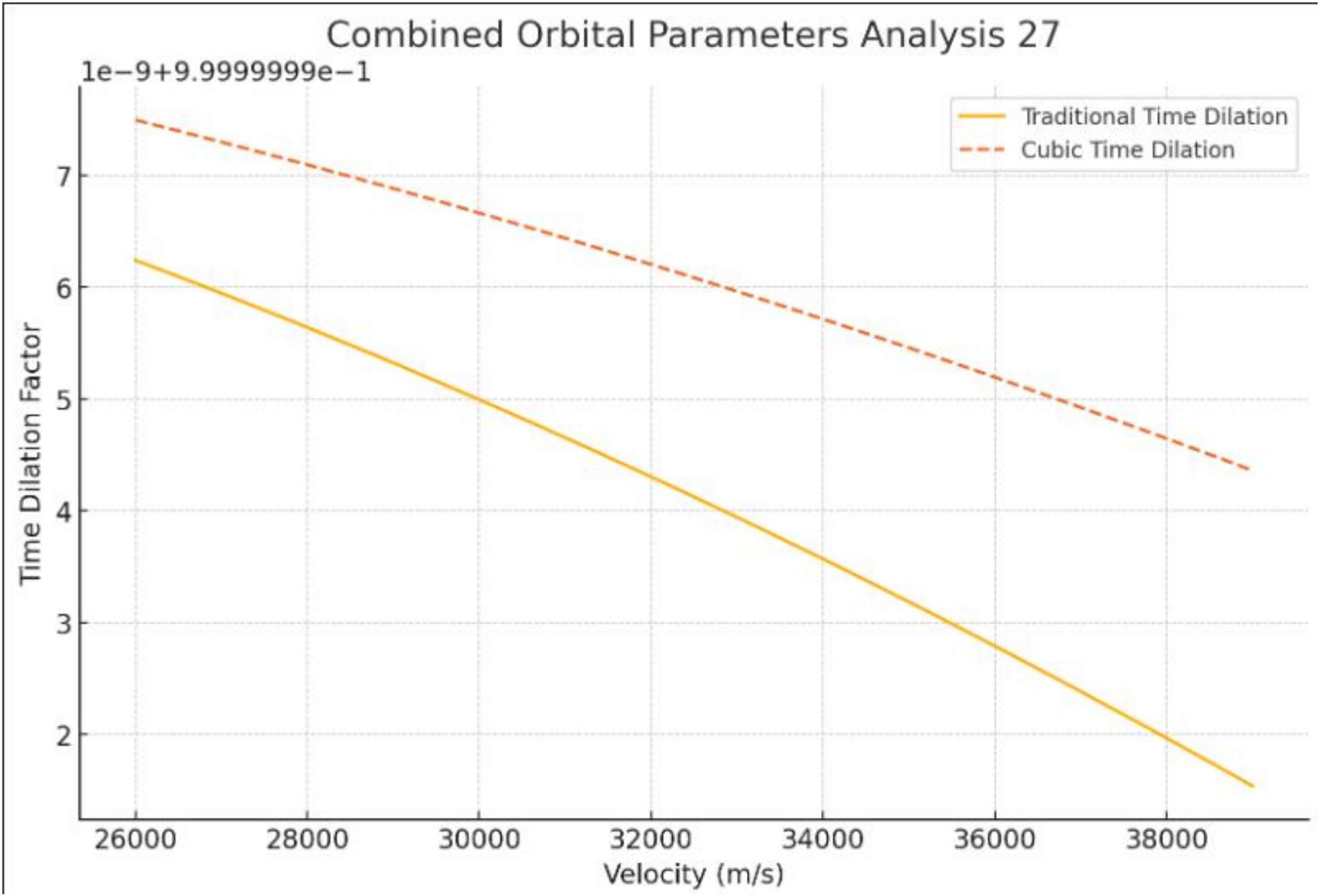

Combined Orbital Parameters Analysis 22
1e−9+9.9999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
8.5
8.0
7.5
7.0
6.5
6.0
5.5
5.0
4.5
22000
24000
26000
28000
30000
32000
Velocity (m/s)

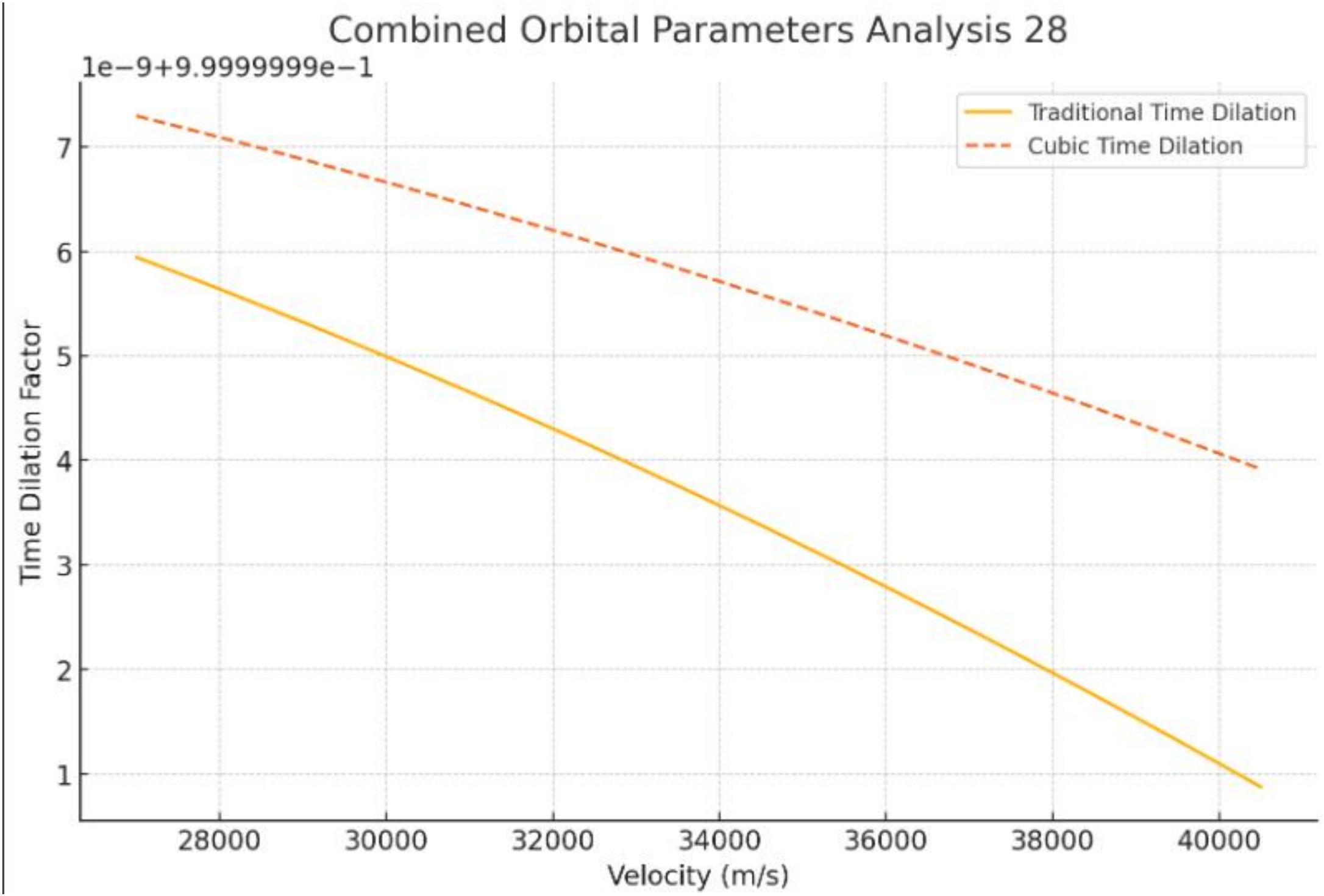

Combined Orbital Parameters Analysis 23
1e−9+9.9999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
8
7
6
5
4
22000
24000
26000
28000
30000
32000
Velocity (m/s)

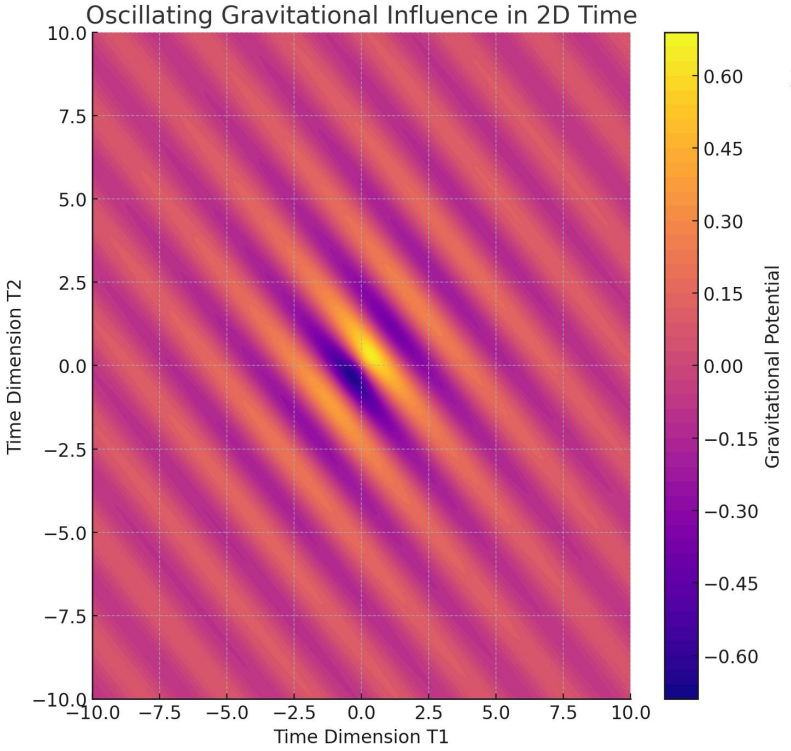

Combined Orbital Parameters Analysis 24
1e−9+9.9999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
8
7
6
5
4
24000
26000
28000
30000
32000
34000
Velocity (m/s)

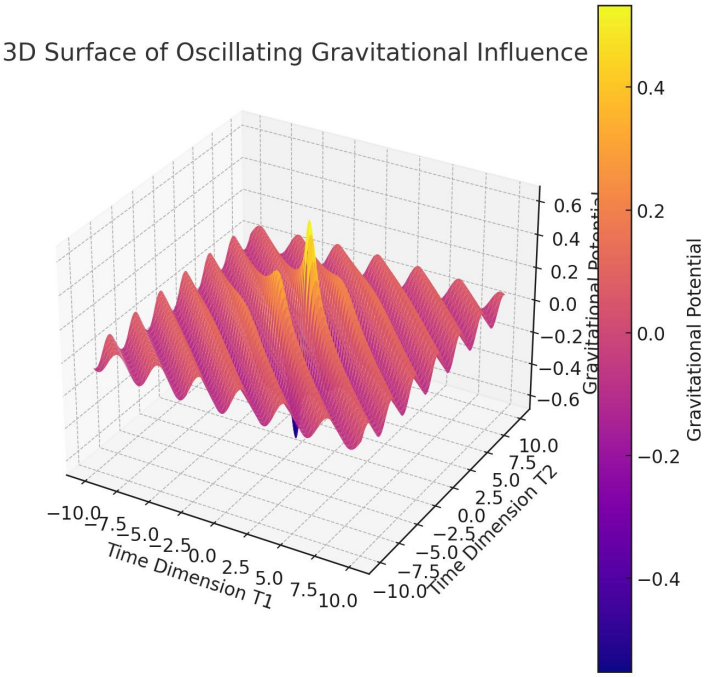

Combined Orbital Parameters Analysis 25
1e−9+9.9999999e−1
8
7
6
5
4
3
Time Dilation Factor
Traditional Time Dilation
Cubic Time Dilation
24000
26000
28000
30000
32000
34000
36000
Velocity (m/s)

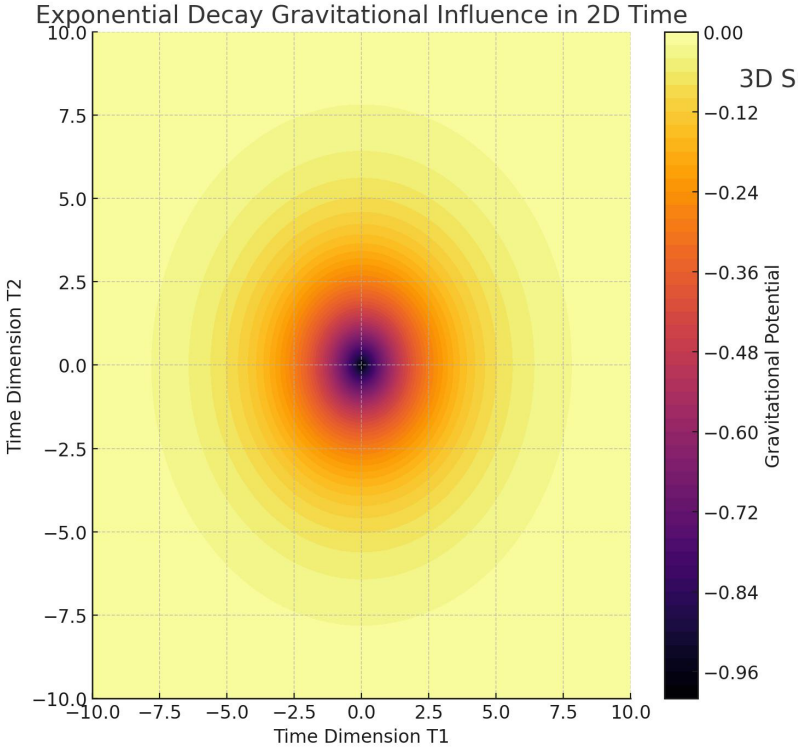

Combined Orbital Parameters Analysis 26
1e−9+9.9999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
7
6
5
4
3
2
26000
28000
30000
32000
34000
36000
38000
Velocity (m/s)

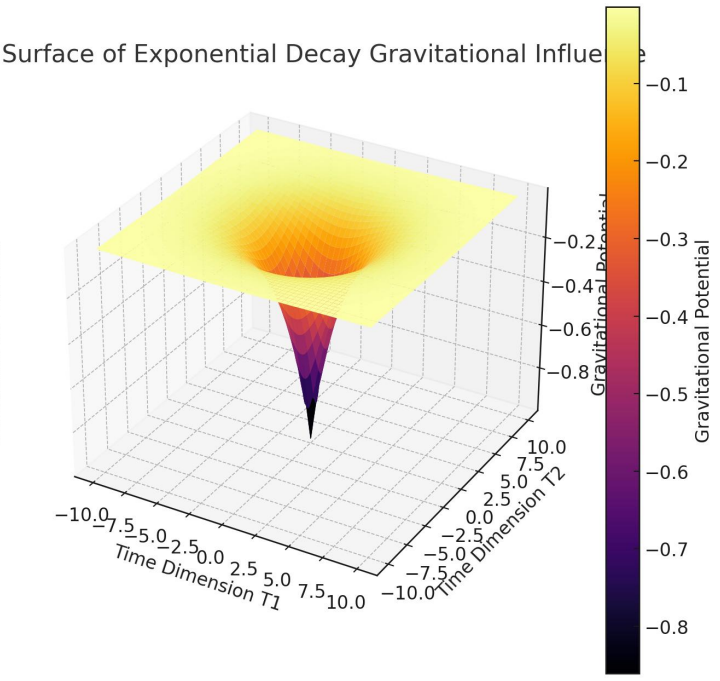

Combined Orbital Parameters Analysis 27
1e−9+9.9999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
7
6
5
4
3
2
26000
28000
30000
32000
34000
36000
38000
Velocity (m/s)


Combined Orbital Parameters Analysis 28
1e−9+9.9999999e−1
Traditional Time Dilation
Cubic Time Dilation
Time Dilation Factor
7
6
5
4
3
2
1
28000
30000
32000
34000
36000
38000
40000
Velocity (m/s)


Oscillating Gravitational Influence in 2D Time
Time Dimension T2
Time Dimension T1
Gravitational Potential


3D Surface of Oscillating Gravitational Influence
Time Dimension T1
Time Dimension T2
Gravitational Potential


Exponential Decay Gravitational Influence in 2D Time
Time Dimension T2
Time Dimension T1
Gravitational Potential


3D Surface of Exponential Decay Gravitational Influe
Time Dimension T1
Time Dimension T2
Gravitational Potential